Cardiff Metropolitan University

B.Sc. (Hons) in Software Engineering

Assignment Cover Sheet

<table>
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<tr><td colspan="6">Declaration</td></tr>
<tr><td colspan="6">I, Siyambalapitiya Ralalage Ayantha Shavinda Dassanayaka certify that the attached material is my original work. No other person’s work or ideas have been used without acknowledgement. Except where I have clearly stated that I have used some of this material elsewhere, I have not presented it for examination / assessment in any other course or unit at this or any other institution</td></tr>
<tr><td>Name/Signature</td><td colspan="2">Siyambalapitiya Ralalage Ayantha Shavinda Dassanayaka</td><td>Date</td><td colspan="2">14/08/2026</td></tr>
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<tr><td colspan="6">Result</td></tr>
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Assessor		Assessor		
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Comments on the Agreed mark				

SIMILARITY REPORT

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Ms. Sanjeewani Ponnamperuma for her guidance, support, and valuable feedback throughout this assignment. Her instruction helped me develop a better understanding of statistical analysis, network analysis, GIS, and Business Intelligence techniques and their practical applications in the aviation sector. I also appreciate the datasets, resources, and guidance provided to support the successful completion of this study.

ABSTRACT

This assignment evaluates the application of statistical, network, geospatial, and Business Intelligence (BI) techniques to support informed decision-making in Sri Lanka’s civil aviation sector. Statistical analysis is used to identify key factors influencing passenger demand and develop regression-based decision models. Social Network Analysis is applied to examine relationships and coordination among aviation stakeholders during high-pressure operational situations. GIS-based spatial analysis is used to identify suitable locations for PSR, SSR, and SMR radar systems at Bandaranaike International Airport. Finally, a BI dashboard is developed to analyse flight operations, delays, passenger activity, international traffic, and potential operational risks. The findings demonstrate how integrated analytical technologies can generate practical intelligence to improve operational efficiency, infrastructure planning, coordination, and strategic decision-making within the aviation sector.

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Task A — Statistical Analysis of Passenger Demand

A.1 Introduction

This section examines the factors associated with passenger demand using the Air Transportation Regression Dataset. The dependent variable is passenger demand, while six explanatory variables are considered: airport traffic, average income, fuel price, average ticket fare, flight frequency, and route distance.

The analysis includes exploratory analysis, outlier detection, distribution and normality testing, Pearson correlation analysis, simple linear regression, multiple linear regression, and regression diagnostic testing. The statistical significance level is $\alpha = 0.05$.

Variable	**Description**	**Unit**
Airport Traffic	Airport traffic handled	Passengers
Average Income	Average income of the population served	USD/year
Fuel Price	Average aviation fuel price	USD/litre
Average Ticket Fare	Average airline ticket fare	USD
Flight Frequency	Number of flights operated	Flights/day
Route Distance	Distance of the route	km
Passenger Demand	Passenger demand	Passengers

Table 1 Dataset Variables

A.2 Exploratory Data Analysis

Exploratory analysis was conducted to examine the distributions, potential outliers, and relationships between the explanatory variables and passenger demand.

A.2.1 Outlier Detection

Potential outliers were identified using the **Interquartile Range (IQR)** method. Observations below Q1 − 1.5(IQR) or above Q3 + 1.5(IQR) were classified as potential outliers.

Variable	**Outlier Count**
Airport Traffic	3
Average Income	2
Fuel Price	1
Average Ticket Fare	0
Flight Frequency	1
Route Distance	2
Passenger Demand	4

Table 2 Potential Outliers

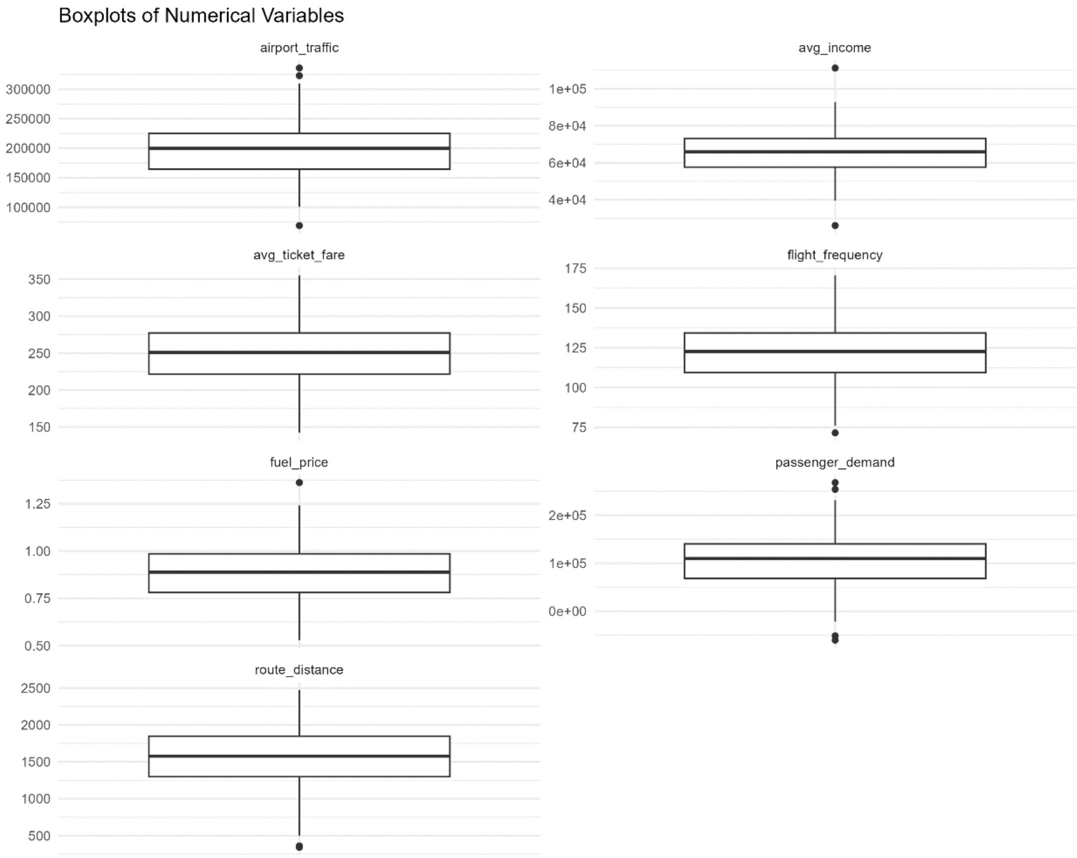

Figure 1 Boxplots of the numerical variables

Passenger demand contained the highest number of potential outliers, with four observations. However, these observations were retained because an IQR-based outlier is not necessarily an erroneous observation. Removing valid extreme values could distort the relationships within the dataset.

A.2.2 Distribution Analysis

Histograms were produced for all seven numerical variables to examine their distributional characteristics.

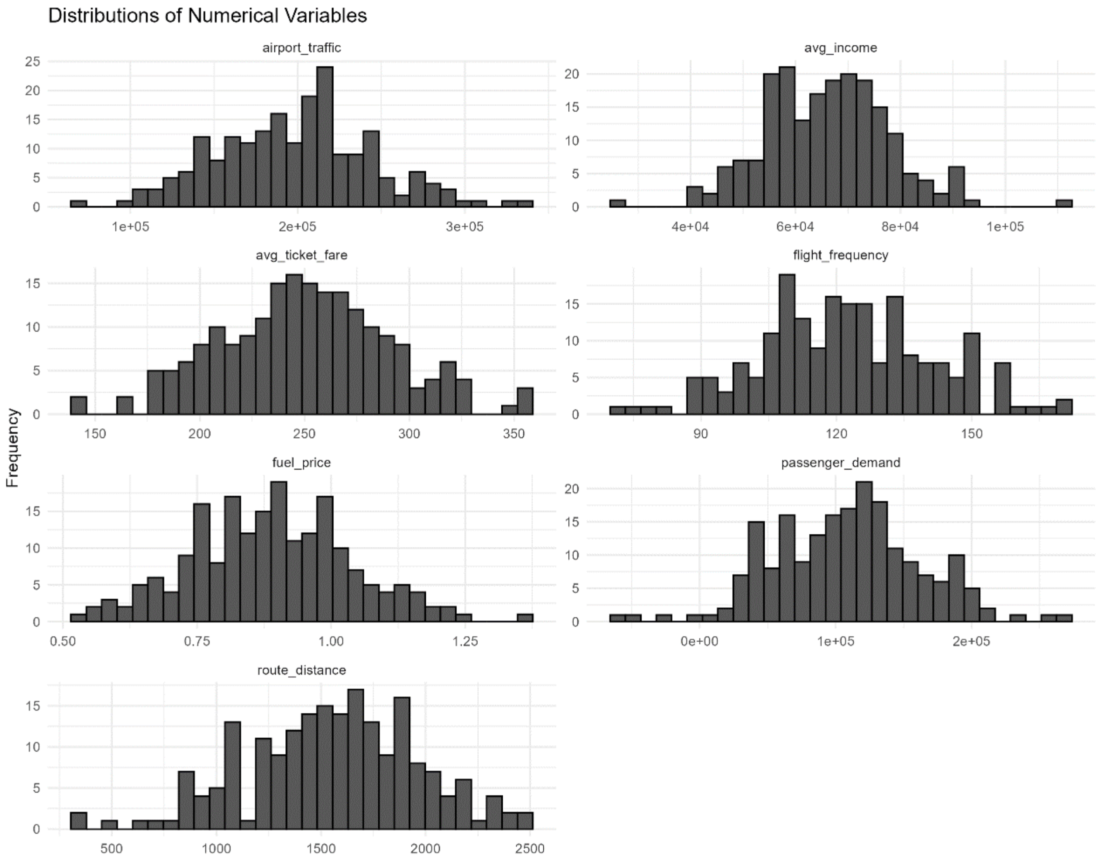

Figure 2 Distributions of the numerical variables

The distributions did not indicate severe irregularities that would prevent further analysis. Formal normality testing was subsequently performed using the Shapiro-Wilk test.

A.2.3 Relationships Between Explanatory Variables and Passenger Demand

Scatterplots with fitted linear regression lines were used to examine the relationships between each explanatory variable and passenger demand.

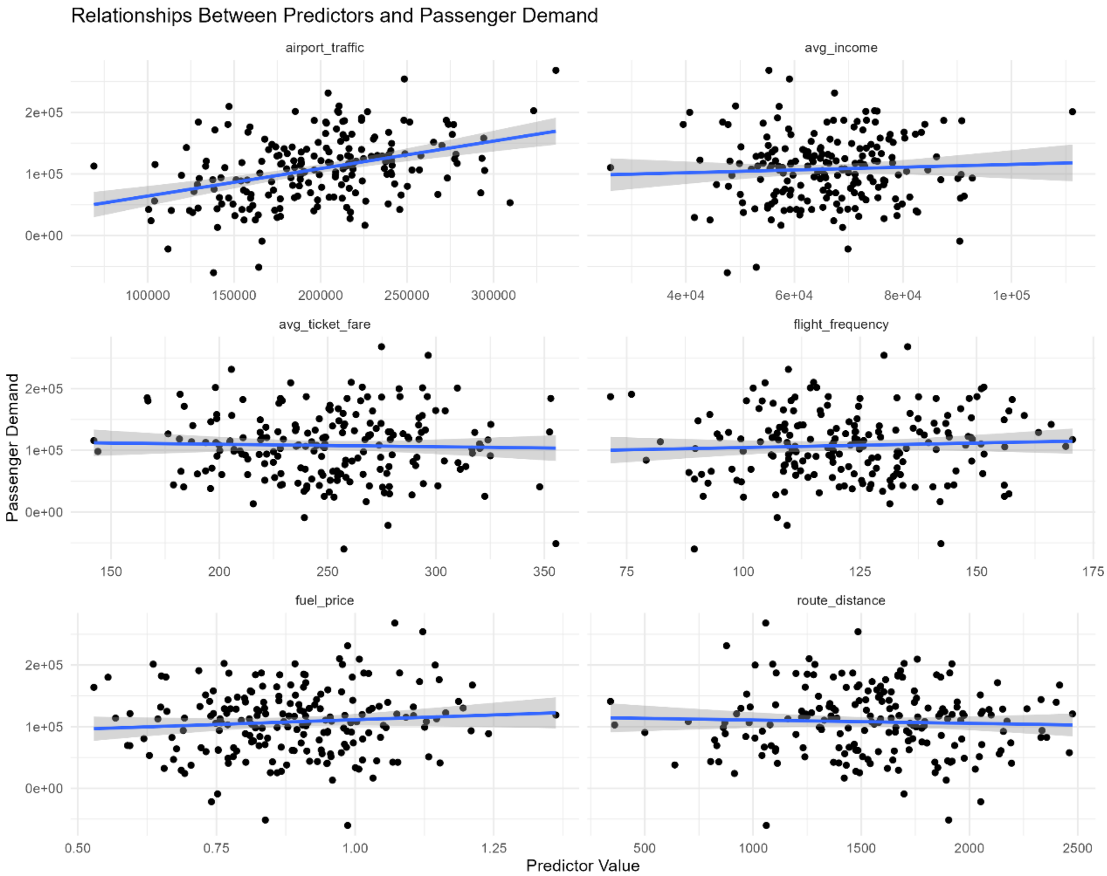

Figure 3 Scatterplots showing the relationships between the six explanatory variables and passenger demand

The six scatterplots show that airport traffic has the clearest positive relationship with passenger demand. The remaining variables show comparatively weak linear relationships.

A.3 Correlation Analysis

Pearson correlation analysis was conducted to quantify the direction and strength of linear relationships between the explanatory variables and passenger demand.

Predictor	**Correlation (r)**	**Interpretation**
Airport Traffic	0.3854	Moderate positive
Average Income	0.0495	Very weak positive
Fuel Price	0.0857	Very weak positive
Average Ticket Fare	-0.0304	Very weak negative
Flight Frequency	0.0522	Very weak positive
Route Distance	-0.0412	Very weak negative

Table 3 Correlation with Passenger Demand

Airport traffic produced the strongest correlation with passenger demand (**r = 0.3854**), indicating a moderate positive linear relationship. The remaining variables demonstrated very weak linear relationships.

Figure 4 Pearson correlation matrix

The explanatory variables also showed relatively weak correlations with one another, providing preliminary evidence that severe multicollinearity may not be a major concern in the multiple regression model

A.4 Hypothesis-Based Statistical Testing

A.4.1 Normality Testing

The **Shapiro–Wilk test** was applied to assess whether the variables were consistent with a normal distribution. The test evaluates the null hypothesis that the sample was drawn from a normally distributed population (Shapiro and Wilk, 1965).

Hypotheses

- **H_0:** The variable follows a normal distribution.
- **H_1:** The variable does not follow a normal distribution.

Table A.4: Shapiro-Wilk Normality Test

Variable	**W**	**p-value**	**Decision**
Airport Traffic	0.9956	0.8290	Fail to reject H_0
Average Income	0.9910	0.2498	Fail to reject H_0
Fuel Price	0.9964	0.9218	Fail to reject H_0
Average Ticket Fare	0.9952	0.7774	Fail to reject H_0
Flight Frequency	0.9940	0.6003	Fail to reject H_0
Route Distance	0.9928	0.4398	Fail to reject H_0
Passenger Demand	0.9927	0.4249	Fail to reject H_0

Table 4 Shapiro-Wilk Normality Test

All p-values exceed 0.05. Therefore, there is **no statistically significant evidence of non-normality** for the variables examined.

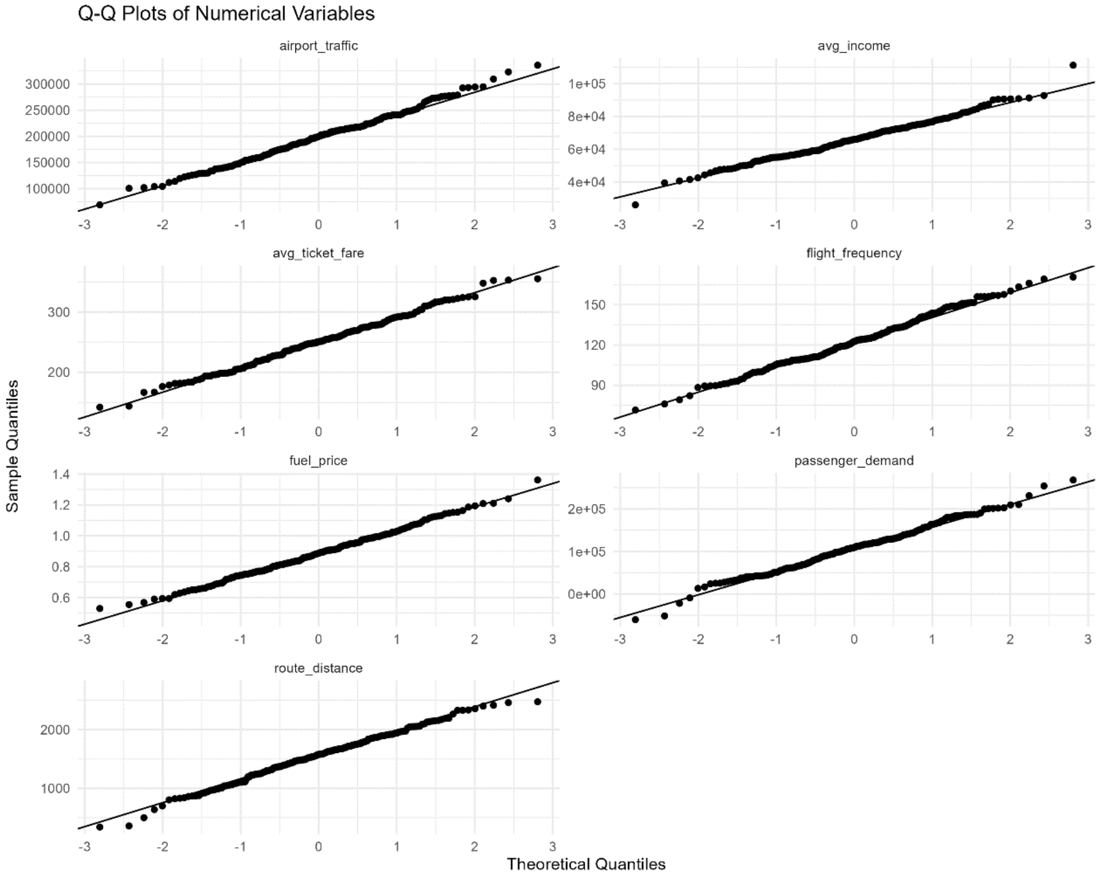

Figure 5 Q-Q plot of passenger demand

A.4.2 Pearson Correlation Hypothesis Testing

Individual Pearson correlation tests were conducted between each explanatory variable and passenger demand.

Hypotheses

- **H_0:** There is no statistically significant linear relationship between the predictor and passenger demand.
- **H_1:** There is a statistically significant linear relationship between the predictor and passenger demand.

Table A.5: Pearson Correlation Tests

Predictor	**r**	**p-value**	**Decision**
Airport Traffic	0.3854	<0.001	Reject H_0
Average Income	0.0495	0.4860	Fail to reject H_0
Fuel Price	0.0857	0.2277	Fail to reject H_0
Average Ticket Fare	-0.0304	0.6692	Fail to reject H_0
Flight Frequency	0.0522	0.4628	Fail to reject H_0
Route Distance	-0.0412	0.5628	Fail to reject H_0

Table 5 Pearson Correlation Tests

Only airport traffic demonstrated a statistically significant individual linear relationship with passenger demand (**$r = 0.3854$, $p < 0.001$**).

The other five explanatory variables did not demonstrate statistically significant individual linear relationships with passenger demand.

A.5 Simple Linear Regression

Six simple linear regression models were developed, with passenger demand as the dependent variable and each explanatory variable considered individually.

Table A.6: Simple Regression Results

Predictor	R^2	Adjusted R^2	p-value
Airport Traffic	0.1485	0.1442	<0.001
Fuel Price	0.0073	0.0023	0.2277
Flight Frequency	0.0027	-0.0023	0.4628
Average Income	0.0025	-0.0026	0.4859
Route Distance	0.0017	-0.0033	0.5628
Average Ticket Fare	0.0009	-0.0041	0.6692

Table 6 Simple Regression Results

Airport traffic produced the strongest simple regression model, explaining **14.85%** of the variation in passenger demand.

The model was statistically significant (**$R^2 = 0.1485$, $p < 0.001$**), while the remaining five models were not statistically significant.

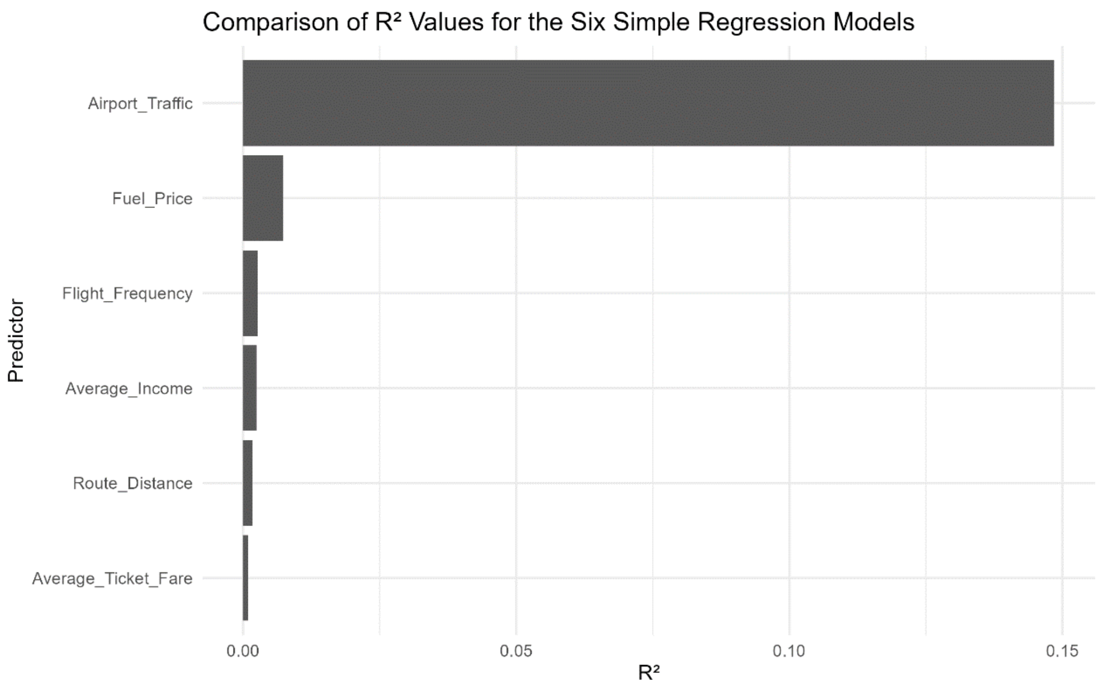

Figure 6 Comparison of R^2 values for the six simple regression models

A.6 Multiple Linear Regression

A multiple linear regression model was developed to examine the combined relationship between the six explanatory variables and passenger demand.

The model was:

Passenger Demand=−37745.0564+0.4759(Airport Traffic)+0.0805(Average Income)+53537.8375(Fuel Price)−111.7682(Average Ticket Fare)+221.2152(Flight Frequency)−0.3826(Route Distance)

A.6.1 Overall Model Fit

Statistic	Value
R^2	0.1789
Adjusted R^2	0.1534
Residual Standard Error	49,554.65
F-statistic	7.0082
Degrees of freedom	6, 193
Model p-value	<0.001

Table 7 Multiple Regression Model Fit

The model explains approximately **17.89%** of the variation in passenger demand.

The overall F-test was statistically significant:

$F(6,193) = 7.008$, $p < 0.001$.

Therefore, the null hypothesis that all regression coefficients are simultaneously equal to zero was rejected. The six predictors collectively provide statistically significant explanatory information about passenger demand.

However, the relatively low R^2 indicates that a substantial proportion of passenger-demand variation remains unexplained.

A.6.2 Regression Coefficients

Predictor	Estimate	p-value	Significance
Airport Traffic	0.4759	<0.001	Significant
Average Income	0.0805	0.7901	Not significant
Fuel Price	53,537.84	0.0268	Significant
Average Ticket Fare	-111.77	0.2084	Not significant
Flight Frequency	221.22	0.2436	Not significant
Route Distance	-0.3826	0.9647	Not significant

Table 8 Regression Coefficients

Airport traffic was statistically significant and positively associated with passenger demand. Holding the other predictors constant, a one-unit increase in airport traffic was associated with an estimated increase of approximately **0.476 units** in passenger demand.

Fuel price was also statistically significant. However, its positive coefficient should be interpreted cautiously because statistical association does not establish causation.

Average income, ticket fare, flight frequency, and route distance were not statistically significant predictors at the 5% level.

A.6.3 Confidence Intervals

Predictor	Lower 95% CI	Upper 95% CI
Airport Traffic	0.3237	0.6281
Average Income	-0.5153	0.6763
Fuel Price	6,204.04	100,871.60
Average Ticket Fare	-286.41	62.87
Flight Frequency	-151.81	594.24
Route Distance	-17.41	16.64

Table 9 95% Confidence Intervals

The confidence intervals for airport traffic and fuel price do not include zero, supporting their statistical significance. The remaining confidence intervals include zero, consistent with their non-significant p-values.

A.7 Regression Diagnostic Analysis

Diagnostic testing was conducted to assess whether the assumptions of the multiple regression model were reasonably satisfied.

The following assumptions were examined:

- Multicollinearity
- Constant variance
- Residual independence
- Residual normality

A.7.1 Multicollinearity

Variance Inflation Factor (VIF) values were calculated.

Predictor	**VIF**
Airport Traffic	1.046
Average Income	1.037
Fuel Price	1.038
Average Ticket Fare	1.057
Flight Frequency	1.060
Route Distance	1.026

Table 10 VIF Results

All VIF values are close to 1 and well below commonly used thresholds. Therefore, there is no evidence of problematic multicollinearity.

A.7.2 Heteroscedasticity

The Breusch-Pagan test was used to assess whether the residual variance was constant.

BP = 4.8406, df = 6, p = 0.5644

Since $p > 0.05$, H_0 was not rejected. Therefore, there is no statistically significant evidence of heteroscedasticity.

A.7.3 Residual Independence

The Durbin-Watson test produced:

DW = 2.0712, p = 0.6972

Since $p > 0.05$ and the statistic is close to 2, there is no statistically significant evidence of residual autocorrelation.

A.7.4 Residual Normality

The Shapiro-Wilk test for the regression residuals produced:

$W = 0.99433$, $p = 0.6501$

Since $p > 0.05$, H_0 was not rejected. Therefore, there is no statistically significant evidence of non-normal residuals.

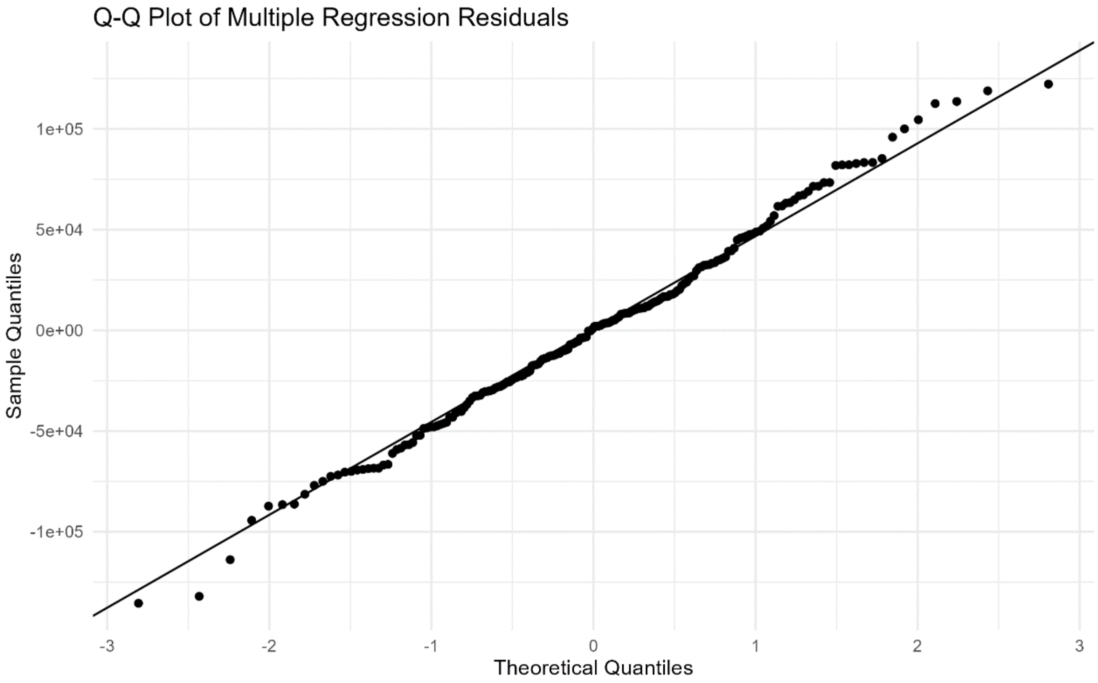

Figure 7 Q-Q plot of multiple regression residuals

A.7.5 Regression Diagnostic Plots

The standard four-panel regression diagnostic plots were generated to visually assess residual behaviour, linearity, influential observations, and variance consistency.

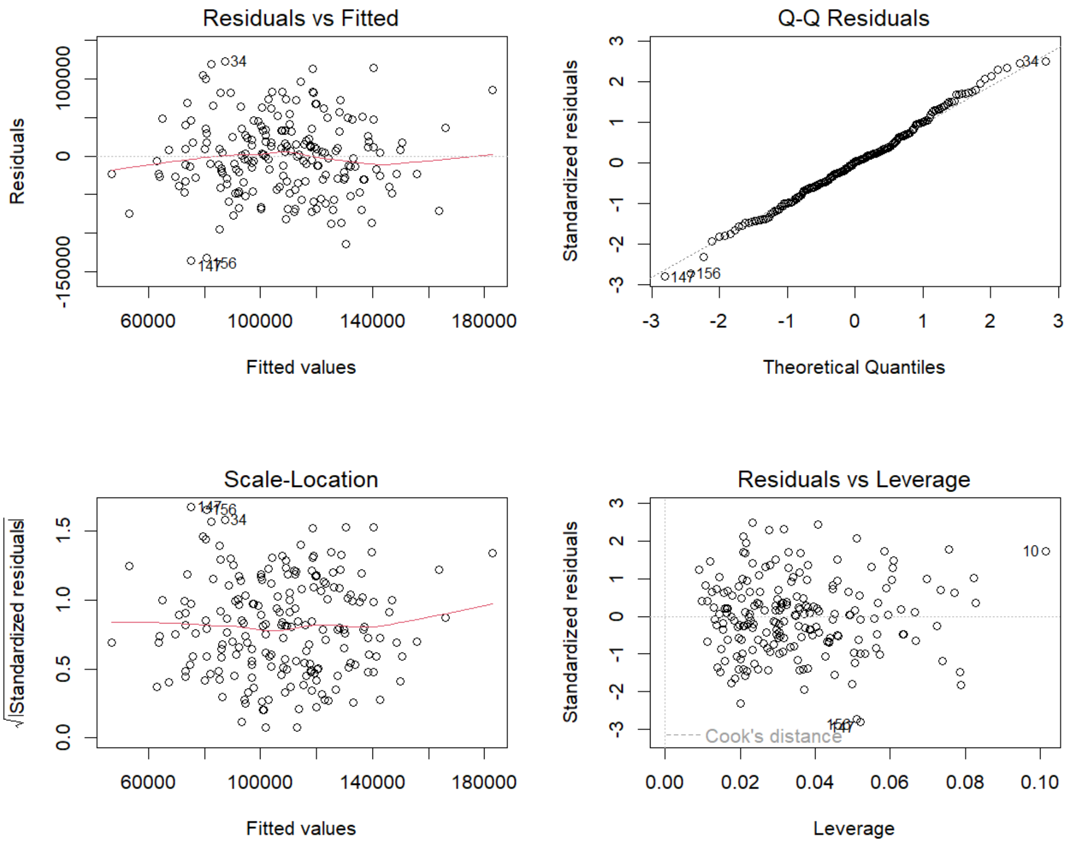

Figure 8 Diagnostic plots for the multiple linear regression model

A residuals-versus-fitted plot was also produced.

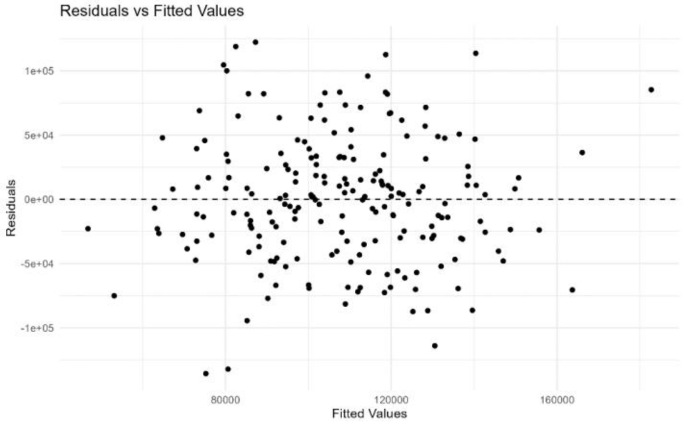

Figure 9 Residuals versus fitted values

The formal tests and graphical diagnostics provide no strong evidence of serious violations of the major regression assumptions.

A.8 Actual Versus Predicted Passenger Demand

The multiple regression model was used to generate predicted passenger-demand values.

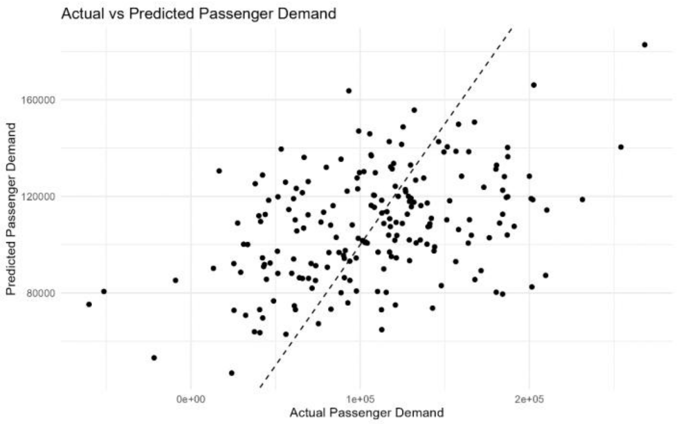

Figure 10 Actual versus predicted passenger demand

The relatively broad spread of observations around the perfect-prediction line is consistent with the model's modest R^2 of 0.1789. Therefore, although the model is statistically significant overall, its predictive explanatory power is limited.

A.9 Critical Discussion

The analysis identifies airport traffic as the most important individual explanatory variable for passenger demand. It produced the strongest Pearson correlation ($r = 0.3854$, $p < 0.001$) and the strongest simple regression model ($R^2 = 0.1485$). This suggests that airport traffic is an important indicator of passenger-demand conditions and may therefore be useful for aviation capacity and operational planning.

The multiple regression model was statistically significant ($F(6,193) = 7.008$, $p < 0.001$) and explained 17.89% of the variation in passenger demand. However, the relatively low R^2 indicates that passenger demand is influenced by many factors not represented in the dataset.

Fuel price became statistically significant in the multiple regression model despite its weak individual Pearson correlation. This demonstrates that relationships can change when predictors are considered jointly. However, the positive coefficient should not be interpreted as a causal effect, as it may reflect characteristics of the observations or other unmeasured factors. Previous research has also found that fuel prices and income can have complex and asymmetric relationships with air transport demand (Wadud, 2014)

The lack of significant effects for average income, ticket fare, flight frequency, and route distance suggests that these variables provide limited independent linear explanatory power within this dataset. This does not necessarily mean that these factors have no real-world influence on aviation demand.

For Sri Lanka's civil aviation sector, the findings suggest that airport traffic should receive particular attention when planning airport capacity, passenger-processing resources, airline operations, and infrastructure requirements. However, decision-makers should avoid relying on this model alone because of its limited explanatory power.

Future models could incorporate tourism activity, seasonality, population size, airline capacity, route availability, airport connectivity, economic conditions, and historical passenger volumes to improve explanatory and predictive performance.

A.10 Conclusion

The statistical analysis examined six explanatory variables associated with passenger demand using correlation and regression techniques.

Airport traffic demonstrated the strongest individual relationship with passenger demand and was statistically significant in both simple and multiple regression. The multiple regression model was also statistically significant overall, explaining 17.89% of the variation in passenger demand.

Fuel price was statistically significant in the multiple regression model, while average income, average ticket fare, flight frequency, and route distance were not statistically significant at the 5% level.

The regression diagnostic tests provided no evidence of serious multicollinearity, heteroscedasticity, residual autocorrelation, or non-normal residuals.

Overall, the findings provide useful evidence for understanding passenger-demand relationships in the dataset. However, the relatively modest R^2 demonstrates that passenger demand is influenced by additional factors not included in the analysis. Therefore, the model should be viewed as an analytical decision-support tool rather than a complete prediction of aviation demand.

TASK B – Aviation Social Network Analysis

B.1 Introduction

Social Network Analysis (SNA) was conducted to examine the relationships and interactions among key stakeholders within the Sri Lankan aviation sector. SNA provides a framework for analysing actors and the relationships between them as a network, allowing the structural position and connectivity of individual stakeholders to be examined (Wasserman and Faust, 1994).

In this analysis, stakeholders are represented as nodes, while relationships between stakeholders are represented as directed and weighted edges. The direction of an edge indicates the source and target of a relationship, while the associated weight represents the relative strength of that relationship. The analysis therefore considers both the structure of the network and the strength of interactions between stakeholders.

The main objectives of the analysis were to:

- construct a directed and weighted network of Sri Lankan aviation stakeholders;
- evaluate the overall structure and connectivity of the network;
- identify influential stakeholders using degree and weighted degree measures;
- identify potential bridge or coordination stakeholders using betweenness centrality;
- analyse the distribution of different relationship types;
- examine shortest paths within the network; and
- assess the potential effect of removing critical stakeholders through vulnerability analysis.

The analysis was implemented in R using the tidyverse, igraph, ggraph, and tidygraph packages. The complete R implementation is provided in Appendix, while only the key analytical code is presented in the main body.

B.2 Dataset and Network Construction

The dataset used for the analysis was the SriLanka_Aviation_SNA_Dataset.csv file. The dataset contains four variables: Source, Target, Relationship, and Weight.

The Source variable identifies the stakeholder initiating or providing the relationship, while Target identifies the stakeholder receiving or being connected by the relationship. The Relationship variable categorises the interaction into one of four relationship types: Commercial, Support, Regulatory, or Operational. The Weight variable represents the relative strength of the relationship.

The dataset contained **28 relationships involving 15 unique stakeholders**. The relationship weights ranged from **1 to 9**, with a mean weight of **4.643**.

Attribute	**Result**
Number of relationships	28
Number of stakeholders	15
Number of relationship types	4
Minimum relationship weight	1
Maximum relationship weight	9
Mean relationship weight	4.643
Missing values	0
Duplicate records	0

Table 11 Dataset characteristics

The four relationship types identified in the dataset were:

- Commercial
- Support
- Regulatory
- Operational

The network was constructed as a directed weighted graph, allowing both the direction and strength of stakeholder relationships to be incorporated into the analysis.

```r
# Create a directed weighted network

aviation_network <- graph_from_data_frame(
  network_data,
  directed = TRUE
)
```

Figure 11 Network Construction Code Snippet

The resulting network contained **15 nodes and 28 directed edges**.

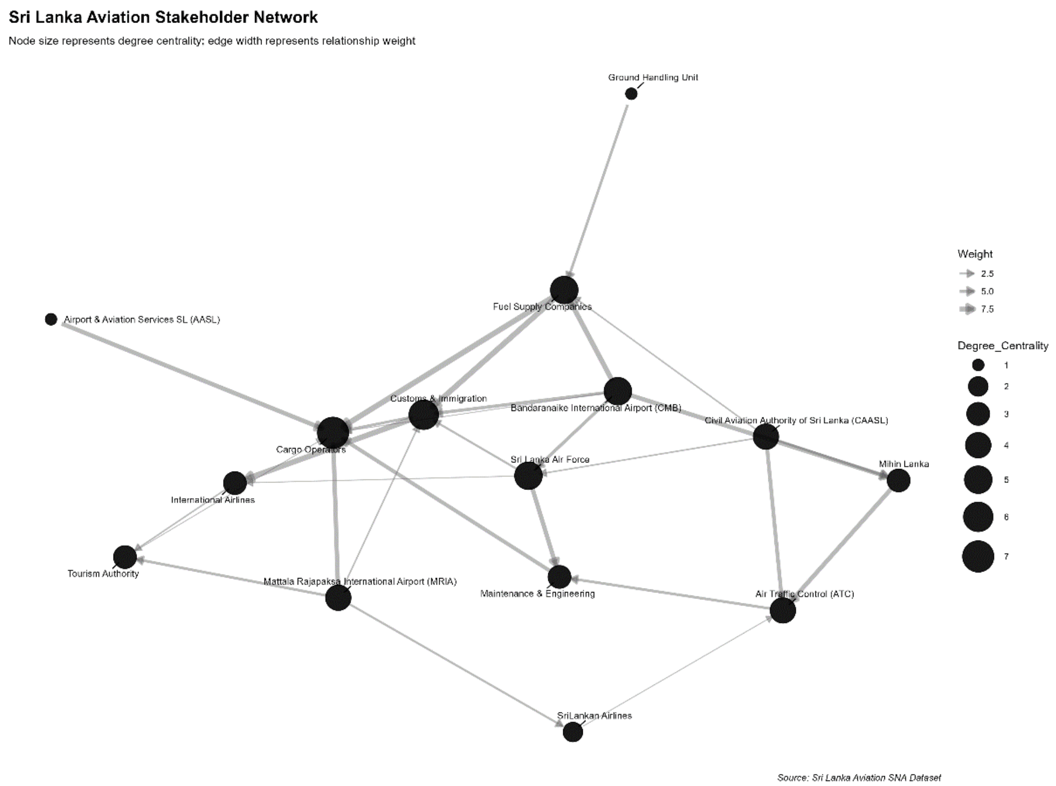

Figure 12 Sri Lanka Aviation Stakeholder Network

B.3 Data Preparation and Quality Assessment

Before constructing the network, the dataset was inspected to determine whether any data-quality problems could affect the analysis. The dataset contained no missing values across the four variables. A total of **zero duplicated rows** were also identified.

The relationship variable contained four distinct categories, while all relationship weights were numeric and fell within the expected range of 1 to 9. The source and target variables were converted to character format and the weight variable was converted to numeric format before network construction.

No incomplete records were identified after the cleaning process. Therefore, no observations had to be removed because of missing or invalid values.

```
>
> colSums(is.na(network_data))
      Source       Target Relationship       Weight
           0            0            0            0
>
> sum(is.na(network_data))
[1] 0
>
> network_data[duplicated(network_data), ]
# A tibble: 0 × 4
# ℹ 4 variables: Source <chr>, Target <chr>, Relationship <chr>, Weight <dbl>
>
> unique(network_data$Relationship)
[1] "Support"      "Commercial"   "Regulatory"   "Operational"
>
> length(unique(network_data$Source))
[1] 14
>
> length(unique(network_data$Target))
[1] 10
>
> range(network_data$Weight, na.rm = TRUE)
[1] 1 9
>
> table(network_data$Relationship)

 Commercial Operational  Regulatory     Support
         10           6           5           7
> |
```

Figure 13 Data Quality Results Code Snippet

The results indicate that the dataset was sufficiently complete for network analysis. The absence of missing values and duplicate records reduces the possibility that incomplete or repeated observations influenced the network measures.

B.4 Network Analysis Methodology

Several SNA measures were applied to evaluate the structure and importance of stakeholders within the network.

B.4.1 Network Density

Network density measures the proportion of possible directed relationships that are actually present in the network. A higher density indicates that a greater proportion of possible connections between stakeholders exists.

The calculated network density was:

0.1333

This means that approximately **13.33% of the possible directed relationships** were represented in the dataset.

```
network_density <- edge_density(
  aviation_network,
  loops = FALSE
)
```

Figure 14 Network Density Code Snippet

B.4.2 Degree Centrality

Degree centrality measures the number of direct connections associated with each stakeholder. Total degree considers both incoming and outgoing relationships.

In addition, in-degree and out-degree were calculated to distinguish between relationships received and relationships initiated by each stakeholder.

Weighted degree, also known as network strength, was also calculated using the relationship weights. This provides an indication of the total strength of a stakeholder's direct relationships rather than simply counting the number of connections.

```
degree_centrality <- degree(
  aviation_network,
  mode = "all"
)


in_degree_centrality <- degree(
  aviation_network,
  mode = "in"
)


out_degree_centrality <- degree(
  aviation_network,
  mode = "out"
)


weighted_degree <- strength(
  aviation_network,
  mode = "all",
  weights = E(aviation_network)$weight
)
```

Figure 15 Degree Centrality Code Snippet

B.4.3 Betweenness Centrality

Betweenness centrality measures how frequently a stakeholder lies on the shortest paths between other stakeholders. A stakeholder with high betweenness can therefore act as an intermediary, bridge, or coordination point within the network.

Because the relationship weights represent relationship strength rather than distance, inverse weights were used when calculating shortest-path-based measures:

d = \frac{1}{w}

This ensures that stronger relationships are treated as shorter network distances.

```
betweenness_centrality <- betweenness(
  aviation_network,
  directed = TRUE,
  weights = 1 / E(aviation_network)$Weight,
  normalized = TRUE
)
```

Figure 16 Betweenness Centrality Code Snippet

B.4.4 Connectivity and Shortest Paths

Weak and strong connected components were calculated to determine whether stakeholders form connected groups.

Weak connectivity ignores edge direction, while strong connectivity considers the direction of relationships. Shortest-path analysis was also conducted using inverse relationship weights to measure weighted network distances.

The analysis produced an average shortest path of **0.5063** and a network diameter of **1.5833**.

These values represent weighted distances based on the inverse relationship weights rather than simple numbers of connections.

B.4.5 Vulnerability Analysis

A vulnerability analysis was conducted by removing selected stakeholders from the network and comparing the resulting network structure with the original network.

Two stakeholder-failure scenarios were examined:

1. Air Traffic Control (ATC) failure
2. Ground Handling Unit failure

The number of nodes, edges, network density, and weakly connected components were compared before and after each removal.

```
if (length(atc_node) > 0) {

  network_without_atc <- delete_vertices(
    aviation_network,
    atc_node
  )

  atc_components <- components(
    network_without_atc,
    mode = "weak"
  )
```

Figure 17 Vulnerability Analysis Code Snippet

The complete vulnerability analysis code is provided in Appendix.

B.5 Results and Analysis

B.5.1 Overall Network Structure

The final network consisted of 15 stakeholders and 28 directed relationships. The network density was 0.1333, indicating that only a relatively small proportion of all possible directed relationships were represented.

The average degree was 3.7333, meaning that each stakeholder had approximately 3.73 direct connections on average.

The average weighted degree was **17.3333**, indicating that the average stakeholder was associated with approximately 17.33 units of relationship strength across its direct connections.

Network Measure	Result
Number of nodes	15
Number of edges	28
Network density	0.1333
Average degree	3.7333
Average weighted degree	17.3333
Weakly connected components	1
Strongly connected components	15
Average shortest path	0.5063
Network diameter	1.5833

Table 12 Overall Network Measures

The network contained **one weakly connected component**, meaning that all 15 stakeholders were connected when relationship direction was ignored. This indicates that the stakeholders form one overall network rather than several isolated groups.

However, the network contained **15 strongly connected components**, with every stakeholder belonging to its own strong component. This indicates that the directed relationships do not form reciprocal cycles in which stakeholders can reach each other through directed paths.

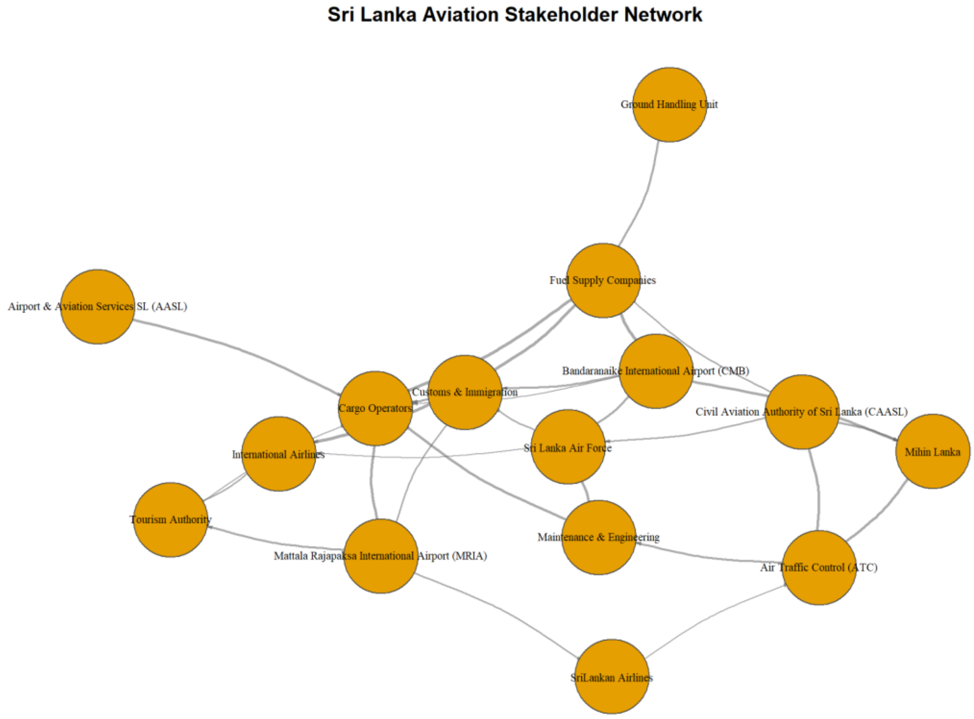

Figure 18 Sri Lankan Stakeholder Network

The visualisation provides an overview of the relationships between stakeholders and demonstrates the directed nature of the network.

B.5.2 Degree Centrality Results

Degree centrality identified Cargo Operators as the stakeholder with the highest total degree centrality, with a value of 7. This was followed by Customs & Immigration, with a degree of 6.

Several stakeholders recorded a degree of 5, including:

- Bandaranaike International Airport (CMB)
- Sri Lanka Air Force
- Fuel Supply Companies

Stakeholder	**Degree**	**In-Degree**	**Out-Degree**	**Weighted Degree**
Cargo Operators	7	7	0	35
Customs & Immigration	6	4	2	32
Bandaranaike International Airport (CMB)	5	0	5	26
Sri Lanka Air Force	5	2	3	18
Fuel Supply Companies	5	3	2	32
Mattala Rajapaksa International Airport (MRIA)	4	0	4	16
Civil Aviation Authority of Sri Lanka (CAASL)	4	0	4	14
Air Traffic Control (ATC)	4	3	1	18
Mihin Lanka	3	2	1	18
Maintenance & Engineering	3	2	1	17

Table 13 Top Stakeholders by Degree Centrality

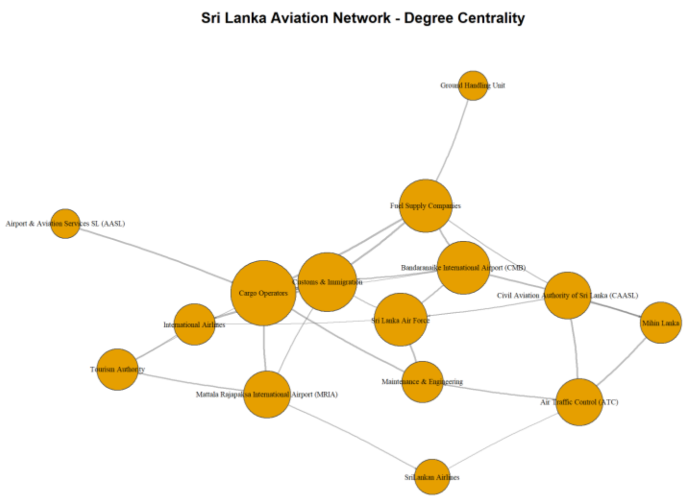

Figure 19 Top Stakeholders by Degree Centrality

Cargo Operators has the highest degree because it has seven direct relationships. However, its degree is entirely based on incoming relationships, with an in-degree of 7 and an out-degree of 0. This suggests that Cargo Operators is highly connected to other stakeholders but does not initiate relationships in the network represented by the dataset.

Customs & Immigration has a degree of 6 and a weighted degree of 32, making it one of the most strongly connected stakeholders. Bandaranaike International Airport has five outgoing relationships and no incoming relationships, indicating its role as an important source of relationships within the constructed network.

The results demonstrate why multiple centrality measures are useful. A high number of connections does not necessarily mean that a stakeholder acts as a bridge between other stakeholders.

B.5.3 Betweenness Centrality Results

Betweenness centrality produced a different ranking from degree centrality.

Customs & Immigration had the highest betweenness centrality at 0.0604, followed by:

1. Fuel Supply Companies – 0.0440
2. Air Traffic Control (ATC) – 0.0385
3. International Airlines – 0.0330
4. Maintenance & Engineering – 0.0275

Stakeholder	**Betweenness Centrality**
Customs & Immigration	0.0604
Fuel Supply Companies	0.0440
Air Traffic Control (ATC)	0.0385
International Airlines	0.0330
Maintenance & Engineering	0.0275

Table 14 Top Stakeholders by Betweenness Centrality

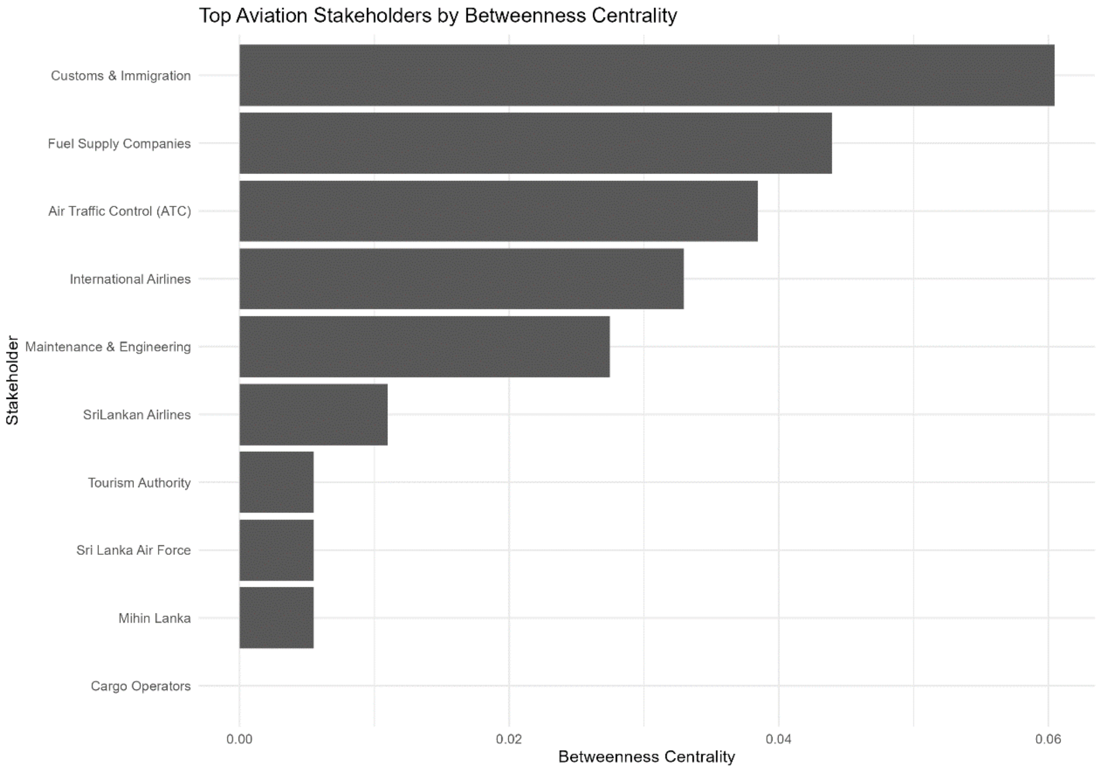

Figure 20 Top Stakeholders by Betweenness Centrality

The results indicate that Customs & Immigration is the most prominent potential intermediary in the network. Its high betweenness suggests that it is positioned on a comparatively large number of weighted shortest paths between other stakeholders.

Fuel Supply Companies and ATC also have relatively high betweenness values. These stakeholders may therefore have an important coordination or intermediary role within the relationships represented by the dataset.

An important observation is that Cargo Operators has the highest degree centrality but a betweenness value of **0**. Therefore, Cargo Operators is highly connected but does not function as a bridge between other stakeholders in the constructed network.

B.5.4 Critical Bridge Stakeholders

The five stakeholders with the highest betweenness centrality were identified as potential critical bridge or coordination stakeholders.

Rank	Stakeholder	Betweenness
1	Customs & Immigration	0.0604
2	Fuel Supply Companies	0.0440
3	Air Traffic Control (ATC)	0.0385
4	International Airlines	0.0330
5	Maintenance & Engineering	0.0275

Table 15 Critical Bridge Stakeholders

These stakeholders may deserve particular attention when considering coordination, information flow, and operational dependencies. However, the results should be interpreted as structural findings within the constructed network rather than as definitive evidence that these organisations are operationally indispensable.

B.5.5 Relationship Type Analysis

Four relationship types were identified in the dataset: Commercial, Support, Operational, and Regulatory.

Commercial relationships were the most frequent, with 10 relationships, followed by Support with 7, Operational with 6, and Regulatory with 5.

Relationship Type	Number of Relationships	Total Weight	Average Weight
Commercial	10	51	5.10
Support	7	27	3.86
Operational	6	27	4.50
Regulatory	5	25	5.00

Table 16 Relationship Type Analysis

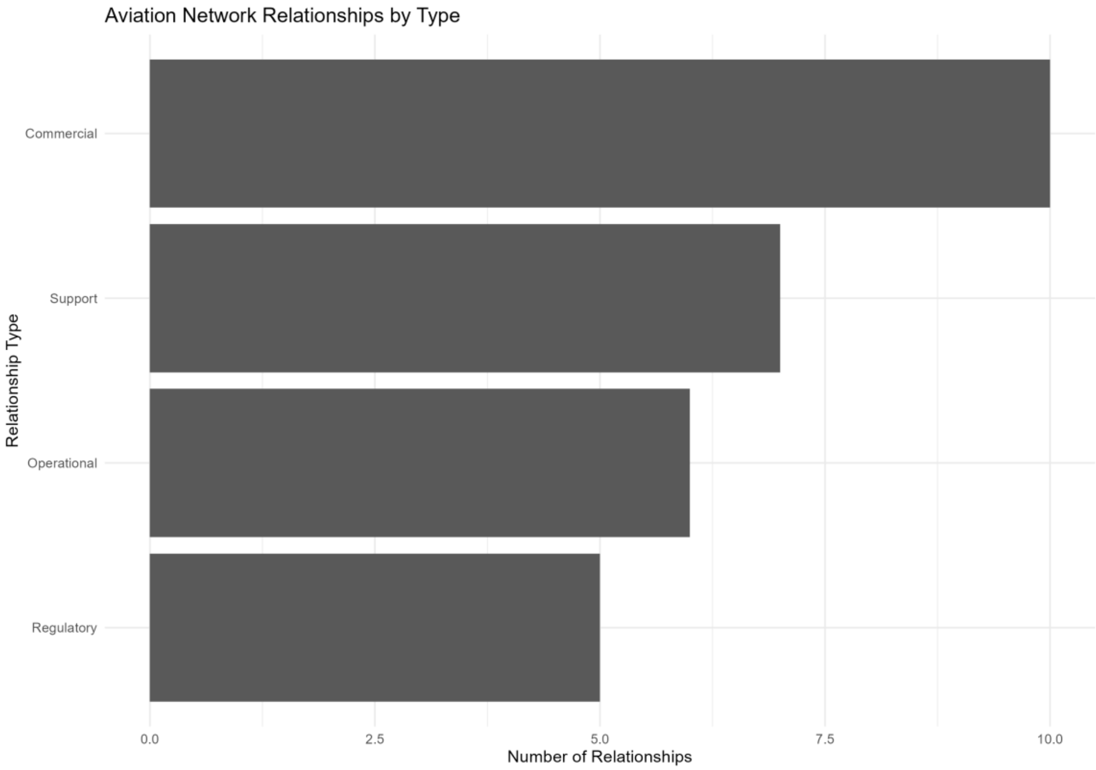

Figure 21 Relationship Types

Commercial relationships represent the largest proportion of recorded relationships and also have the highest total relationship weight of **51**. Regulatory relationships are less frequent but have an average weight of **5.00**, indicating that although fewer regulatory relationships were recorded, their average strength is relatively high.

This demonstrates that relationship frequency and relationship strength provide different perspectives. A relationship category can occur less frequently while still having relatively strong individual relationships.

B.5.6 Network Connectivity

The network contained one weakly connected component containing all 15 stakeholders. This indicates that the stakeholders are part of one overall connected structure when the direction of relationships is ignored.

In contrast, the network contained 15 strongly connected components. Each stakeholder formed its own strong component, indicating that reciprocal directed connectivity is limited.

Measure	Result
Weakly connected components	1
Strongly connected components	15
Nodes in largest weak component	15

Table 17 Connectivity Analysis

The difference between weak and strong connectivity is important. Although stakeholders are connected through the overall network, the direction of relationships prevents the formation of strongly interconnected groups.

B.5.7 Shortest Path Analysis

Weighted shortest-path analysis produced an average shortest path value of **0.5063** and a network diameter of **1.5833**.

The analysis used inverse relationship weights as network distances. Therefore, stronger relationships resulted in shorter distances.

Measure	Result
Average weighted shortest path	0.5063
Network diameter	1.5833

Table 18 Shortest Path Results

The relatively low weighted distance indicates that strong relationships can provide relatively short routes between stakeholders in the weighted network. However, these values should not be interpreted as physical travel distances or actual operational time because they are based on the relationship weights assigned in the dataset.

B.5.8 Vulnerability Analysis

The vulnerability analysis examined the effect of removing ATC and Ground Handling Unit from the network.

Scenario	Nodes	Edges	Density	Weak Components
Original Network	15	28	0.1333	1
ATC Failure	14	24	0.1319	1
Ground Handling Failure	14	27	0.1484	1

Table 19 Network Vulnerability Comparison

Removing ATC reduced the network from 15 to 14 nodes and from 28 to 24 relationships. The network density decreased slightly from **0.1333 to 0.1319**. However, the number of weakly connected components remained at one.

This indicates that removing ATC did not disconnect the network into separate weakly connected groups, although several direct relationships were removed.

The removal of Ground Handling Unit reduced the number of nodes from 15 to 14 and the number of edges from 28 to 27. The network remained a single weakly connected component. Interestingly, network density increased to **0.1484**, because one node with relatively few connections was removed while most of the remaining relationships remained intact.

Therefore, neither failure scenario caused complete network fragmentation based on weak connectivity. However, this does not imply that the stakeholders are operationally unimportant. The analysis only measures structural connectivity within the relationships represented in the dataset.

B.6 Discussion

The SNA results reveal several important features of the Sri Lankan aviation stakeholder network. The network is relatively sparse, with a density of **0.1333**, indicating that relationships are concentrated around particular stakeholders rather than being evenly distributed. Network density represents the proportion of observed connections relative to the maximum possible number of connections (Wasserman and Faust, 1994).

Cargo Operators has the highest degree centrality (**7**), making it the most connected stakeholder. However, its zero outgoing relationships and betweenness centrality indicate that high connectivity does not necessarily imply an intermediary role. Degree and betweenness capture different structural aspects of network position(Freeman, 1977)

Customs & Immigration has the highest betweenness centrality, together with relatively high degree and weighted degree. It therefore appears to be an important potential bridge for coordination. Fuel Supply Companies and ATC also show relatively high betweenness, suggesting important intermediary positions.

The network remains weakly connected, meaning all stakeholders form one overall structure, while the lack of large strongly connected components indicates that relationships are mainly directional rather than reciprocal.

The vulnerability analysis shows that removing either ATC or Ground Handling Unit does not fragment the network, as both scenarios retain a single weakly connected component. This suggests the presence of alternative connections. However, connectivity alone does not fully represent operational resilience; centrality, relationship strength, and aviation-sector knowledge should be considered together when identifying critical stakeholders.

B.7 Recommendations

Based on the SNA findings, several recommendations can be proposed.

1. Strengthen coordination with high-betweenness stakeholders

Customs & Immigration, Fuel Supply Companies, ATC, International Airlines, and Maintenance & Engineering have relatively high betweenness centrality. Coordination procedures involving these stakeholders should therefore be clearly documented and regularly reviewed.

2. Reduce dependency on individual coordination points

Although the network remains weakly connected after the simulated ATC and Ground Handling failures, contingency mechanisms should still be maintained. Alternative communication and coordination channels can reduce the potential impact of disruptions.

3. Monitor high-strength relationships

The highest-weight relationships should receive particular attention because they represent strong dependencies within the constructed network.

A1 | Source

	A	B	C	D	E
1	Source	Target	Relationsl	Weight	
2	Fuel Supp	Customs &	Operation	9	
3	Fuel Supp	Cargo Ope	Commerci	9	
4	Customs &	Internatio	Commerci	9	
5	Bandaran	Fuel Supp	Regulator	8	
6	Bandaran	Mihin Lan	Support	7	
7	Mattala R	Cargo Ope	Commerci	7	
8	Airport &	Cargo Ope	Operation	7	
9	Mihin Lan	Air Traffic	Support	7	
10	Sri Lanka	Maintena	Regulator	7	
11	Civil Aviat	Air Traffic	Commerci	6	
12	Maintena	Cargo Ope	Support	6	
13	Bandaran	Sri Lanka	Commerci	5	
14	Bandaran	Customs &	Commerci	5	
15	Mattala R	Tourism A	Regulator	4	
16	Civil Aviat	Mihin Lan	Operation	4	
17	Ground H	Fuel Supp	Operation	4	
18	Air Traffic	Maintena	Commerci	4	
19	Customs &	Cargo Ope	Regulator	4	
20	Mattala R	SriLankan	Commerci	3	
21	Sri Lanka	Customs &	Support	3	
22	Mattala R	Customs &	Operation	2	
23	Civil Aviat	Sri Lanka	Commerci	2	
24	Civil Aviat	Fuel Supp	Regulator	2	
25	Internatio	Tourism A	Support	2	
26	Bandaran	Cargo Ope	Operation	1	
27	SriLankan	Air Traffic	Support	1	
28	Sri Lanka	Internatio	Support	1	
29	Tourism A	Cargo Ope	Commerci	1	
30					

Table 20 Highest-Weight Relationships

The analysis identified several relationships with a maximum weight of **9**, including relationships involving Fuel Supply Companies, Customs & Immigration, and Cargo Operators.

4. Consider both connectivity and relationship strength

Decision-makers should avoid relying exclusively on the number of direct connections. The difference between Cargo Operators' high degree centrality and zero betweenness demonstrates that the number of connections alone does not fully explain structural importance.

5. Maintain alternative stakeholder relationships

Since the network is weakly connected but not strongly connected, strengthening reciprocal communication and coordination between key stakeholders could improve resilience and reduce excessive dependence on one-directional relationships.

B.8 Conclusion

The Social Network Analysis successfully modelled the Sri Lankan aviation stakeholder ecosystem as a directed and weighted network consisting of 15 stakeholders and 28 relationships.

The analysis identified a network density of 0.1333, an average degree of 3.7333, and an average weighted degree of 17.3333. All stakeholders belonged to a single weakly connected component, while the network contained 15 strong components, demonstrating that the network is connected overall but contains limited reciprocal directed connectivity.

Degree centrality identified Cargo Operators as the most connected stakeholder, while betweenness centrality identified Customs & Immigration as the strongest potential bridge stakeholder. Fuel Supply Companies and ATC also demonstrated relatively high intermediary importance.

The relationship analysis showed that Commercial relationships were the most frequent, while several Regulatory relationships had relatively high average weights. The vulnerability analysis demonstrated that removing ATC or Ground Handling Unit did not disconnect the network into multiple weak components, although the removal of ATC resulted in a larger reduction in relationships.

Overall, the analysis demonstrates the value of SNA for examining stakeholder structure, identifying influential and intermediary actors, and assessing network-level vulnerability (Wasserman and Faust, 1994).

However, the findings should be interpreted within the scope of the available dataset, as network measures reflect the relationships and weights represented in the dataset rather than providing a complete representation of real-world operational dependencies.

Task C – Geospatial Analysis for Radar Site Selection at Bandaranaike International Airport

C.1 Introduction

Bandaranaike International Airport (CMB) is a major aviation facility containing runways, taxiways, aprons, buildings and other infrastructure. Selecting suitable locations for surveillance radar therefore requires consideration of spatial constraints, existing infrastructure, potential obstructions and operational requirements.

This task used QGIS, PostgreSQL/PostGIS and Google Earth/KML/KMZ data to develop a georeferenced and digitized GIS representation of CMB and identify suitable candidate areas for Primary Surveillance Radar (PSR), Secondary Surveillance Radar (SSR) and Surface Movement Radar (SMR).

The analysis followed the required **EPSG:5234** coordinate reference system and used the SL_BIA_Aerial_Info PostgreSQL/PostGIS database. Spatial buffers and overlay analysis were applied to identify candidate areas based on the specified radar-placement criteria.

C.2 Objectives

The main objectives of the analysis were to:

- Georeference the supplied aerial imagery.
- Digitize relevant airport and surrounding infrastructure.
- Use EPSG:5234 as the project coordinate reference system.
- Integrate GPS information from KML/KMZ data.
- Store the spatial information in the SL_BIA_Aerial_Info database.
- Identify suitable areas for SMR using the RCP and Control Tower constraints.
- Identify suitable areas for PSR/SSR using the specified RCP distance constraints.
- Analyse potential building and infrastructure obstructions.
- Calculate the resulting suitable areas.
- Produce a professional final GIS map showing the analysis results and proposed radar locations.

C.3 Software and Data

The analysis was carried out using:

- **QGIS** – georeferencing, digitization, spatial analysis and map production.
- **PostgreSQL/PostGIS** – spatial database management.
- **Google Earth** – GPS/reference information and KML/KMZ data.
- **Georeferenced aerial imagery** – base map and digitization source.

The project used **EPSG:5234 – Kandawala / Sri Lanka Grid**. The assignment requires the aerial imagery to be georeferenced before digitization and the resulting spatial information to be supported by the specified geospatial database.

C.4 Georeferencing and Digitization

The supplied aerial image was first georeferenced so that its features corresponded to their correct geographic locations. This established the spatial reference required for accurate digitization and distance and area calculations.

After georeferencing, relevant airport and surrounding features were digitized in QGIS. These included:

- Buildings
- Runways
- Taxiways
- Aprons
- Parking areas
- Roads
- Railway features
- Airport-related boundaries
- Sri Lanka Air Force area
- Radar-related reference locations

Appropriate attribute information was maintained for the digitized layers. Polygon areas were calculated in square metres where required.

This process transformed the aerial image from a visual reference into structured GIS data that could be used for quantitative spatial analysis.

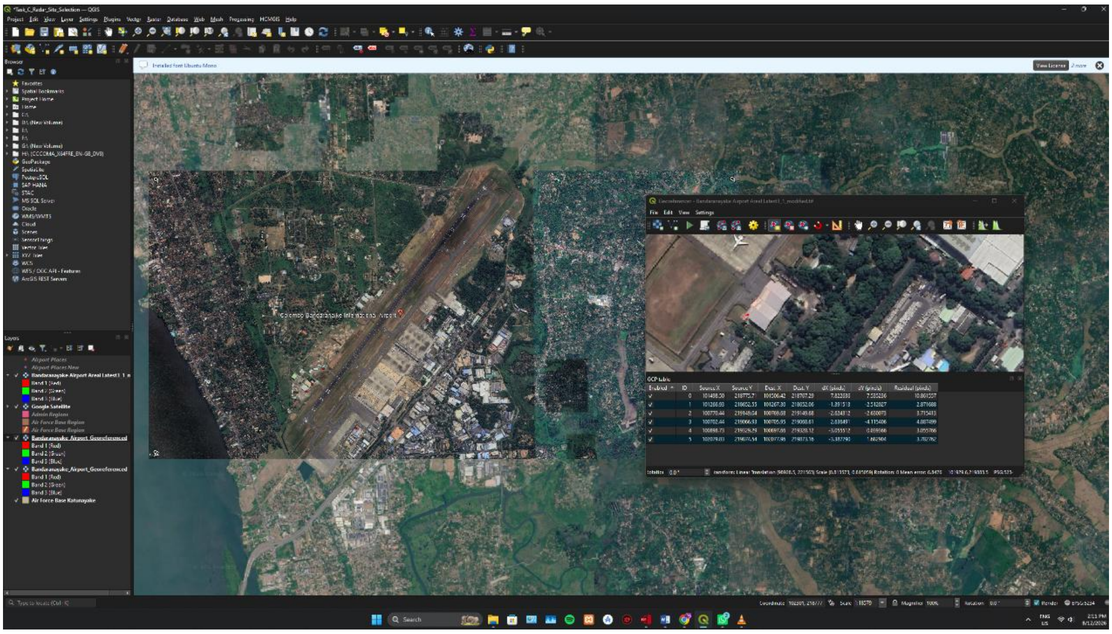

Figure 22 Georeferenced aerial imagery

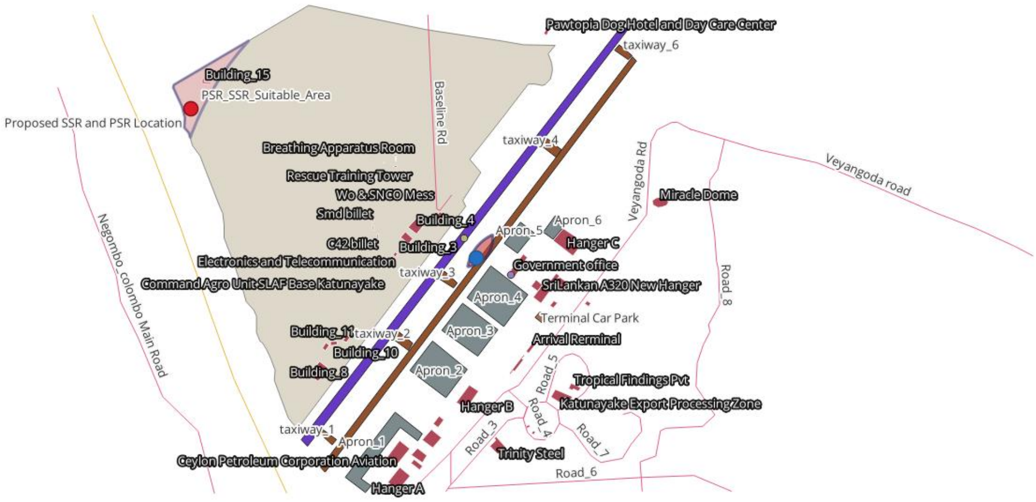

Figure 23 Digitized airport infrastructure

C.5 GPS and Spatial Database Integration

GPS/reference information required for the radar analysis was obtained from KML/KMZ data and incorporated into QGIS. Important reference locations included the Runway Centre Point (RCP) and Control Tower.

A PostgreSQL/PostGIS database named **SL_BIA_Aerial_Info** was established to store the spatial information generated during the project.

The spatial database provided a centralized method for managing the digitized layers and analysis-related spatial information.

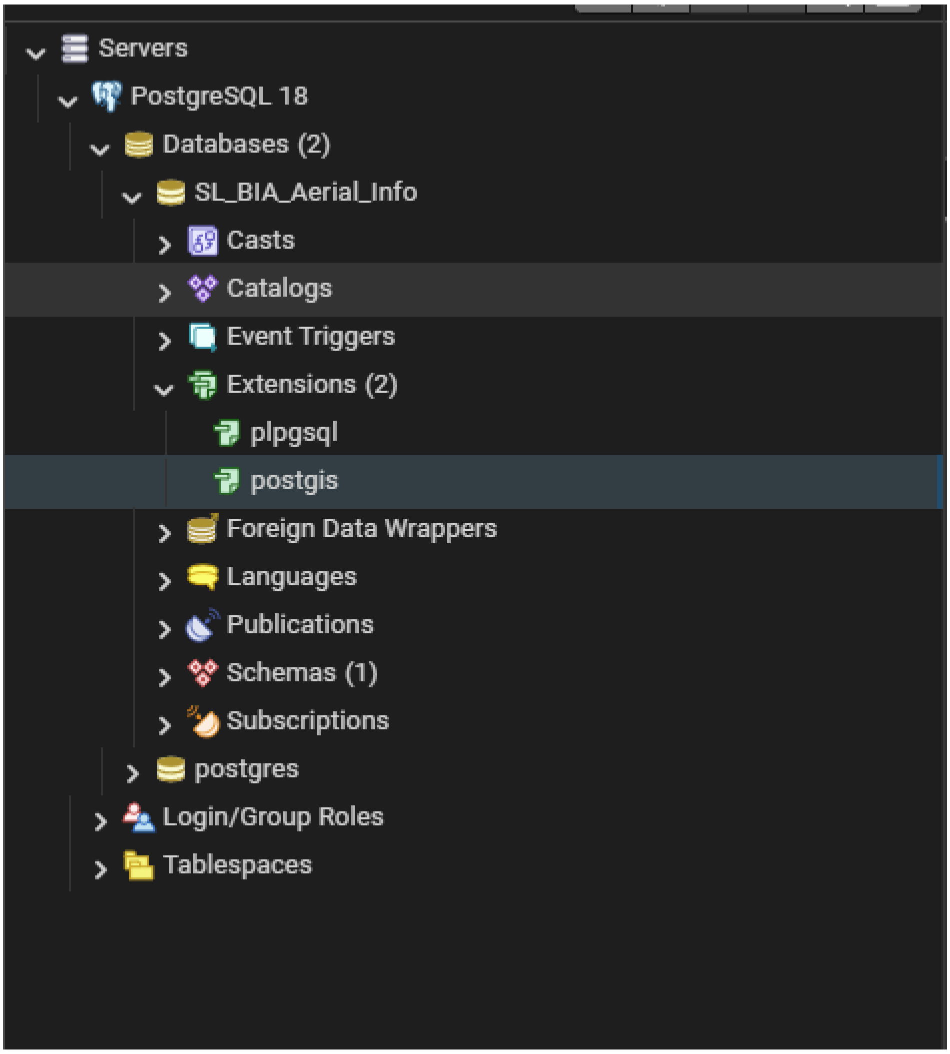

Figure 24 SL_BIA_Aerial_Info PostgreSQL/PostGIS database

C.6 Radar Site Selection Criteria

The spatial analysis followed the radar-placement criteria specified in the task.

C.6.1 SMR Criteria

Surface Movement Radar (SMR) is intended to monitor aircraft surface movements around the airport, including runways, taxiways and apron areas.

The specified constraints were approximately:

- 200 m from the RCP
- 300 m from the Control Tower

C.6.2 PSR/SSR Criteria

PSR and SSR were considered as a co-located radar installation for aircraft surveillance.

The specified criteria were:

- Approximately **2 km from the RCP**
- Within a maximum of **3 km from the RCP**
- Preferably within the adjacent **Sri Lanka Air Force premises**

The guide also identifies line-of-sight, electromagnetic interference, obstruction avoidance, aviation safety, maintenance access and power availability as important considerations.

Therefore, the GIS analysis was used to identify **candidate suitable areas**, rather than treating a buffer result as automatic approval for construction.

C.7 SMR Spatial Analysis

The SMR analysis was performed using QGIS buffer and overlay operations.

A **200 m buffer** was generated around the RCP and a **300 m buffer** around the Control Tower. These spatial constraints were combined to identify the area satisfying the required SMR location criteria.

The resulting SMR suitable area was:

17,500.70 m²

Figure 25 SMR buffer analysis

C.8 SMR Obstruction Analysis

The SMR suitable area was compared with the digitized building layer.

The analysis identified:

Buildings within SMR suitable area: 0

This is favourable because no building footprint was identified within the calculated candidate area.

However, a taxiway is present within/overlapping the suitable area. This is an important operational consideration because SMR is specifically intended to monitor aircraft surface movement.

The nearby runway and apron areas must also be considered when selecting the exact radar installation point. Therefore, although the area satisfies the applied spatial constraints and has no identified building obstruction, the final location would require additional aviation-safety and engineering assessment.

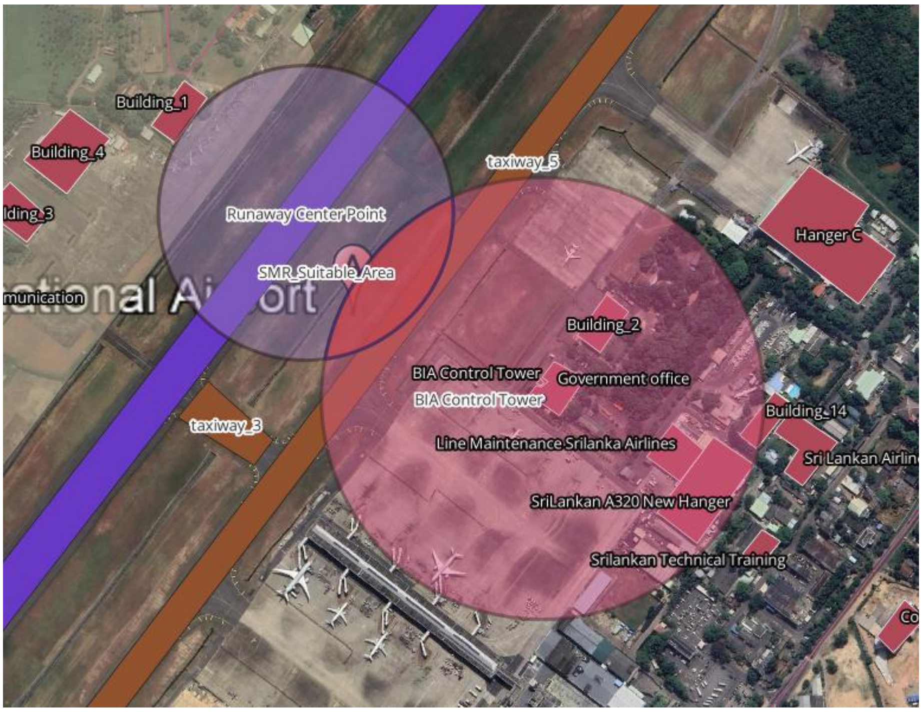

Figure 26 SMR suitable area with taxiway and airport infrastructure

C.9 PSR/SSR Spatial Analysis

For PSR/SSR, a 2 km RCP buffer was generated to represent the approximate preferred distance and a 3 km RCP buffer to represent the maximum distance constraint.

The resulting region was then evaluated in relation to the surrounding land use, including the Sri Lanka Air Force area and existing airport infrastructure.

The analysis produced a PSR/SSR suitable area of:

91,314.90 m²

The area was then examined for potential physical obstructions, particularly buildings.

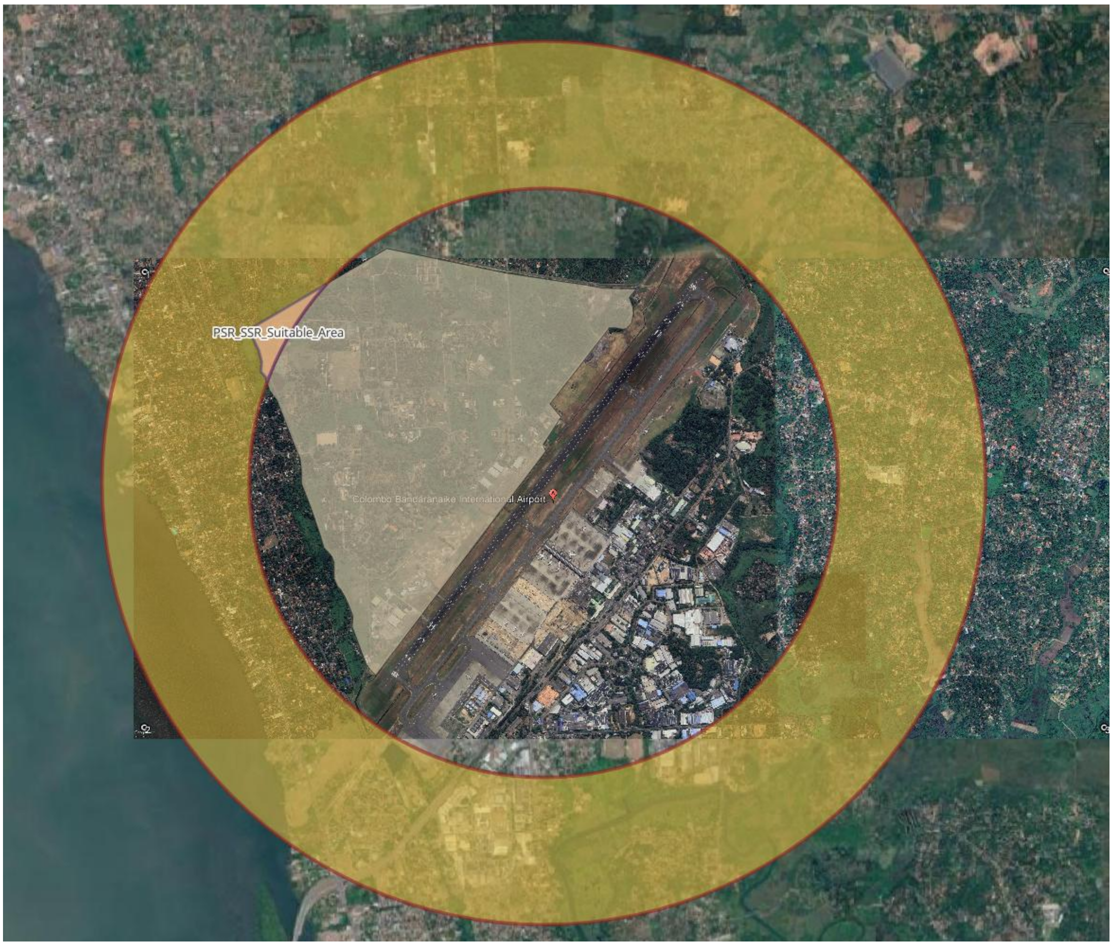

Figure 27 2 km and 3 km RCP intersection

Figure 28 PSR/SSR suitable area

C.10 PSR/SSR Obstruction Analysis

One building was identified within the PSR/SSR suitable area.

The measured building area was:

1,095.85 m²

The preliminary area remaining after accounting for this building was:

91,314.90 − 1,095.85 = 90,219.05 m²

Measure	Result
Suitable area	91,314.90 m²
Buildings	1
Building area	1,095.85 m²
Preliminary available area	90,219.05 m²

Table 21 PSR/SSR Suitability Results

The result indicates that most of the identified PSR/SSR candidate region is not occupied by the identified building footprint. Nevertheless, the building must be considered during final site selection because radar performance depends on unobstructed line-of-sight.

When Using Select By Location Research Tool it Highlighted the Obstructed Building clearly.

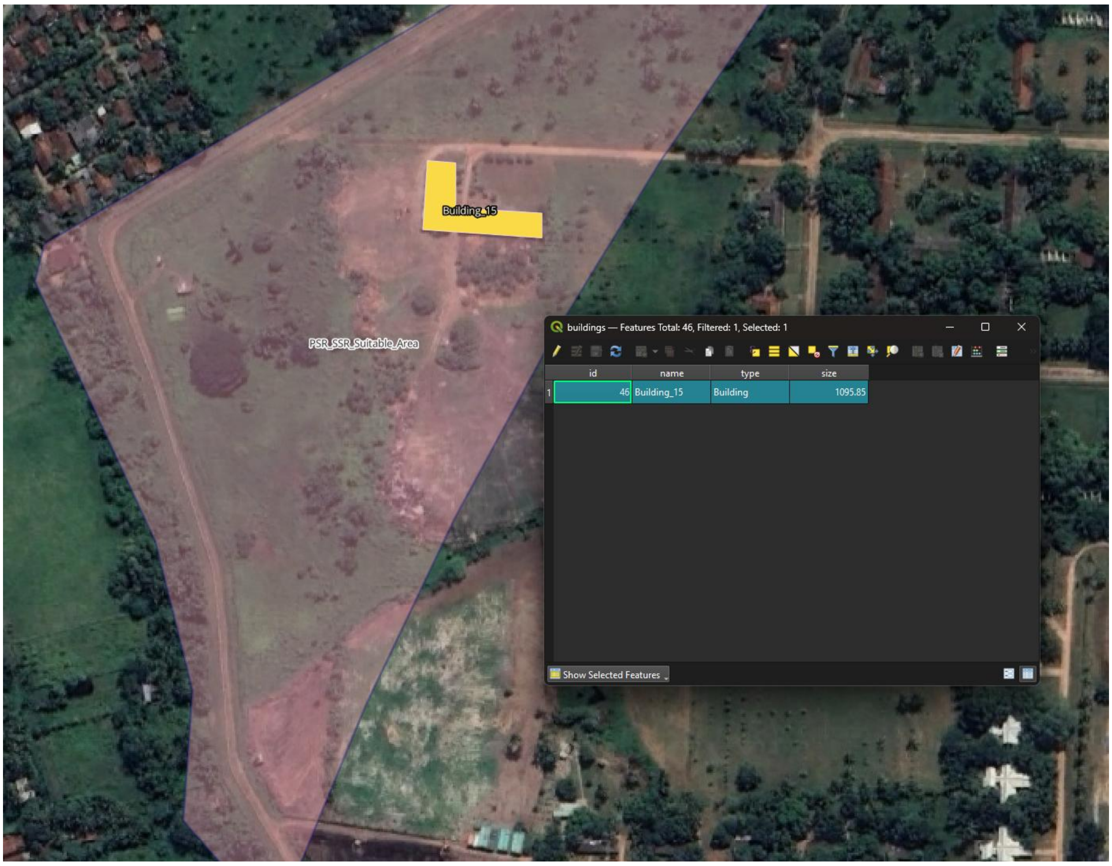

Figure 29 Identified building within PSR/SSR suitable area

C.11 Summary of Results

Radar system	Suitable area	Building obstruction	Key observation
SMR	17,500.70 m²	0	Taxiway present/overlapping
PSR/SSR	91,314.90 m²	1	Building area = 1,095.85 m²

Table 22 Summary of Radar Suitability Results

The SMR analysis produced a smaller candidate region but no identified building obstruction. The taxiway remains an important operational consideration.

The PSR/SSR analysis produced a substantially larger candidate region. One building was identified, leaving a preliminary available area of 90,219.05 m² after subtracting its footprint.

C.12 Interpretation and Practical Implications

The results demonstrate how GIS can support aviation infrastructure planning by converting radar-placement requirements into measurable spatial constraints.

For SMR, the 17,500.70 m^2 suitable area satisfies the applied RCP and Control Tower distance constraints. The absence of buildings is favourable; however, the presence of a taxiway means that the exact installation point must be selected carefully to avoid interference with aircraft movement and airport operations.

For PSR/SSR, the 91,314.90 m^2 candidate area provides a relatively large region for further investigation. The identified building occupies 1,095.85 m^2, leaving a preliminary area of 90,219.05 m^2. The final radar point should therefore be positioned within an unobstructed portion of the candidate area while maintaining the required distance from the RCP.

The analysis demonstrates that spatial suitability should not be considered independently from operational requirements. Factors such as radar line-of-sight, building height, terrain, electromagnetic interference, maintenance access, electrical infrastructure and aviation safety requirements should be evaluated before final implementation.

C.13 Limitations

Several limitations should be considered when interpreting the results.

First, the analysis depends on the available aerial imagery and digitized spatial information. Changes to airport infrastructure may require the GIS database to be updated.

Second, the building analysis was based on building footprints. The footprint alone does not provide complete information about building height or its effect on radar line-of-sight.

Third, electromagnetic interference cannot be fully assessed using the available GIS layers. A technical radio-frequency assessment would be required before implementation.

Finally, the identified areas represent preliminary candidate areas rather than final construction approvals. Detailed engineering, aviation safety, obstruction, electromagnetic and operational assessments would be required before a final radar site is selected.

C.14 Final Map

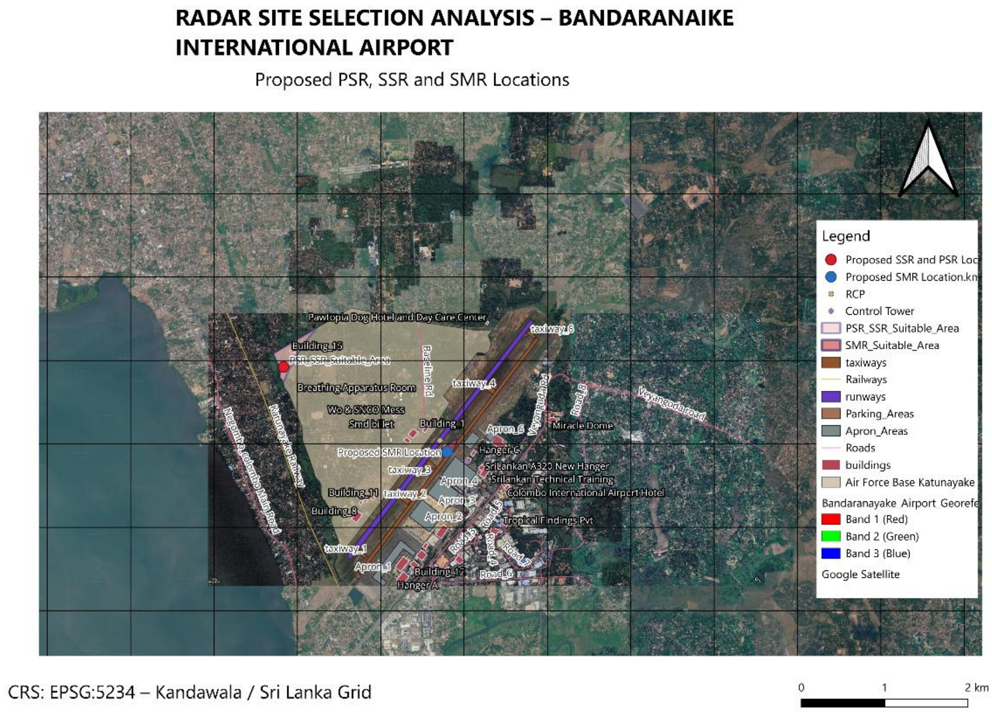

Figure 30 Final Map

A final A3 landscape map was produced in QGIS to communicate the results of the spatial analysis.

The final map includes:

- Aerial imagery
- Digitized airport infrastructure
- Buildings
- Runway
- Taxiways
- Apron and parking areas
- Roads and railway features
- RCP and Control Tower
- PSR/SSR buffers

- SMR buffers
- PSR/SSR suitable area
- SMR suitable area
- Proposed PSR location
- Proposed SSR location
- Proposed SMR location
- Legend
- North arrow
- Scale bar
- Coordinate grid
- CRS information
- Suitability summary

Neutral colours were used for general airport infrastructure, while contrasting colours were used for the radar suitability areas and proposed radar locations. This provides a clear visual hierarchy and allows the main analytical results to be distinguished from supporting infrastructure.

C.15 Conclusion

The task successfully developed a georeferenced and digitized GIS representation of Bandaranaike International Airport and applied spatial analysis to identify candidate areas for PSR, SSR and SMR radar systems.

The analysis used EPSG:5234, integrated GPS/KML/KMZ information, and stored the spatial data within the SL_BIA_Aerial_Info PostgreSQL/PostGIS database.

The SMR analysis identified 17,500.70 m^2 of suitable area with no identified buildings, although a taxiway occurs within/overlaps the candidate area. The PSR/SSR analysis identified 91,314.90 m^2 of suitable area containing one building with an area of 1,095.85 m^2, resulting in a preliminary available area of 90,219.05 m^2 after accounting for that footprint.

Overall, the results demonstrate that QGIS-based spatial analysis can provide useful decision support for aviation radar planning. The identified areas provide suitable candidates for further technical investigation, while final radar installation decisions should incorporate detailed engineering, aviation-safety, electromagnetic and operational assessments.

Task D – CMB Airport Operations Control Business Intelligence Dashboard

D.1 Introduction

CMB Airport Operations Control Business Intelligence Dashboard provide an interactive method of presenting operational data to support monitoring, analysis and decision-making. For airport operations, a dashboard can bring together information about flight movements, passenger volumes, delays, operational issues and risk indicators in a single interface.

For this task, a BI dashboard was developed for Bandaranaike International Airport (CMB) using the provided BIA_CMB_Dataset.csv dataset. The dashboard is designed to support airport managers and air traffic controllers by providing an overview of flight operations, passenger traffic, delays, international routes and operational risks.

The dataset contains 40 flight records and includes operational, passenger, weather, radar and geographical information. Key variables include airline, route, flight type, flight status, delay duration, aircraft capacity, weather condition, radar information, alert status, operational issue, turnaround time, taxi time, queue position, passenger count and load factor.

The dashboard was developed using Microsoft Power BI, with interactive visualisations and slicers allowing users to explore the operational information from different perspectives.

D.2 Dataset Preparation

Before creating the dashboard, the provided flight dataset was imported into Power BI and reviewed using the Power Query/Data Model environment.

The dataset contains the following main categories of information:

- **Flight information:** Flight ID, airline, route and flight type.
- **Operational information:** Scheduled and estimated time, gate, terminal, runway and flight status.
- **Delay information:** Delay duration in minutes.
- **Aircraft information:** Aircraft type and aircraft capacity.
- **Weather information:** Weather condition, temperature and wind speed.
- **Radar and location information:** PSR track ID, SSR code, SMR location, latitude and longitude.
- **Risk information:** Alert status and issue type.
- **Performance information:** Turnaround time, taxi time and queue position.
- **Passenger information:** Passenger count and load factor.

The original dataset was retained without removing operational variables because the different fields provide useful information for the dashboard. Calculated aggregations were handled within Power BI visualisations.

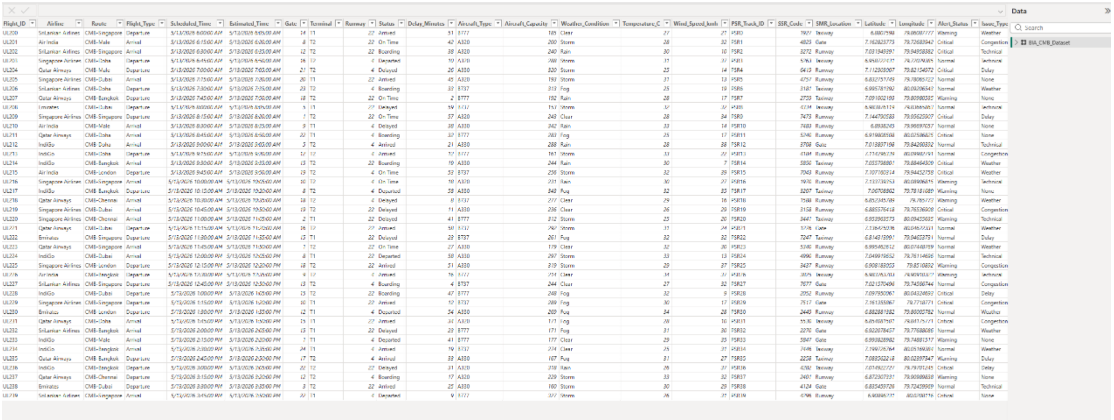

Figure 31 Dataset Preparation

D.3 BI Dashboard Design

The dashboard was designed as a single operational monitoring interface rather than as a collection of unrelated charts. The layout uses KPI cards, charts, slicers and a geographical map to allow users to quickly identify important operational conditions.

The dashboard contains:

1. KPI cards for key operational indicators.
2. Flight Status Summary.
3. Average Delay by Airline.
4. Operational Risk analysis.
5. Passenger Traffic by Airline.
6. International Traffic by Route.
7. CMB Flight Activity and Risk Map.
8. Interactive slicers for filtering the dashboard.

The use of different visualisation types allows numerical indicators, categorical comparisons and geographical information to be presented together.

Figure 32 CMB Airport Operations Control Business Intelligence Dashboard

D.4 Key Performance Indicators

Several KPI cards were included at the top of the dashboard to provide an immediate overview of airport activity.

The dataset contains **40 flights** and a total passenger count of **8,974 passengers**. The average recorded flight delay is **31.4 minutes**. The dashboard also includes a critical-alert indicator, which identifies flights requiring greater operational attention.

The KPI section allows users to understand the overall operational situation without having to examine individual charts.

Key indicators from the dataset

KPI	Value
Total Flights	40
Total Passengers	8,974
Average Delay	31.4 minutes
Critical Alerts	13
Average Turnaround Time	75.7 minutes
Average Taxi Time	15.9 minutes
Average Load Factor	82.2%

Table 23 Key indicators from the dataset

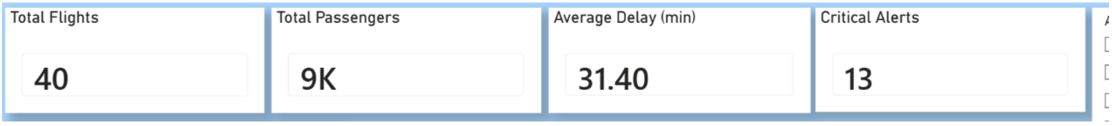

Figure 33 KPI from dashboard

The KPI section is particularly useful for airport management because it provides a high-level summary before more detailed analysis is performed.

D.5 Flight Status Summary

A donut chart was used to display the distribution of flight statuses.

The 40 flights consist of:

Flight Status	Number of Flights
Arrived	12
Delayed	9
Boarding	7
On Time	6
Departed	6

Table 24 Flight Status Summary Table

The results show that Arrived is the largest status category, representing 12 of the 40 flights. However, 9 flights were recorded as Delayed, indicating that delays represent an important operational consideration within the dataset.

The visualisation enables airport managers to quickly determine the current distribution of flight activities and identify whether a significant proportion of flights are experiencing delays.

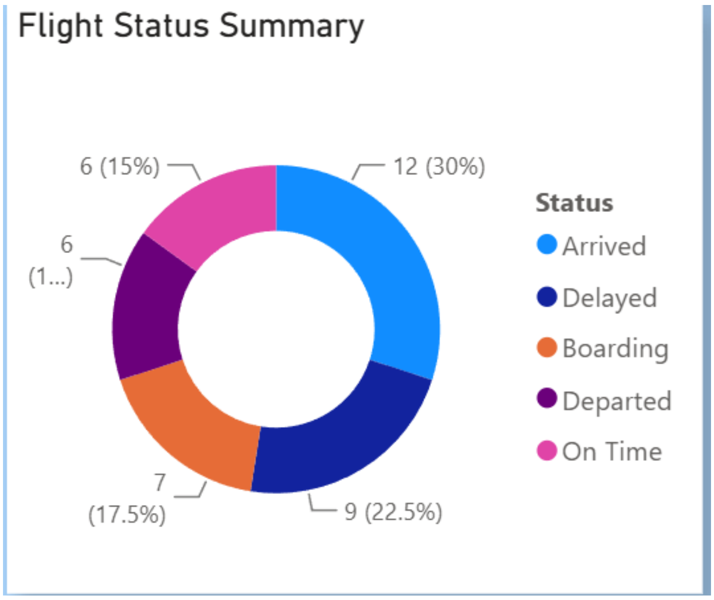

Figure 34 Flight Status Summary donut chart

D.6 Delay Analysis by Airline

The **Average Delay by Airline** clustered column chart was used to compare delay performance across airlines.

The calculated average delays are:

Airline	Average Delay (minutes)
Emirates	40.2
Air India	37.2
IndiGo	34.0
Singapore Airlines	30.6
Qatar Airways	26.3
SriLankan Airlines	26.3

Table 25 Delay Analysis by Airline table

The results indicate that Emirates has the highest average recorded delay at approximately 40.2 minutes, followed by Air India at approximately 37.2 minutes. Qatar Airways and SriLankan Airlines have the lowest average delay at approximately 26.3 minutes.

This comparison provides an operational performance indicator that could help airport management investigate whether particular airlines, routes or operational conditions are associated with higher delays.

However, the results should not automatically be interpreted as airline performance problems because the dataset also contains weather, congestion, technical and other operational factors that may contribute to delays.

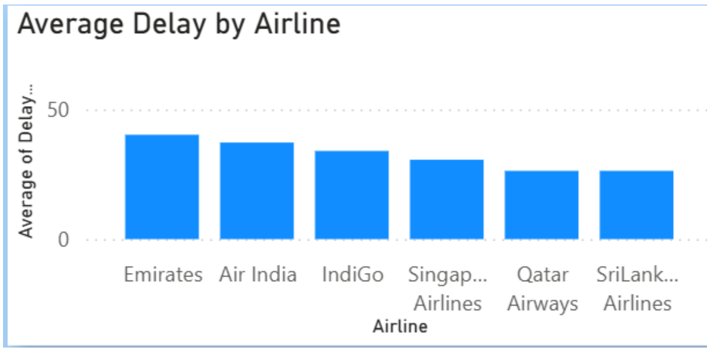

Figure 35 Average Delay by Airline Clustured Column Chart

D.7 Operational Risk Analysis

The **Operational Risk** clustered bar chart was included to identify the main types of operational issues recorded in the dataset.

The issue distribution is:

Issue Type	Number of Flights
Weather	10
Technical	9
None	8
Congestion	7
Delay	6

Table 26 Issue distribution Table

Weather-related issues are the most frequently recorded operational issue, with **10 flights**, followed by technical issues with **9 flights** and congestion with **7 flights**.

This indicates that operational performance is influenced by multiple factors rather than delays being caused by a single issue. Weather conditions can affect runway and taxiway operations, while technical problems and congestion may increase turnaround and movement times.

The visualisation therefore provides an early indication of areas where operational resources and monitoring may need to be prioritised.

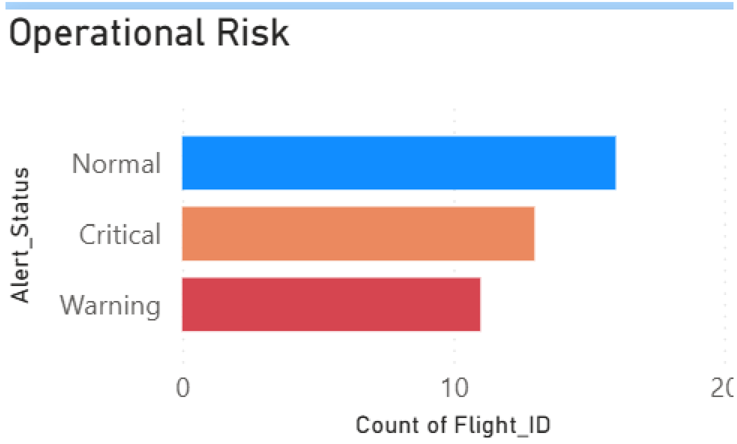

Figure 36 Operational Risk Analysis Clustered Bar Chart

D.8 Passenger Traffic by Airline

Passenger traffic was analysed using the Passenger Traffic by Airline clustered column chart.

The total passenger count across the dataset is 8,974 passengers.

The passenger distribution by airline is:

Airline	Passengers
IndiGo	1,902
Qatar Airways	1,969
Singapore Airlines	1,786
SriLankan Airlines	1,432
Air India	1,094
Emirates	791

Table 27 passenger distribution by airline Table

Qatar Airways has the highest passenger volume in the dataset with 1,969 passengers, closely followed by IndiGo with 1,902 passengers. Emirates has the lowest passenger volume with 791 passengers.

Passenger traffic is important for airport management because higher passenger volumes can increase demand for gates, terminal facilities, security processing and other airport resources.

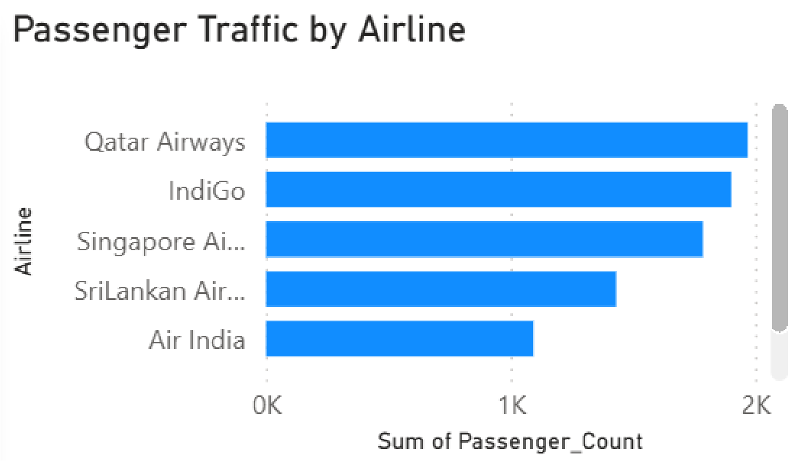

Figure 37 Passenger Traffic by Airline Clustered bar Chart

D.9 International Traffic by Route

The dashboard also includes an International Traffic by Route chart. The route information uses Colombo as the CMB origin/destination point. For example, CMB-Singapore represents the Colombo–Singapore route, rather than treating CMB and Singapore as separate airports in the analysis.

The passenger distribution by route is:

Route	Flights	Passengers	Average Delay
CMB-Singapore	9	2,042	26.6 min
CMB-Bangkok	8	1,764	25.1 min
CMB-Dubai	7	1,295	43.3 min
CMB-Doha	6	1,461	23.7 min
CMB-Male	4	1,056	36.8 min
CMB-Chennai	3	613	22.0 min
CMB-London	3	743	52.7 min

Table 28 passenger distribution by route

The CMB-Singapore route has the highest passenger traffic, with 2,042 passengers across nine flights. However, the CMB-London route records the highest average delay at approximately 52.7 minutes, although it contains only three flights.

This demonstrates why route-level analysis is useful. A route with lower passenger volume can still experience significant operational delays and therefore require attention.

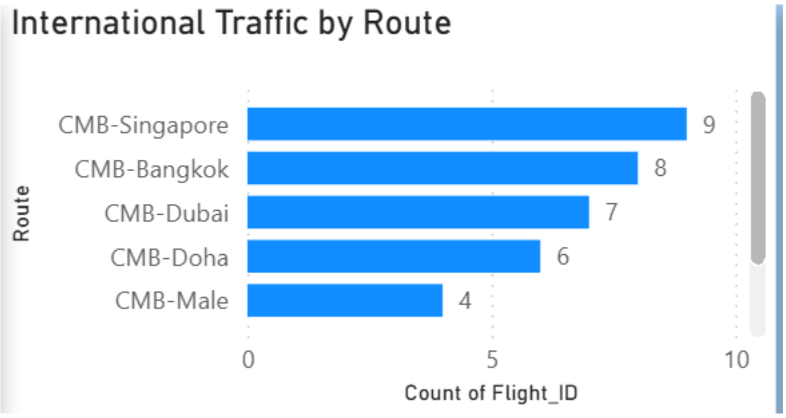

Figure 38 International Traffic by Route Clustered Bar Chart

D.10 CMB Flight Activity and Risk Map

A geographical map was incorporated into the dashboard using the latitude and longitude information provided in the dataset.

The map displays the geographical positions associated with individual flight records and allows users to visually identify the spatial distribution of flight activity and operational risk indicators.

The map can be interpreted together with the Alert_Status and other operational variables to identify locations associated with Normal, Warning and Critical conditions.

The dataset contains:

- 16 Normal alerts
- 11 Warning alerts
- 13 Critical alerts

Therefore, 13 out of 40 flight records are classified as Critical, representing a substantial proportion of the records requiring attention.

The geographical visualisation complements the numerical charts because it allows operational information to be viewed spatially rather than only as aggregated values.

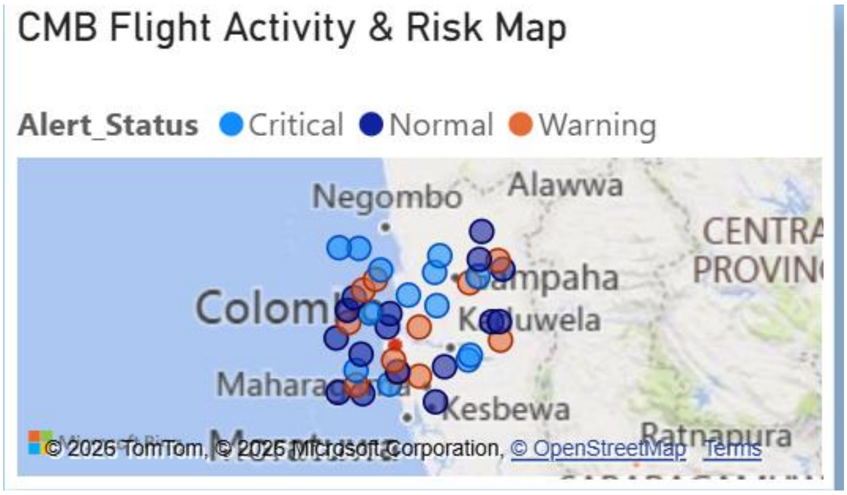

Figure 39 CMB Flight Activity and Risk Map

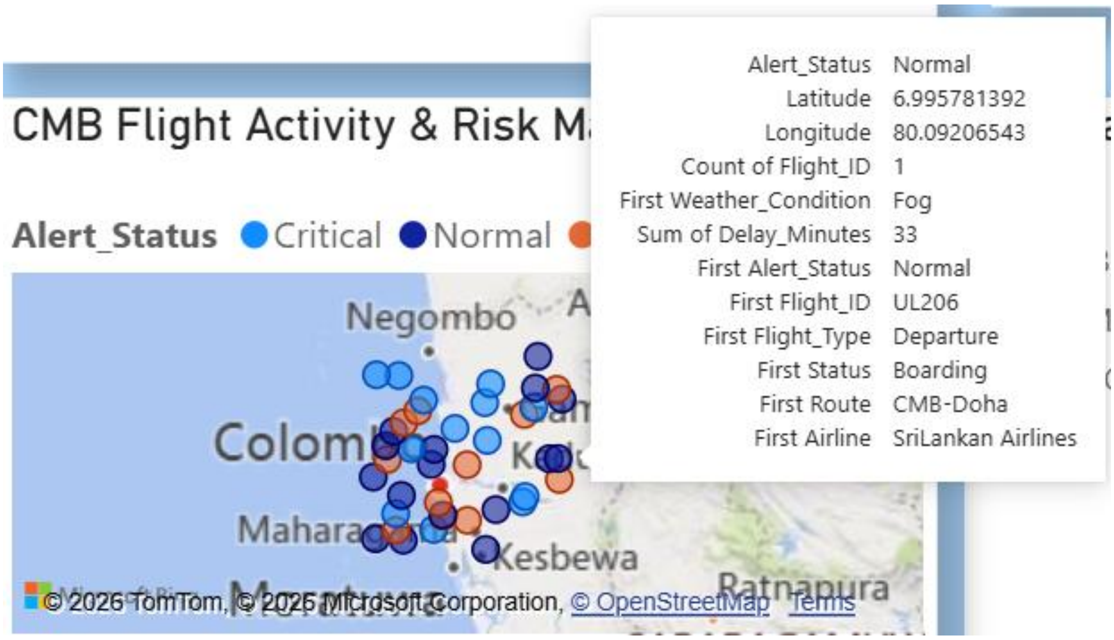

Figure 40 CMB Flight Activity and Risk Map tooltips

D.11 Interactive Dashboard Features

Interactivity was included to improve the usefulness of the dashboard.

The dashboard contains slicers positioned on the right-hand side, allowing users to filter the displayed information according to selected criteria such as:

- Airline
- Flight Type
- Route
- Flight Status
- Alert Status
- Weather Condition

When a filter is selected, the relevant charts and KPI values update according to the selected data.

For example, selecting a particular airline allows the user to examine its passenger traffic, average delay, flight status and operational risk information without manually filtering the underlying dataset.

This makes the dashboard more suitable for decision-support purposes than a static collection of charts.

Figure 41 dashboard before choosing a flight type

Figure 42 dashboard after choosing a flight type

D.12 Critical Discussion

The BI dashboard provides useful insights into CMB airport operations. The overall average delay is 31.4 minutes, with Emirates recording the highest airline average delay (40.2 minutes) and the CMB-London route the highest route average (52.7 minutes).

Operational risk is also significant, with 13 Critical and 11 Warning alerts. Weather is the most common operational issue, followed by technical issues, highlighting potential causes of delays and bottlenecks.

Passenger traffic totals 8,974, with Qatar Airways and IndiGo carrying the highest volumes. The CMB-Singapore route has the highest passenger traffic at 2,042 passengers, which may require greater resource allocation.

The map further supports analysis by showing flight locations alongside operational risk information. However, the dataset contains only 40 records, so the findings cannot represent long-term airport performance. The dashboard should therefore be viewed as a BI decision-support prototype rather than a live operational system.

Overall, the dashboard effectively combines flight, passenger, delay, risk and geographical information to support more informed airport decision-making.

D.13 Dashboard Evaluation Against Requirements

The developed dashboard addresses the main requirements of the task.

Requirement	**Dashboard Implementation**
Monitor flight operations	Flight Status Summary and KPI cards
Analyse delays	Average Delay by Airline
Track passenger flow	Passenger Traffic by Airline
Analyse international traffic	International Traffic by Route
Identify operational risks	Operational Risk chart
Monitor critical events	Critical Alert KPI
Geographic analysis	CMB Flight Activity and Risk Map
Interactive analysis	Airline, route, status and other slicers
Easy interpretation	KPI cards and clearly labelled visualisations

Table 29 Dashboard Evaluation Against Requirements table

The combination of these elements provides airport managers with both a high-level operational overview and the ability to investigate specific areas through interactive filtering.

D.14 Conclusion

A comprehensive Power BI dashboard was developed using the provided CMB flight dataset to support airport operational monitoring and decision-making.

The dashboard integrates KPIs, flight status analysis, delay analysis, passenger traffic, international route analysis, operational risk information, geographical mapping and interactive filtering within a single interface.

The analysis identified an overall average delay of 31.4 minutes, total passenger traffic of 8,974, and 13 Critical alert records. Weather was the most frequently recorded operational issue, while differences in delay performance were observed across both airlines and international routes.

The dashboard therefore demonstrates how Business Intelligence can transform raw airport flight data into an accessible decision-support tool. While the relatively small dataset limits the ability to make long-term operational conclusions, the dashboard provides an effective prototype that could be extended with live flight feeds and historical datasets for more advanced airport performance monitoring.

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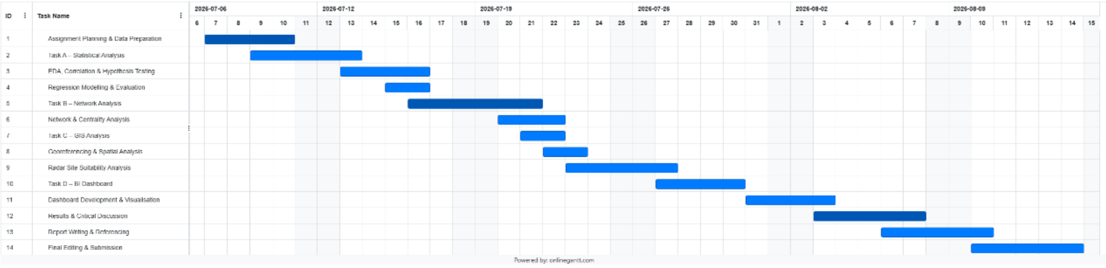
APPENDICES

Gantt Chart



TASK A

Workspace and All Source Files (All Tasks) -
https://drive.google.com/drive/folders/173plHFpqVbHkjYgYrtDGaXVSu6hH_rXT?usp=sharing

Appendix A.1- Outlier Analysis

- Outlier counts

Variable	Lower_Bo	Upper_Bo	Outlier_Count
airport_tr	74295.11	315491.1	3
avg_incon	34454.71	96522.01	2
fuel_price	0.473576	1.293156	1
avg_ticket	138.2425	360.9059	0
flight_frec	71.90547	171.8428	1
route_dist	471.7943	2675.942	2
passenger	-39000.8	247912.4	4

Figure 43 Outlier counts

- Seven boxplots

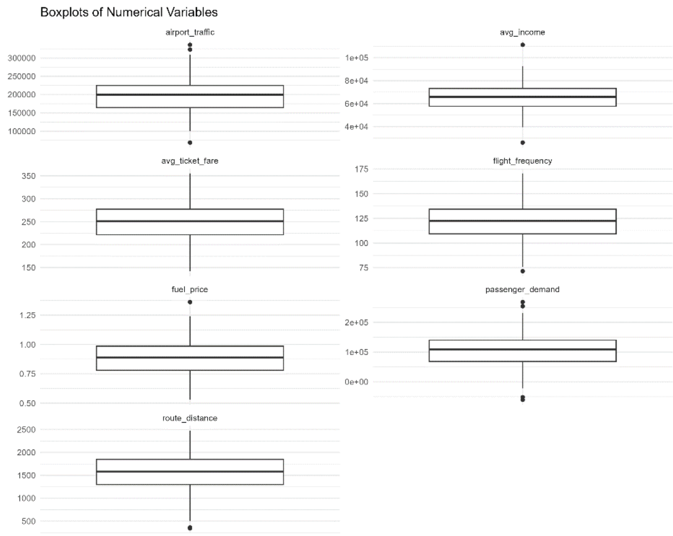


Figure 44 Seven boxplots

A.2 - Distribution Analysis

- Distribution of Numerical Variables

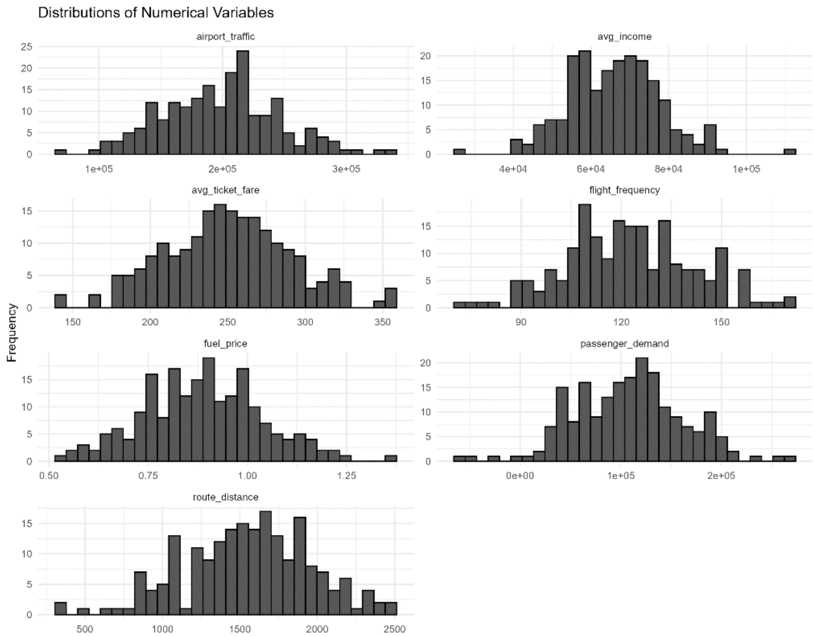


Figure 45 Distribution of Numerical Variables

A.3 - Relationship Analysis

- Relationships Between Predictors and Passenger Demand

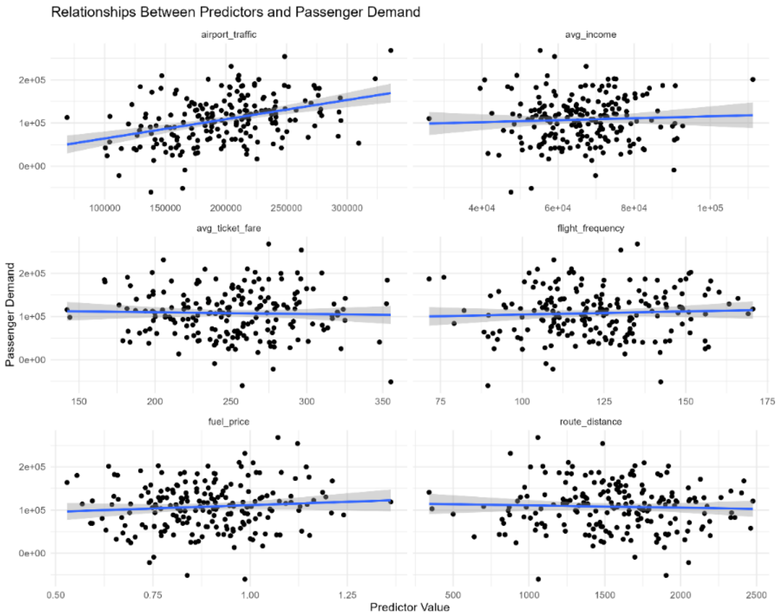


Figure 46 Relationships Between Predictors and Passenger Demand Scatter plots

A.4 - Correlation Analysis

- Correlation matrix

A	B	C	D	E	F	G	H	I	J
	airport_tr	avg_incon	fuel_price	avg_ticket	flight_fre	route_dist	passenger_demand		
airport_tr	1	0.095147	-0.1342	0.065398	-0.03121	-0.09099	0.385401		
avg_incon	0.095147	1	-0.03263	-0.1014	-0.14204	-0.03898	0.049549		
fuel_price	-0.1342	-0.03263	1	0.105959	0.025853	-0.05651	0.085676		
avg_ticket	0.065398	-0.1014	0.105959	1	0.171162	0.003465	-0.0304		
flight_fre	-0.03121	-0.14204	0.025853	0.171162	1	0.108882	0.052206		
route_dist	-0.09099	-0.03898	-0.05651	0.003465	0.108882	1	-0.04116		
passenger	0.385401	0.049549	0.085676	-0.0304	0.052206	-0.04116	1		

Figure 47 Correlation matrix

- Demand correlations

	A	B	C
1	Predictor	Correlation	
2	airport_tr	0.385401	
3	avg_incon	0.049549	
4	fuel_price	0.085676	
5	avg_ticket	-0.0304	
6	flight_frec	0.052206	
7	route_dist	-0.04116	
8			
9			

Figure 48 Demand correlations

- Correlation matrix visualisation

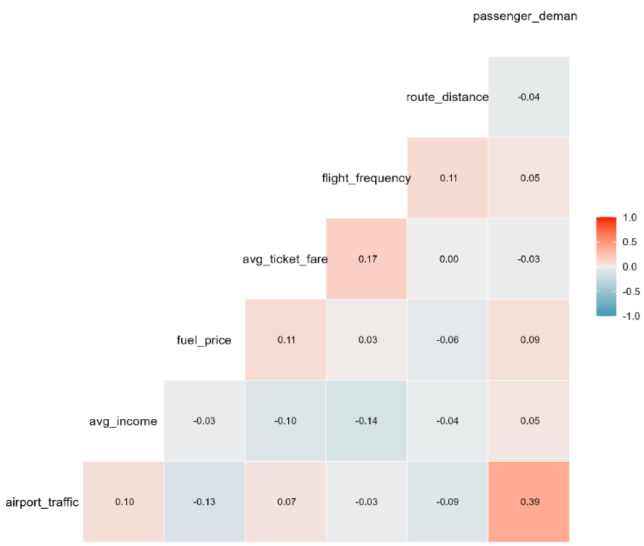


Figure 49 Correlation matrix visualisation

A.5 - Normality Analysis

- Seven Q-Q plots

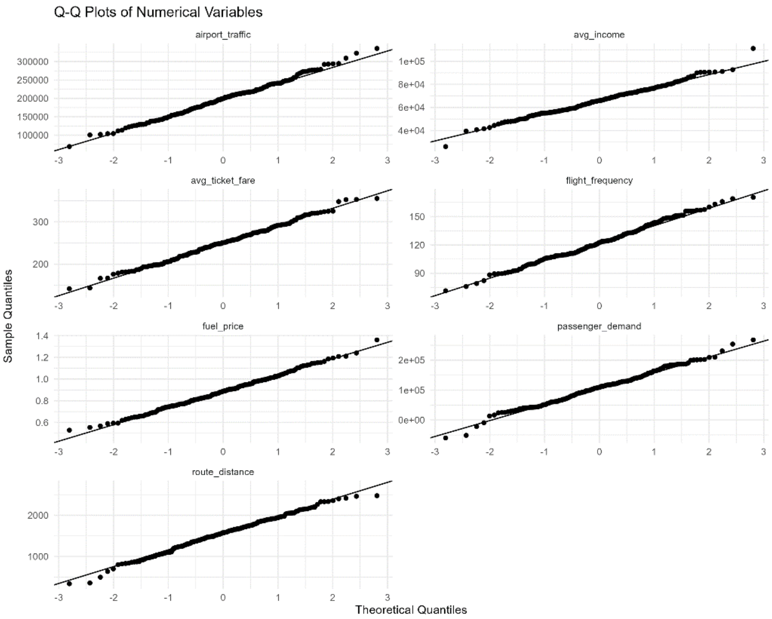


Figure 50 Seven Q-Q plots

A.6 -Simple Regression

- Six regression summaries

```
Call:
lm(formula = passenger_demand ~ route_distance, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-170698  -41978    2295   33646  157655

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)    116167.628  14910.763   7.791 3.67e-13 ***
route_distance     -5.379      9.279  -0.580    0.563
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 53950 on 198 degrees of freedom
Multiple R-squared:  0.001694,	Adjusted R-squared:  -0.003348
F-statistic: 0.3361 on 1 and 198 DF,  p-value: 0.5628
```

Figure 51 regression summaries 1

```
Call:
lm(formula = passenger_demand ~ fuel_price, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-171128  -37345     950   32046  154610

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    80358      23005   3.493 0.000589 ***
fuel_price     30946      25575   1.210 0.227717
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 53790 on 198 degrees of freedom
Multiple R-squared:  0.00734,	Adjusted R-squared:  0.002327
F-statistic: 1.464 on 1 and 198 DF,  p-value: 0.2277
```

Figure 52 regression summaries 2

```
Call:
lm(formula = passenger_demand ~ flight_frequency, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-163187  -37944    1816   32253  158458

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       89786.9    24797.9   3.621 0.000373 ***
flight_frequency    147.1      199.9   0.736 0.462842
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 53920 on 198 degrees of freedom
Multiple R-squared:  0.002725,	Adjusted R-squared:  -0.002311
F-statistic: 0.5411 on 1 and 198 DF,  p-value: 0.4628
```

Figure 53 regression summaries 3

```
Call:
lm(formula = passenger_demand ~ avg_ticket_fare, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-167766  -39458    1938   32247  161299 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)     117860.69   23792.86   4.954 1.56e-06 ***
avg_ticket_fare    -40.14      93.80  -0.428    0.669    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 53970 on 198 degrees of freedom
Multiple R-squared:  0.0009239, Adjusted R-squared:  -0.004122 
F-statistic: 0.1831 on 1 and 198 DF,  p-value: 0.6692
```

Figure 54 regression summaries 4

```
Call:
lm(formula = passenger_demand ~ avg_income, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-163904  -39056    3439   32650  162735 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 9.293e+04  2.165e+04   4.293 2.76e-05 ***
avg_income  2.253e-01  3.228e-01   0.698    0.486    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 53930 on 198 degrees of freedom
Multiple R-squared:  0.002455,  Adjusted R-squared:  -0.002583 
F-statistic: 0.4873 on 1 and 198 DF,  p-value: 0.4859
```

Figure 55 regression summaries 5

```
Call:
lm(formula = passenger_demand ~ airport_traffic, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-144234  -31671       9   32371  124420 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)     1.954e+04  1.543e+04   1.267    0.207    
airport_traffic 4.459e-01  7.587e-02   5.877 1.75e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49820 on 198 degrees of freedom
Multiple R-squared:  0.1485,	Adjusted R-squared:  0.1442 
F-statistic: 34.54 on 1 and 198 DF,  p-value: 1.746e-08
```

Figure 56 regression summaries 6

- Simple regression comparison

	A	B	C	D	E
	Predictor	R_Square	Adjusted_	P_Value	
	Airport_Tr	0.148534	0.144234	1.75E-08	
	Fuel_Price	0.00734	0.002327	0.227717	
	Flight_Fre	0.002725	-0.00231	0.462842	
	Average_I	0.002455	-0.00258	0.485947	
	Route_Dis	0.001694	-0.00335	0.562774	
	Average_T	9.24E-04	-0.00412	0.669187	

Figure 57 Simple regression comparison

Appendix A.7 - Multiple Regression

- Regression summary

```
Call:
lm(formula = passenger_demand ~ airport_traffic + avg_income +
    fuel_price + avg_ticket_fare + flight_frequency + route_distance,
    data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-135506  -30459    1322   31761  122302

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)      -3.775e+04  4.729e+04  -0.798   0.4258
airport_traffic   4.759e-01  7.717e-02   6.167    4e-09 ***
avg_income        8.052e-02  3.021e-01   0.267   0.7901
fuel_price        5.354e+04  2.400e+04   2.231   0.0268 *
avg_ticket_fare  -1.118e+02  8.855e+01  -1.262   0.2084
flight_frequency  2.212e+02  1.891e+02   1.170   0.2436
route_distance   -3.826e-01  8.632e+00  -0.044   0.9647
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 49550 on 193 degrees of freedom
Multiple R-squared:  0.1789,	Adjusted R-squared:  0.1534
F-statistic: 7.008 on 6 and 193 DF,  p-value: 9.266e-07
```

Figure 58 Regression summary

- Coefficients

Variable	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept	-37745.1	47291.43	-0.79814	0.425772
airport_tra	0.475886	0.077171	6.166604	4.00E-09
avg_incon	0.080516	0.302075	0.266545	0.790104
fuel_price	53537.84	23998.9	2.230845	0.026843
avg_ticket	-111.768	88.54582	-1.26226	0.208377
flight_frec	221.2152	189.1313	1.169638	0.243589
route_dist	-0.38257	8.632086	-0.04432	0.964696

Figure 59 Coefficients

- Confidence intervals

Variable	Lower_95_	Upper_95_CI
(Intercept	-131019	55529.32
airport_tr	0.323678	0.628093
avg_incon	-0.51528	0.676308
fuel_price	6204.044	100871.6
avg_ticket	-286.41	62.87348
flight_frec	-151.814	594.2447
route_dist	-17.4079	16.64277

Figure 60 Confidence intervals

- Model fit

R_Square	Adjusted_	Residual_	F_Statistic	Numerato	Denomina	Model_P_Value
0.178895	0.153369	49554.65	7.00819	6	193	9.27E-07

Figure 61 Model fi

- ANOVA

A	B	C	D	E	F
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
airport_tra	1	8.57E+10	8.57E+10	34.91276	1.53E-08
avg_incon	1	96624950	96624950	0.039348	0.84297
fuel_price	1	1.11E+10	1.11E+10	4.537326	0.03443
avg_ticket	1	2.91E+09	2.91E+09	1.185602	0.277576
flight_frec	1	3.37E+09	3.37E+09	1.372135	0.24289
route_dist	1	4823439	4823439	0.001964	0.964696
Residuals	193	4.74E+11	2.46E+09	NA	NA

Figure 62 ANOVA

Appendix A.8 - Regression Diagnostics

- Four diagnostic plots

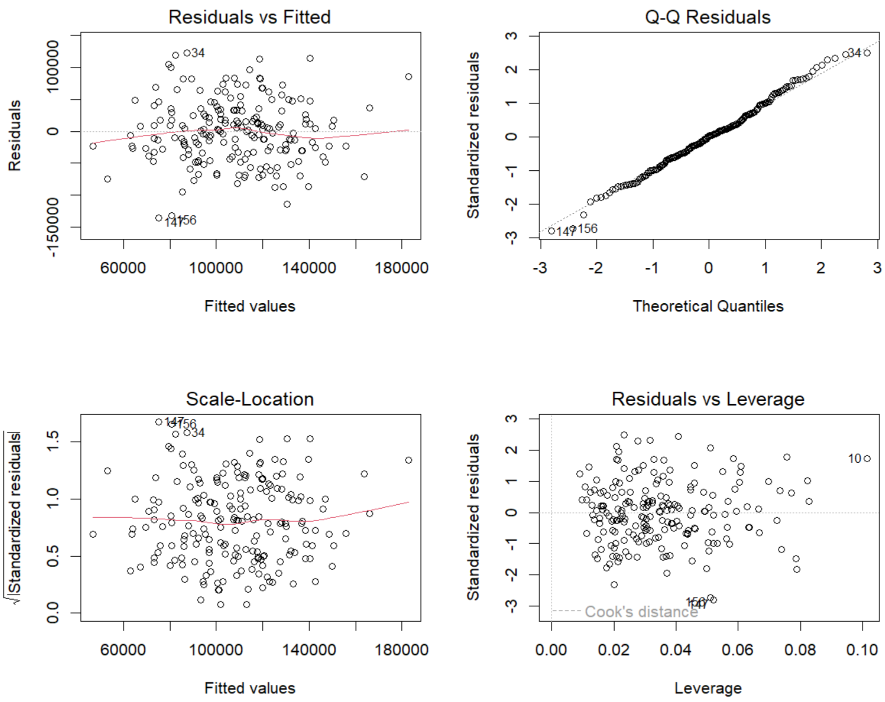


Figure 63 Four diagnostic plots

- VIF

Variable	VIF
airport_tr	1.04578
avg_incon	1.037321
fuel_price	1.037669
avg_ticket	1.056764
flight_fre	1.059561
route_dis	1.025608

Figure 64 VIF

- Breusch-Pagan

```
	studentized Breusch-Pagan test

data:  model
BP = 4.8406, df = 6, p-value = 0.5644
```

Figure 65 Breusch-Pagan

- Durbin-Watson

```
	Durbin-Watson test

data:  model
DW = 2.0712, p-value = 0.6972
alternative hypothesis: true autocorrelation is greater than 0
```

Figure 66 Durbin-Watson

Appendix A.9 - R Script Code

```r
# ============================================================
# 1. PACKAGES
# ============================================================

library(tidyverse)
library(car)
library(lmtest)
library(GGally)


# ============================================================
# 2. PROJECT PATHS
# ============================================================

task_path <- "C:/Users/Ayant/Desktop/Analytics_Business_Intelligence/Analytics_Business_Intelligence/Task A"

data_path <- file.path(task_path, "Data")
output_path <- file.path(task_path, "Outputs")

if (!dir.exists(output_path)) {
  dir.create(output_path, recursive = TRUE)
}


# ============================================================
# 3. IMPORT DATA
# ============================================================

data <- read_csv(
  file.path(data_path, "Air_Transport_Data.csv"),
  show_col_types = FALSE
)
```

Figure 67 R Script Code 1

```r
# ============================================================
# 4. DATA QUALITY AND DESCRIPTIVE STATISTICS
# ============================================================

missing_table <- data.frame(
  Variable = names(data),
  Missing_Count = colSums(is.na(data)),
  Missing_Percentage = colMeans(is.na(data)) * 100
)

write_csv(
  missing_table,
  file.path(output_path, "01_missing_value_analysis.csv")
)


data_quality <- data.frame(
  Variable = names(data),
  Data_Type = sapply(data, function(x) class(x)[1]),
  Unique_Values = sapply(
    data,
    function(x) length(unique(x))
  )
)

write_csv(
  data_quality,
  file.path(output_path, "02_data_quality_check.csv")
)


descriptive_statistics <- data %>%
  summarise(
    across(
      everything(),
      list(
        Mean = ~mean(.x, na.rm = TRUE),
        Median = ~median(.x, na.rm = TRUE),
        SD = ~sd(.x, na.rm = TRUE),
        Minimum = ~min(.x, na.rm = TRUE),
        Maximum = ~max(.x, na.rm = TRUE)
      )
    )
  )

write_csv(
  descriptive_statistics,
  file.path(output_path, "03_descriptive_statistics.csv")
)
```

Figure 68 R Script Code 2

```r
# ============================================================
# 5. OUTLIER ANALYSIS
# ============================================================

outlier_info <- function(x) {

  Q1 <- quantile(
    x,
    0.25,
    na.rm = TRUE
  )

  Q3 <- quantile(
    x,
    0.75,
    na.rm = TRUE
  )

  IQR_value <- IQR(
    x,
    na.rm = TRUE
  )

  lower <- Q1 - 1.5 * IQR_value
  upper <- Q3 + 1.5 * IQR_value

  data.frame(
    Lower_Bound = lower,
    Upper_Bound = upper,
    Outlier_Count = sum(
      x < lower | x > upper,
      na.rm = TRUE
    )
  )
}


outlier_table <- bind_rows(
  lapply(
    data,
    outlier_info
  ),
  .id = "Variable"
)

write_csv(
  outlier_table,
  file.path(
    output_path,
    "04_outlier_analysis.csv"
  )
)
```

Figure 69 R Script Code 3

```r
# ============================================================
# 6. BOXPLOTS
# ============================================================

data_long <- data %>%
  pivot_longer(
    cols = everything(),
    names_to = "Variable",
    values_to = "Value"
  )


boxplot_all <- ggplot(
  data_long,
  aes(
    x = Variable,
    y = Value
  )
) +
  geom_boxplot() +
  facet_wrap(
    ~ Variable,
    scales = "free",
    ncol = 2
  ) +
  labs(
    title = "Boxplots of Numerical Variables",
    x = "",
    y = ""
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank()
  )


ggsave(
  file.path(
    output_path,
    "05_boxplots_all_variables.png"
  ),
  boxplot_all,
  width = 10,
  height = 8,
  dpi = 300
)
```

Figure 70 R Script Code 4

```r
# ============================================================
# 7. DISTRIBUTION ANALYSIS
# ============================================================

histogram_all <- ggplot(
  data_long,
  aes(
    x = Value
  )
) +
  geom_histogram(
    bins = 30,
    color = "black"
  ) +
  facet_wrap(
    ~ Variable,
    scales = "free",
    ncol = 2
  ) +
  labs(
    title = "Distributions of Numerical Variables",
    x = "",
    y = "Frequency"
  ) +
  theme_minimal()


ggsave(
  file.path(
    output_path,
    "06_histograms_all_variables.png"
  ),
  histogram_all,
  width = 10,
  height = 8,
  dpi = 300
)
```

Figure 71 R Script Code 5

```r
# ============================================================
# 8. SCATTERPLOT ANALYSIS
# ============================================================

predictors <- c(
  "airport_traffic",
  "avg_income",
  "fuel_price",
  "avg_ticket_fare",
  "flight_frequency",
  "route_distance"
)


scatter_data <- data %>%
  pivot_longer(
    cols = all_of(predictors),
    names_to = "Predictor",
    values_to = "Value"
  )


scatter_all <- ggplot(
  scatter_data,
  aes(
    x = Value,
    y = passenger_demand
  )
) +
  geom_point() +
  geom_smooth(
    method = "lm",
    se = TRUE
  ) +
  facet_wrap(
    ~ Predictor,
    scales = "free_x",
    ncol = 2
  ) +
  labs(
    title = "Relationships Between Predictors and Passenger Demand",
    x = "Predictor Value",
    y = "Passenger Demand"
  ) +
  theme_minimal()


ggsave(
  file.path(
    output_path,
    "07_scatterplots_all_predictors.png"
  ),
  scatter_all,
  width = 10,
  height = 8,
  dpi = 300
)
```

Figure 72 R Script Code 6

```r
# -----------------------------------------------------------------
# 9. CORRELATION ANALYSIS
# =================================================================

correlation_matrix <- cor(
  data,
  use = "complete.obs",
  method = "pearson"
)


write.csv(
  correlation_matrix,
  file.path(
    output_path,
    "08_correlation_matrix.csv"
  )
)


demand_correlations <- data.frame(
  Predictor = predictors,
  Correlation = sapply(
    data[predictors],
    function(x) {
      cor(
        x,
        data$passenger_demand,
        use = "complete.obs"
      )
    }
  )
)


write_csv(
  demand_correlations,
  file.path(
    output_path,
    "09_demand_correlations.csv"
  )
)


correlation_plot <- GGally::ggcorr(
  data,
  method = c(
    "pairwise",
    "pearson"
  ),
  label = TRUE,
  label_round = 2,
  label_size = 3,
  layout.exp = 2
)


ggsave(
  file.path(
    output_path,
    "10_correlation_matrix_plot.png"
  ),
  correlation_plot,
  width = 10,
  height = 8,
  dpi = 300
)
```

Figure 73 R Script Code 7

```r
# ============================================================
# 10. NORMALITY HYPOTHESIS TESTING
# ============================================================

normality_test <- function(
    x,
    variable_name
) {

  test <- shapiro.test(x)

  data.frame(
    Variable = variable_name,
    W = as.numeric(test$statistic),
    P_Value = test$p.value,
    Decision = ifelse(
      test$p.value < 0.05,
      "Reject H0 - Non-normal distribution",
      "Fail to reject H0 - No significant evidence of non-normality"
    )
  )
}


normality_results <- bind_rows(

  normality_test(
    data$airport_traffic,
    "Airport Traffic"
  ),

  normality_test(
    data$avg_income,
    "Average Income"
  ),

  normality_test(
    data$fuel_price,
    "Fuel Price"
  ),

  normality_test(
    data$avg_ticket_fare,
    "Average Ticket Fare"
  ),

  normality_test(
    data$flight_frequency,
    "Flight Frequency"
  ),

  normality_test(
    data$route_distance,
    "Route Distance"
  ),

  normality_test(
    data$passenger_demand,
    "Passenger Demand"
  )
)


write_csv(
  normality_results,
  file.path(
    output_path,
    "11_normality_test_results.csv"
  )
)
```

Figure 74 R Script Code 8

==
11. Q-Q PLOTS
==

qq_all <- ggplot(
  data_long,
  aes(
    sample = Value
)
) +
  stat_qq() +
  stat_qq_line() +
  facet_wrap(
    ~ Variable,
    scales = "free",
    ncol = 2
) +
  labs(
    title = "Q-Q Plots of Numerical Variables",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles"
) +
  theme_minimal()

ggsave(
  file.path(
    output_path,
    "12_qq_plots_all_variables.png"
  ),
  qq_all,
  width = 10,
  height = 8,
  dpi = 300
)
```

*Figure 75 R Script Code 9*
```

```r
# 12. PEARSON CORRELATION HYPOTHESIS TESTING
# ==============================================================

correlation_test <- function(
    x,
    y,
    predictor_name
) {

  test <- cor.test(
    x,
    y,
    method = "pearson"
  )

  data.frame(
    Predictor = predictor_name,
    Correlation = as.numeric(test$estimate),
    P_Value = test$p.value,
    Confidence_Lower = test$conf.int[1],
    Confidence_Upper = test$conf.int[2],
    Decision = ifelse(
      test$p.value < 0.05,
      "Reject H0 - Significant correlation",
      "Fail to reject H0 - No significant correlation"
    )
  )
}


correlation_tests <- bind_rows(

  correlation_test(
    data$airport_traffic,
    data$passenger_demand,
    "Airport Traffic"
  ),

  correlation_test(
    data$avg_income,
    data$passenger_demand,
    "Average Income"
  ),

  correlation_test(
    data$fuel_price,
    data$passenger_demand,
    "Fuel Price"
  ),

  correlation_test(
    data$avg_ticket_fare,
    data$passenger_demand,
    "Average Ticket Fare"
  ),

  correlation_test(
    data$flight_frequency,
    data$passenger_demand,
    "Flight Frequency"
  ),

  correlation_test(
    data$route_distance,
    data$passenger_demand,
    "Route Distance"
  )
)


write_csv(
  correlation_tests,
  file.path(
    output_path,
    "13_pearson_correlation_tests.csv"
```

Figure 76 R Script Code 10

```r
# ============================================================
# 13. SIMPLE LINEAR REGRESSION
# ============================================================

simple_models <- list(

  Airport_Traffic = lm(
    passenger_demand ~ airport_traffic,
    data = data
  ),

  Average_Income = lm(
    passenger_demand ~ avg_income,
    data = data
  ),

  Fuel_Price = lm(
    passenger_demand ~ fuel_price,
    data = data
  ),

  Average_Ticket_Fare = lm(
    passenger_demand ~ avg_ticket_fare,
    data = data
  ),

  Flight_Frequency = lm(
    passenger_demand ~ flight_frequency,
    data = data
  ),

  Route_Distance = lm(
    passenger_demand ~ route_distance,
    data = data
  )
)


simple_regression_results <- bind_rows(

  lapply(
    names(simple_models),
    function(x) {

      model_x <- simple_models[[x]]

      model_summary <- summary(model_x)

      data.frame(
        Predictor = x,
        R_Squared = model_summary$r.squared,
        Adjusted_R_Squared = model_summary$adj.r.squared,
        P_Value = coef(model_summary)[2, 4]
      )
    }
  )

) %>%
  arrange(
    desc(R_Squared)
  )


write_csv(
  simple_regression_results,
  file.path(
    output_path,
    "14_simple_regression_comparison.csv"
  )
)
```

Figure 77 R Script Code 11

```r
# R² comparison for the six simple regression models

r_squared_plot <- ggplot(
  simple_regression_results,
  aes(
    x = reorder(
      Predictor,
      R_Squared
    ),
    y = R_Squared
  )
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Comparison of R² Values for the Six Simple Regression Models",
    x = "Predictor",
    y = "R²"
  ) +
  theme_minimal()


ggsave(
  file.path(
    output_path,
    "15_simple_regression_r_squared_comparison.png"
  ),
  r_squared_plot,
  width = 8,
  height = 5,
  dpi = 300
)


# Save individual simple regression summaries

for (model_name in names(simple_models)) {

  capture.output(
    summary(
      simple_models[[model_name]]
    ),
    file = file.path(
      output_path,
      paste0(
        "16_simple_regression_",
        tolower(model_name),
        ".txt"
      )
    )
  )
}
```

Figure 78 R Script Code 12

```r
# ============================================================
# 14. MULTIPLE LINEAR REGRESSION
# ============================================================

model <- lm(
  passenger_demand ~
    airport_traffic +
    avg_income +
    fuel_price +
    avg_ticket_fare +
    flight_frequency +
    route_distance,
  data = data
)


model_summary <- summary(model)


capture.output(
  model_summary,
  file = file.path(
    output_path,
    "22_multiple_regression_summary.txt"
  )
)


# ============================================================
# 15. REGRESSION COEFFICIENTS
# ============================================================

coefficient_table <- as.data.frame(
  model_summary$coefficients
)


coefficient_table$variable <-
  rownames(coefficient_table)


rownames(coefficient_table) <- NULL


coefficient_table <- coefficient_table %>%
  select(
    variable,
    Estimate,
    `Std. Error`,
    `t value`,
    `Pr(>|t|)`
  )


write_csv(
  coefficient_table,
  file.path(
    output_path,
    "23_regression_coefficients.csv"
  )
)
```

Figure 79 R Script Code 13

```r
# ==========================================================
# 16. CONFIDENCE INTERVALS
# ==========================================================

confidence_intervals <- as.data.frame(
  confint(model)
)


confidence_intervals$Variable <-
  rownames(confidence_intervals)


rownames(confidence_intervals) <- NULL


names(confidence_intervals)[1:2] <- c(
  "Lower_95_CI",
  "Upper_95_CI"
)


confidence_intervals <- confidence_intervals %>%
  select(
    Variable,
    Lower_95_CI,
    Upper_95_CI
  )


write_csv(
  confidence_intervals,
  file.path(
    output_path,
    "24_regression_confidence_intervals.csv"
  )
)


# ----------------------------------------------------------
```

Figure 80 R Script Code 14

```r
# ============================================================
# 17. MODEL FIT
# ============================================================

f_value <- model_summary$fstatistic[1]

f_num_df <- model_summary$fstatistic[2]

f_den_df <- model_summary$fstatistic[3]


model_p_value <- pf(
  f_value,
  f_num_df,
  f_den_df,
  lower.tail = FALSE
)


model_fit <- data.frame(
  R_Squared = model_summary$r.squared,
  Adjusted_R_Squared = model_summary$adj.r.squared,
  Residual_Standard_Error = model_summary$sigma,
  F_Statistic = f_value,
  Numerator_DF = f_num_df,
  Denominator_DF = f_den_df,
  Model_P_Value = model_p_value
)


write_csv(
  model_fit,
  file.path(
    output_path,
    "25_model_fit_statistics.csv"
  )
)
```

Figure 81 R Script Code 15

```r
# ==============================================================
# 18. REGRESSION ANOVA
# ==============================================================

anova_results <- anova(model)


write.csv(
  anova_results,
  file.path(
    output_path,
    "26_regression_anova.csv"
  )
)


# ==============================================================
# 19. REGRESSION DIAGNOSTICS
# ==============================================================

png(
  file.path(
    output_path,
    "27_multiple_regression_diagnostic_plots.png"
  ),
  width = 1200,
  height = 1000,
  res = 150
)


par(
  mfrow = c(
    2,
    2
  )
)


plot(model)


dev.off()


par(
  mfrow = c(
    1,
    1
  )
)


# ==============================================================
# 20. MULTICOLLINEARITY - VIF
# ==============================================================

vif_results <- car::vif(model)


vif_table <- data.frame(
  Variable = names(vif_results),
  VIF = as.numeric(vif_results)
)


write_csv(
  vif_table,
  file.path(
    output_path,
    "28_vif_multicollinearity.csv"
  )
)
```

Figure 82 R Script Code 16

```r
# ============================================================
# 21. HETEROSCEDASTICITY - BREUSCH-PAGAN
# ============================================================

bp_test <- bptest(model)


capture.output(
  bp_test,
  file = file.path(
    output_path,
    "29_breusch_pagan_test.txt"
  )
)


# ============================================================
# 22. AUTOCORRELATION - DURBIN-WATSON
# ============================================================

dw_test <- dwtest(model)


capture.output(
  dw_test,
  file = file.path(
    output_path,
    "30_durbin_watson_test.txt"
  )
)


# ============================================================
# 23. RESIDUAL NORMALITY
# ============================================================

residual_normality <- shapiro.test(
  residuals(model)
)


capture.output(
  residual_normality,
  file = file.path(
    output_path,
    "31_residual_normality_test.txt"
  )
)
```

Figure 83 R Script Code 17

```r
# ============================================================
# 24. RESIDUAL Q-Q PLOT
# ============================================================

residual_qq <- ggplot(
  data.frame(
    Residuals = residuals(model)
  ),
  aes(
    sample = Residuals
  )
) +
  stat_qq() +
  stat_qq_line() +
  labs(
    title = "Q-Q Plot of Multiple Regression Residuals",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_minimal()


ggsave(
  file.path(
    output_path,
    "32_regression_residual_qq_plot.png"
  ),
  residual_qq,
  width = 8,
  height = 5,
  dpi = 300
)


# ============================================================
# 25. RESIDUALS VS FITTED
# ============================================================

residual_fitted <- ggplot(
  data.frame(
    Fitted = fitted(model),
    Residuals = residuals(model)
  ),
  aes(
    x = Fitted,
    y = Residuals
  )
) +
  geom_point() +
  geom_hline(
    yintercept = 0,
    linetype = "dashed"
  ) +
  labs(
    title = "Residuals vs Fitted Values",
    x = "Fitted Values",
    y = "Residuals"
  ) +
  theme_minimal()


ggsave(
  file.path(
    output_path,
    "33_residuals_vs_fitted.png"
  ),
  residual_fitted,
  width = 8,
  height = 5,
  dpi = 300
)
```

Figure 84 R Script Code 18

```r
# ============================================================
# 26. ACTUAL VS PREDICTED VALUES
# ============================================================

data_with_predictions <- data %>%
  mutate(
    Predicted_Passenger_Demand = predict(model),
    Regression_Residual = residuals(model)
  )


write_csv(
  data_with_predictions,
  file.path(
    output_path,
    "34_regression_predictions.csv"
  )
)


actual_predicted_plot <- ggplot(
  data_with_predictions,
  aes(
    x = passenger_demand,
    y = Predicted_Passenger_Demand
  )
) +
  geom_point() +
  geom_abline(
    slope = 1,
    intercept = 0,
    linetype = "dashed"
  ) +
  labs(
    title = "Actual vs Predicted Passenger Demand",
    x = "Actual Passenger Demand",
    y = "Predicted Passenger Demand"
  ) +
  theme_minimal()


ggsave(
  file.path(
    output_path,
    "35_actual_vs_predicted_demand.png"
  ),
  actual_predicted_plot,
  width = 8,
  height = 5,
  dpi = 300
)
```

Figure 85 R Script Code 19

```r
# ============================================================
# 27. REGRESSION EQUATION
# ============================================================

coefficients <- coef(model)


regression_equation <- paste0(
  "Passenger Demand = ",
  round(
    coefficients[1],
    4
  ),
  " + (",
  round(
    coefficients[2],
    4
  ),
  " x Airport Traffic)",
  " + (",
  round(
    coefficients[3],
    4
  ),
  " x Average Income)",
  " + (",
  round(
    coefficients[4],
    4
  ),
  " x Fuel Price)",
  " + (",
  round(
    coefficients[5],
    4
  ),
  " x Average Ticket Fare)",
  " + (",
  round(
    coefficients[6],
    4
  ),
  " x Flight Frequency)",
  " + (",
  round(
    coefficients[7],
    4
  ),
  " x Route Distance)"
)


writeLines(
  regression_equation,
  file.path(
    output_path,
    "36_regression_equation.txt"
  )
)
```

Figure 86 R Script Code 20

```r
# ============================================================
# 28. FINAL REGRESSION TABLE
# ============================================================

final_model_results <- coefficient_table %>%
  left_join(
    confidence_intervals,
    by = "Variable"
  )


write_csv(
  final_model_results,
  file.path(
    output_path,
    "37_final_regression_model_results.csv"
  )
)


# ============================================================
# 29. SAVE ANALYSIS OBJECTS
# ============================================================

save(
  data,
  model,
  simple_models,
  correlation_matrix,
  correlation_tests,
  normality_results,
  simple_regression_results,
  final_model_results,
  file = file.path(
    output_path,
    "38_task_A_analysis_objects.RData"
  )
)


# ============================================================
# 30. COMPLETION MESSAGE
# ============================================================

cat(
  "\nTASK A ANALYSIS COMPLETED SUCCESSFULLY\n"
)

cat(
  "Outputs saved to:\n"
)

cat(
  output_path,
  "\n"
)

cat(
  "Number of output files:",
  length(
    list.files(output_path)
  ),
  "\n"
)
```

Figure 87 R Script Code 21

Appendix B

Task B

B.1 Complete R Implementation

```
# ============================================================
# 1. LOAD REQUIRED PACKAGES
# ============================================================

library(tidyverse)
library(igraph)
library(ggraph)
library(tidygraph)


# ============================================================
# 2. DEFINE FILE AND OUTPUT LOCATIONS
# ============================================================

data_folder <- "C:/Users/Ayant/Desktop/Analytics_Business_Intelligence/Analytics_Business_Intelligence/Task B/Data"

data_file <- file.path(
  data_folder,
  "SriLanka_Aviation_SNA_Dataset.csv"
)

output_folder <- "C:/Users/Ayant/Desktop/Analytics_Business_Intelligence/Analytics_Business_Intelligence/Task B/Outputs"

if (!dir.exists(output_folder)) {
  dir.create(output_folder, recursive = TRUE)
}


# ============================================================
# 3. CHECK AVAILABLE FILES
# ============================================================

list.files(data_folder)


# ============================================================
# 4. IMPORT THE DATASET
# ============================================================

network_data <- read_csv(
  data_file,
  show_col_types = FALSE
)


# ============================================================
# 5. BASIC DATA INSPECTION
# ============================================================

head(network_data)

tail(network_data)

str(network_data)

dim(network_data)

names(network_data)

summary(network_data)
```

Figure 88 Task B R Script Code 1

```r
# ============================================================
# 6. CHECK DATA QUALITY
# ============================================================

colSums(is.na(network_data))

sum(is.na(network_data))

network_data[duplicated(network_data), ]

unique(network_data$Relationship)

length(unique(network_data$Source))

length(unique(network_data$Target))

range(network_data$Weight, na.rm = TRUE)

table(network_data$Relationship)



# ============================================================
# 7. PREPARE NETWORK DATA
# ============================================================

network_data <- network_data %>%
  mutate(
    Source = as.character(Source),
    Target = as.character(Target),
    Relationship = as.character(Relationship),
    Weight = as.numeric(Weight)
  )


network_data <- network_data %>%
  filter(
    !is.na(Source),
    !is.na(Target),
    !is.na(Relationship),
    !is.na(Weight)
  )


head(network_data)

# ============================================================
# 8. CREATE THE AVIATION NETWORK
# ============================================================

aviation_network <- graph_from_data_frame(
  network_data,
  directed = TRUE
)


aviation_network

vcount(aviation_network)

ecount(aviation_network)

V(aviation_network)$name
```

Figure 89 Task B R Script Code 2

```r
# ==============================================================
# 9. BASIC NETWORK PROPERTIES
# ==============================================================

network_density <- edge_density(
  aviation_network,
  loops = FALSE
)

network_density


components(
  aviation_network,
  mode = "weak"
)$no


average_degree <- mean(
  degree(
    aviation_network,
    mode = "all"
  )
)

average_degree


average_weighted_degree <- mean(
  strength(
    aviation_network,
    mode = "all",
    weights = E(aviation_network)$weight
  )
)

average_weighted_degree


#
```

Figure 90 Task B R Script Code 3

```r
# ============================================================
# 10. DEGREE CENTRALITY
# ============================================================

degree_centrality <- degree(
  aviation_network,
  mode = "all"
)


in_degree_centrality <- degree(
  aviation_network,
  mode = "in"
)


out_degree_centrality <- degree(
  aviation_network,
  mode = "out"
)


weighted_degree <- strength(
  aviation_network,
  mode = "all",
  weights = E(aviation_network)$weight
)


degree_results <- data.frame(
  Node = names(degree_centrality),
  Degree_Centrality = as.numeric(degree_centrality),
  In_Degree = as.numeric(in_degree_centrality),
  Out_Degree = as.numeric(out_degree_centrality),
  Weighted_Degree = as.numeric(weighted_degree)
)


degree_results <- degree_results %>%
  arrange(desc(Degree_Centrality))


degree_results


write_csv(
  degree_results,
  file.path(
    output_folder,
    "degree_centrality_results.csv"
  )
)
```

Figure 91 Task B R Script Code 4

```r
# 11. BETWEENNESS CENTRALITY
# ============================================================

betweenness_centrality <- betweenness(
  aviation_network,
  directed = TRUE,
  weights = 1 / E(aviation_network)$Weight,
  normalized = TRUE
)


betweenness_results <- data.frame(
  Node = names(betweenness_centrality),
  Betweenness_Centrality = as.numeric(betweenness_centrality)
)


betweenness_results <- betweenness_results %>%
  arrange(desc(Betweenness_Centrality))


betweenness_results


write_csv(
  betweenness_results,
  file.path(
    output_folder,
    "betweenness_centrality_results.csv"
  )
)


# ============================================================
# 12. COMBINE CENTRALITY MEASURES
# ============================================================

centrality_results <- degree_results %>%
  left_join(
    betweenness_results,
    by = "Node"
  ) %>%
  arrange(
    desc(Degree_Centrality)
  )


centrality_results


write_csv(
  centrality_results,
  file.path(
    output_folder,
    "complete_centrality_results.csv"
  )
)


"
```

Figure 92 Task B R Script Code 5

```r
# ============================================================
# 13. IDENTIFY MOST INFLUENTIAL NODES
# ============================================================

top_degree_nodes <- centrality_results %>%
  arrange(desc(Degree_Centrality)) %>%
  slice_head(n = 10)


top_degree_nodes


top_betweenness_nodes <- centrality_results %>%
  arrange(desc(Betweenness_Centrality)) %>%
  slice_head(n = 10)


top_betweenness_nodes


write_csv(
  top_degree_nodes,
  file.path(
    output_folder,
    "top_degree_centrality_nodes.csv"
  )
)


write_csv(
  top_betweenness_nodes,
  file.path(
    output_folder,
    "top_betweenness_centrality_nodes.csv"
  )
)


# ============================================================
# 14. IDENTIFY CRITICAL BRIDGE NODES
# ============================================================

critical_bridge_nodes <- centrality_results %>%
  arrange(desc(Betweenness_Centrality)) %>%
  slice_head(n = 5)


critical_bridge_nodes


write_csv(
  critical_bridge_nodes,
  file.path(
    output_folder,
    "critical_bridge_nodes.csv"
  )
)
```

Figure 93 Task B R Script Code 6

```r
# ==============================================================
# 15. NETWORK GRAPH - CLEAN VERSION
# ==============================================================

set.seed(123)

network_layout <- layout_with_fr(
  aviation_network,
  niter = 3000
)

png(
  file.path(
    output_folder,
    "network_graph_basic.png"
  ),
  width = 2200,
  height = 1700,
  res = 200
)

plot(
  aviation_network,
  layout = network_layout,
  vertex.size = 18,
  vertex.label.cex = 0.65,
  vertex.label.color = "black",
  vertex.frame.color = "grey30",
  edge.arrow.size = 0.25,
  edge.curved = 0.15,
  edge.width = 1 + E(aviation_network)$weight / 4,
  edge.color = adjustcolor("grey40", alpha.f = 0.55),
  asp = 0,
  main = "Sri Lanka Aviation Stakeholder Network"
)

dev.off()
```

Figure 94 Task B R Script Code 7

```r
# ============================================================
# 16. NETWORK GRAPH WITH DEGREE CENTRALITY
# ============================================================

V(aviation_network)$Degree_Centrality <- degree_centrality


node_sizes <- 12 +
  (
    degree_centrality /
      max(degree_centrality)
  ) * 32


png(
  file.path(
    output_folder,
    "network_graph_degree_centrality.png"
  ),
  width = 2400,
  height = 1800,
  res = 200
)

set.seed(123)

plot(
  aviation_network,
  layout = network_layout,
  vertex.size = node_sizes,
  vertex.label.cex = 0.65,
  vertex.label.color = "black",
  vertex.frame.color = "grey30",
  edge.arrow.size = 0.22,
  edge.curved = 0.18,
  edge.width = 1 + E(aviation_network)$weight / 4,
  edge.color = adjustcolor("grey40", alpha.f = 0.50),
  asp = 0,
  main = "Sri Lanka Aviation Network - Degree Centrality"
)

dev.off()
```

Figure 95 Task B R Script Code 8

```r
# ============================================================
# 17. NETWORK GRAPH USING GGRAPH
# ============================================================

V(aviation_network)$Degree_Centrality <- degree_centrality


network_plot <- ggraph(
  as_tbl_graph(aviation_network),
  layout = network_layout
) +

  geom_edge_link(
    aes(
      width = weight
    ),
    alpha = 0.45,
    colour = "grey40",
    curvature = 0.15,
    arrow = arrow(
      length = unit(2.5, "mm"),
      type = "closed"
    ),
    end_cap = circle(4, "mm"),
    start_cap = circle(4, "mm")
  ) +

  geom_node_point(
    aes(
      size = Degree_Centrality
    ),
    alpha = 0.9
  ) +

  geom_node_text(
    aes(
      label = name
    ),
    repel = TRUE,
    size = 3.2,
    box.padding = 0.8,
    point.padding = 0.5,
    force = 2,
    max.overlaps = Inf
  ) +

  scale_edge_width(
    range = c(0.4, 2.5)
  ) +

  scale_size(
    range = c(5, 14)
  ) +

  labs(
    title = "Sri Lanka Aviation Stakeholder Network",
    subtitle = "Node size represents degree centrality; edge width represents relationship weight",
    caption = "Source: Sri Lanka Aviation SNA Dataset"
  ) +

  theme_graph() +

  theme(
    plot.title = element_text(
      size = 16,
      face = "bold"
    ),
    plot.subtitle = element_text(
      size = 11
    ),
    plot.margin = margin(
      20,
      30,
```

Figure 96 Task B R Script Code 9

```r
# ============================================================
# 18. TOP NODES - DEGREE CENTRALITY GRAPH
# ============================================================

degree_plot_data <- centrality_results %>%
  arrange(desc(Degree_Centrality)) %>%
  slice_head(n = 10)


degree_plot <- ggplot(
  degree_plot_data,
  aes(
    x = reorder(
      Node,
      Degree_Centrality
    ),
    y = Degree_Centrality
  )
) +

  geom_col() +

  coord_flip() +

  labs(
    title = "Top Aviation Stakeholders by Degree Centrality",
    x = "Stakeholder",
    y = "Degree Centrality"
  ) +

  theme_minimal()


ggsave(
  file.path(
    output_folder,
    "degree_centrality_chart.png"
  ),
  degree_plot,
  width = 10,
  height = 7,
  dpi = 300
)
```

Figure 97 Task B R Script Code 10

```r
# ============================================================
# 19. TOP NODES - BETWEENNESS CENTRALITY GRAPH
# ============================================================

betweenness_plot_data <- centrality_results %>%
  arrange(desc(Betweenness_Centrality)) %>%
  slice_head(n = 10)


betweenness_plot <- ggplot(
  betweenness_plot_data,
  aes(
    x = reorder(
      Node,
      Betweenness_Centrality
    ),
    y = Betweenness_Centrality
  )
) +

  geom_col() +

  coord_flip() +

  labs(
    title = "Top Aviation Stakeholders by Betweenness Centrality",
    x = "Stakeholder",
    y = "Betweenness Centrality"
  ) +

  theme_minimal()


ggsave(
  file.path(
    output_folder,
    "betweenness_centrality_chart.png"
  ),
  betweenness_plot,
  width = 10,
  height = 7,
  dpi = 300
)
```

Figure 98 Task B R Script Code 11

```r
# ============================================================
# 20. RELATIONSHIP TYPE ANALYSIS
# ============================================================

relationship_summary <- network_data %>%
  group_by(Relationship) %>%
  summarise(
    Number_of_Relationships = n(),
    Total_Weight = sum(Weight),
    Average_Weight = mean(Weight),
    .groups = "drop"
  ) %>%
  arrange(
    desc(Number_of_Relationships)
  )


relationship_summary


write_csv(
  relationship_summary,
  file.path(
    output_folder,
    "relationship_type_summary.csv"
  )
)


# ============================================================
# 21. RELATIONSHIP TYPE GRAPH
# ============================================================

relationship_plot <- ggplot(
  relationship_summary,
  aes(
    x = reorder(
      Relationship,
      Number_of_Relationships
    ),
    y = Number_of_Relationships
  )
) +

  geom_col() +

  coord_flip() +

  labs(
    title = "Aviation Network Relationships by Type",
    x = "Relationship Type",
    y = "Number of Relationships"
  ) +

  theme_minimal()


ggsave(
  file.path(
    output_folder,
    "relationship_type_chart.png"
  ),
  relationship_plot,
  width = 10,
  height = 7,
  dpi = 300
)
```

Figure 99 Task B R Script Code 12

```r
# ============================================================
# 22. NETWORK CONNECTIVITY ANALYSIS
# ============================================================

weak_components <- components(
  aviation_network,
  mode = "weak"
)


weak_components


strong_components <- components(
  aviation_network,
  mode = "strong"
)


strong_components


weak_components$no

strong_components$no


component_results <- data.frame(
  Node = V(aviation_network)$name,
  Weak_Component = weak_components$membership,
  Strong_Component = strong_components$membership
)


component_results


write_csv(
  component_results,
  file.path(
    output_folder,
    "network_component_analysis.csv"
  )
)
```

Figure 100 Task B R Script Code 13

```r
# ============================================================
# 23. SHORTEST PATH ANALYSIS
# ============================================================

edge_distance <- 1 /
  E(aviation_network)$weight


average_path_length <- mean_distance(
  aviation_network,
  directed = TRUE,
  weights = edge_distance
)


average_path_length


network_diameter <- diameter(
  aviation_network,
  directed = TRUE,
  weights = edge_distance
)


network_diameter


path_analysis_results <- data.frame(
  Average_Shortest_Path = average_path_length,
  Network_Diameter = network_diameter
)


write_csv(
  path_analysis_results,
  file.path(
    output_folder,
    "shortest_path_analysis.csv"
  )
)
```

Figure 101 Task B R Script Code 14

```r
# 24. VULNERABILITY ANALYSIS - ATC
# ============================================================

atc_node <- V(
  aviation_network
)[
  grepl(
    "Air Traffic Control",
    name,
    ignore.case = TRUE
  )
]


atc_node$name


if (length(atc_node) > 0) {

  network_without_atc <- delete_vertices(
    aviation_network,
    atc_node
  )

  atc_components <- components(
    network_without_atc,
    mode = "weak"
  )

  atc_components$no

} else {

  network_without_atc <- NULL

  atc_components <- NULL

}
```

Figure 102 Task B R Script Code 15

```r
# ============================================================
# 25. VULNERABILITY ANALYSIS - GROUND HANDLING
# ============================================================

ground_node <- V(
  aviation_network
)[
  grepl(
    "Ground Handling",
    name,
    ignore.case = TRUE
  )
]


ground_node$name


if (length(ground_node) > 0) {

  network_without_ground <- delete_vertices(
    aviation_network,
    ground_node
  )

  ground_components <- components(
    network_without_ground,
    mode = "weak"
  )

  ground_components$no

} else {

  network_without_ground <- NULL

  ground_components <- NULL

}
```

Figure 103 Task B R Script Code 16

```r
# ============================================================
# 26. VULNERABILITY COMPARISON
# ============================================================

original_nodes <- vcount(
  aviation_network
)


original_edges <- ecount(
  aviation_network
)


original_density <- edge_density(
  aviation_network,
  loops = FALSE
)


original_components <- components(
  aviation_network,
  mode = "weak"
)$no


if (!is.null(network_without_atc)) {

  atc_nodes <- vcount(
    network_without_atc
  )

  atc_edges <- ecount(
    network_without_atc
  )

  atc_density <- edge_density(
    network_without_atc,
    loops = FALSE
  )

  atc_component_count <- components(
    network_without_atc,
    mode = "weak"
  )$no

} else {

  atc_nodes <- NA
  atc_edges <- NA
  atc_density <- NA
  atc_component_count <- NA

}
```

Figure 104 Task B R Script Code 17

```r
if (!is.null(network_without_ground)) {

  ground_nodes <- vcount(
    network_without_ground
  )

  ground_edges <- ecount(
    network_without_ground
  )

  ground_density <- edge_density(
    network_without_ground,
    loops = FALSE
  )

  ground_component_count <- components(
    network_without_ground,
    mode = "weak"
  )$no

} else {

  ground_nodes <- NA
  ground_edges <- NA
  ground_density <- NA
  ground_component_count <- NA

}


vulnerability_results <- data.frame(

  Scenario = c(
    "Original Network",
    "ATC Failure",
    "Ground Handling Failure"
  ),

  Number_of_Nodes = c(
    original_nodes,
    atc_nodes,
    ground_nodes
  ),

  Number_of_Edges = c(
    original_edges,
    atc_edges,
    ground_edges
  ),

  Network_Density = c(
    original_density,
    atc_density,
    ground_density
  ),

  Number_of_Weak_Components = c(
    original_components,
    atc_component_count,
    ground_component_count
  )
)


vulnerability_results


write_csv(
  vulnerability_results,
  file.path(
    output_folder,
    "network_vulnerability_analysis.csv"
  )
```

Figure 105 Task B R Script Code 18

```r
# ==============================================================
# 27. IDENTIFY HIGH-WEIGHT RELATIONSHIPS
# ==============================================================

high_weight_relationships <- network_data %>%
  arrange(
    desc(weight)
  )


head(
  high_weight_relationships,
  10
)


write_csv(
  high_weight_relationships,
  file.path(
    output_folder,
    "highest_weight_relationships.csv"
  )
)


# ==============================================================
# 28. NETWORK SUMMARY
# ==============================================================

network_summary <- data.frame(

  Number_of_Nodes =
    vcount(aviation_network),

  Number_of_Edges =
    ecount(aviation_network),

  Network_Density =
    network_density,

  Average_Degree =
    average_degree,

  Average_Weighted_Degree =
    average_weighted_degree,

  Weakly_Connected_Components =
    components(
      aviation_network,
      mode = "weak"
    )$no,

  Strongly_Connected_Components =
    components(
      aviation_network,
      mode = "strong"
    )$no,

  Average_Shortest_Path =
    average_path_length,

  Network_Diameter =
    network_diameter
)


network_summary


write_csv(
  network_summary,
  file.path(
    output_folder,
    "network_summary.csv"
  )
```

Figure 106 Task B R Script Code 19

```r
# ============================================================
# 29. SAVE CLEANED NETWORK DATA
# ============================================================

write_csv(
  network_data,
  file.path(
    output_folder,
    "cleaned_SriLanka_Aviation_SNA_Dataset.csv"
  )
)


# ============================================================
# 30. FINAL OUTPUT MESSAGE
# ============================================================

cat(
  "\n============================================================\n",
  "TASK B ANALYSIS COMPLETED\n",
  "============================================================\n",
  "Number of nodes: ",
  vcount(aviation_network),
  "\nNumber of relationships: ",
  ecount(aviation_network),
  "\nNetwork density: ",
  round(network_density, 4),
  "\nAverage degree: ",
  round(average_degree, 4),
  "\nAverage weighted degree: ",
  round(average_weighted_degree, 4),
  "\nWeak components: ",
  components(
    aviation_network,
    mode = "weak"
  )$no,
  "\nStrong components: ",
  components(
    aviation_network,
    mode = "strong"
  )$no,
  "\n\nResults and figures have been saved to:\n",
  output_folder,
  "\n============================================================\n"
)
```

Figure 107 Task B R Script Code 20

Appendix C
TASK C
C.1 QGIS Project Setup and Coordinate Reference System

- QGIS project overview

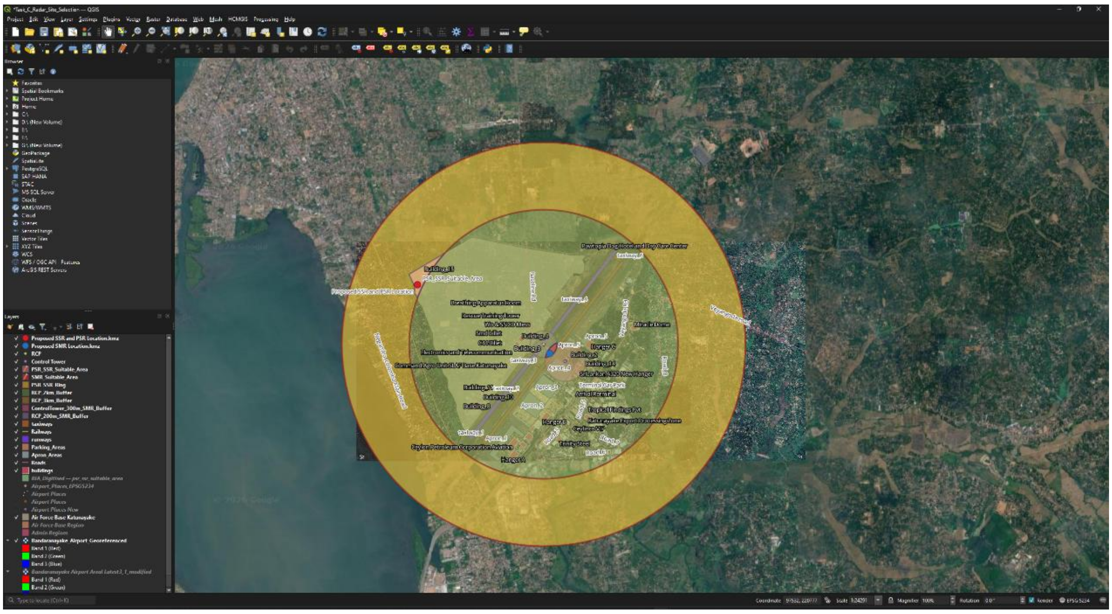


Figure 108 QGIS project overview

- Project CRS showing **EPSG:5234**

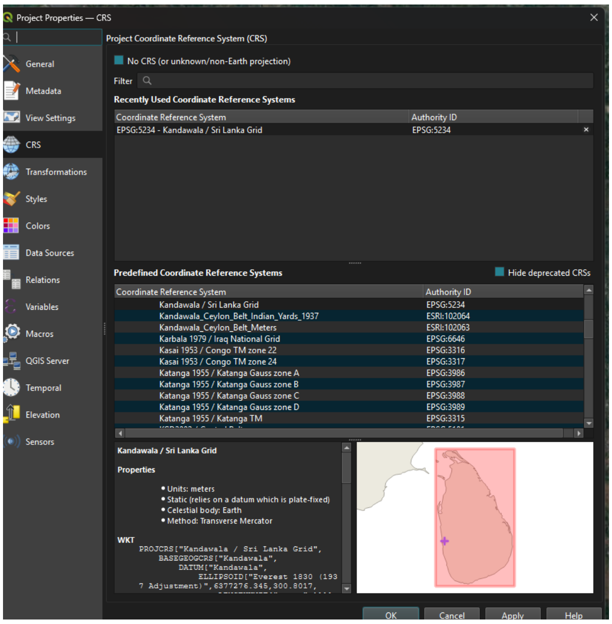


Figure 109 Project CRS showing EPSG:5234

- Layers panel showing the main GIS layers

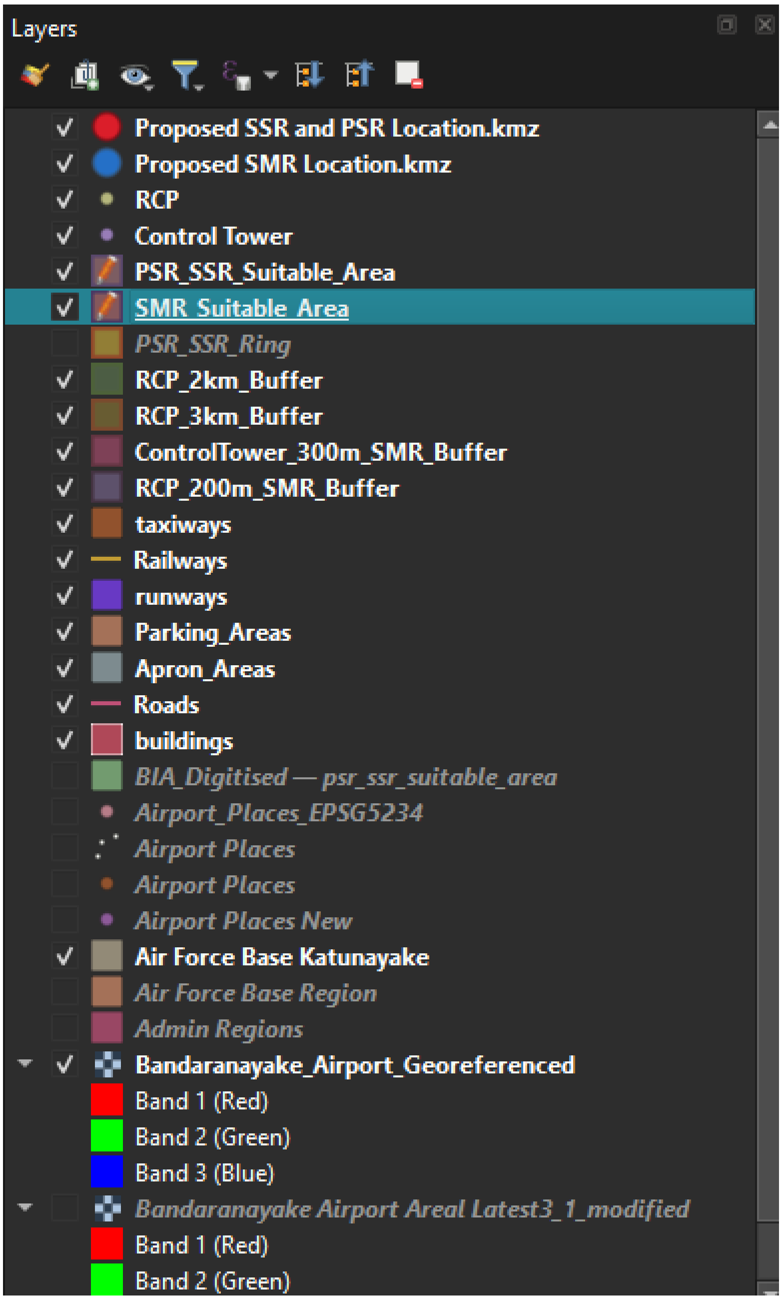


Figure 110 Layers panel showing the main GIS layers

C.2 Georeferencing

- Georeferencing setup

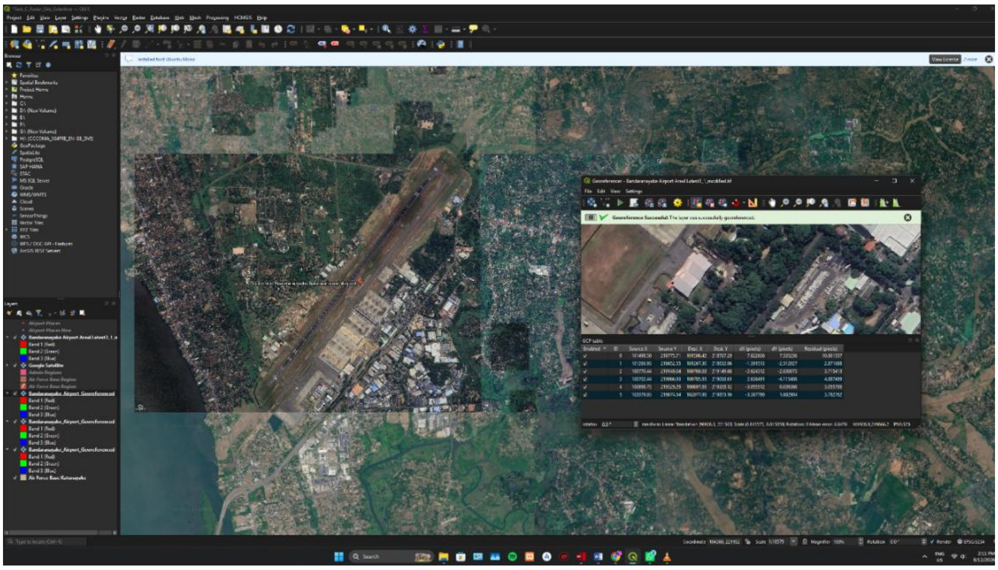


Figure 111 Georeferencing setup

- Final georeferenced aerial imagery

Figure 112 Final georeferenced aerial imagery

C.3 Digitized Airport Features

- Buildings

Figure 113 Buildings Layer

- Runways

Figure 114 Runways Layer

- Taxiways

Figure 115 Taxiways Layer

- Aprons

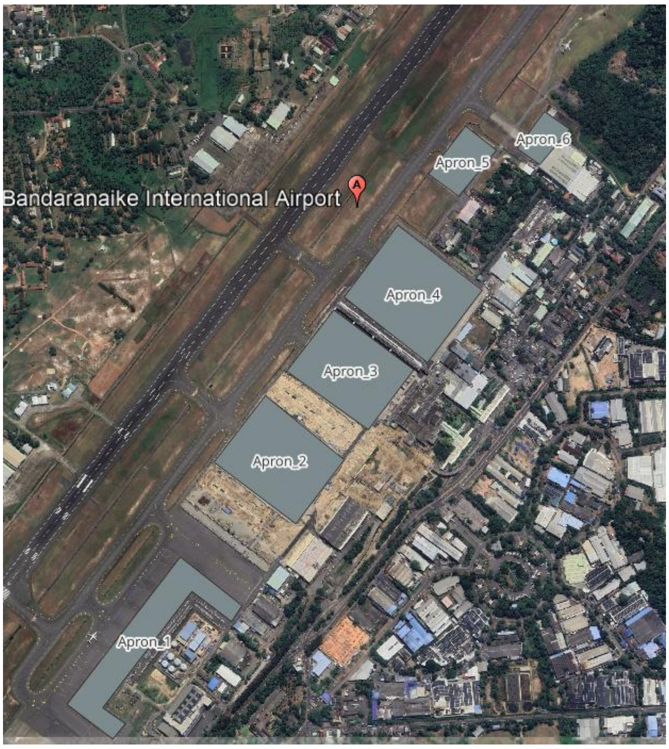


Figure 116 Aprons Layer

- Roads

Figure 117 Roads layer

- Railways

Figure 118 Railways

C.4 Google Earth and KML/KMZ Data

- Google Earth reference locations

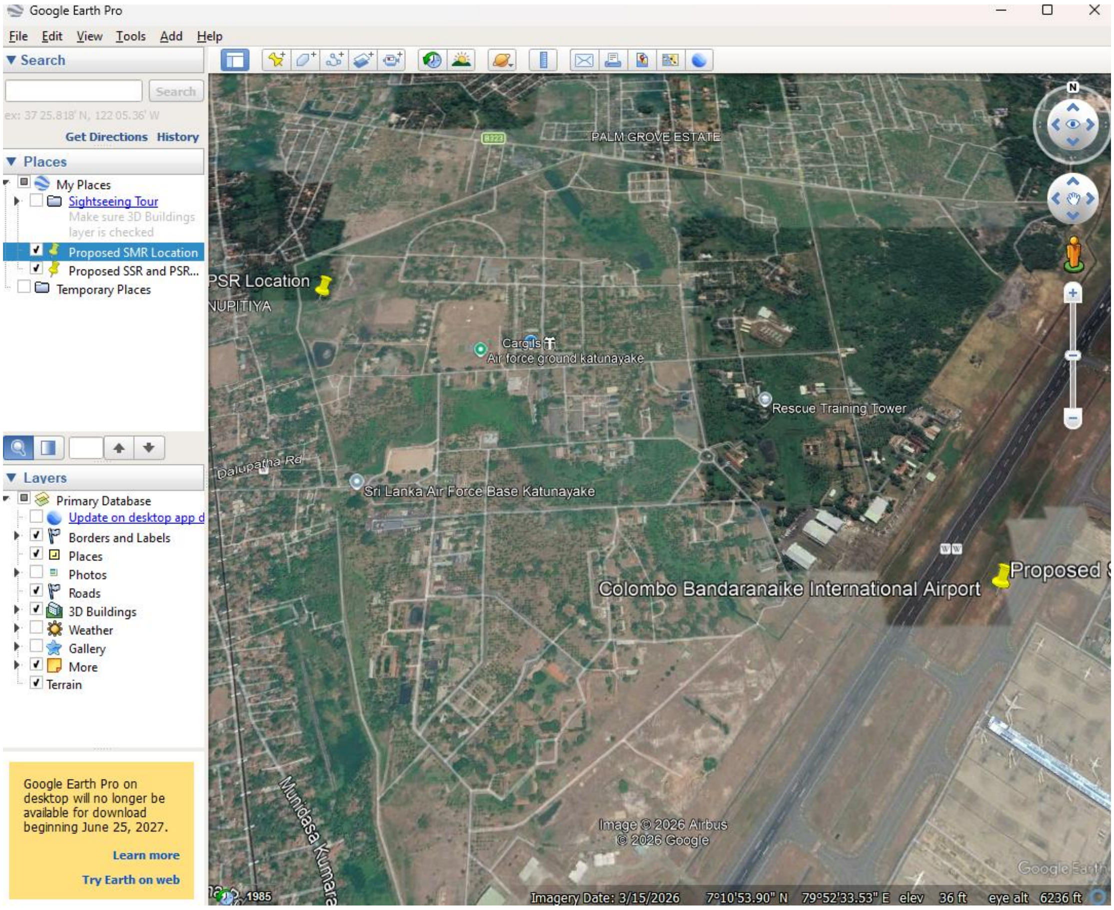
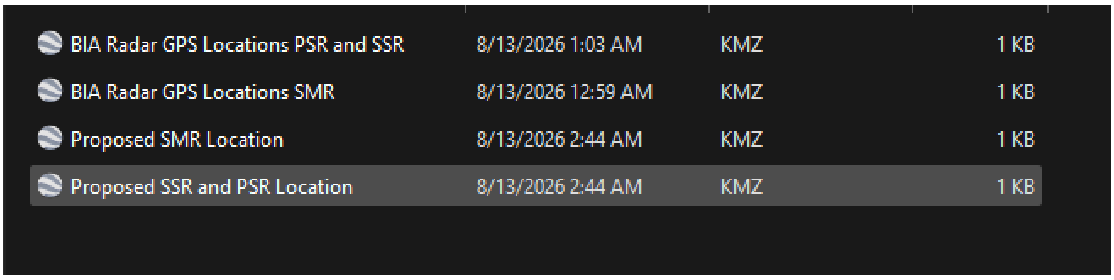


Figure 119 Google Earth reference locations

- KML/KMZ data used in the analysis



BIA Radar GPS Locations PSR and SSR	8/13/2026 1:03 AM	KMZ	1 KB
BIA Radar GPS Locations SMR	8/13/2026 12:59 AM	KMZ	1 KB
Proposed SMR Location	8/13/2026 2:44 AM	KMZ	1 KB
Proposed SSR and PSR Location	8/13/2026 2:44 AM	KMZ	1 KB

Figure 120 KML/KMZ data used in the analysis

- Imported KML/KMZ layers in QGIS

Figure 121 Imported KML/KMZ layers in QGIS

- RCP and Control Tower reference points

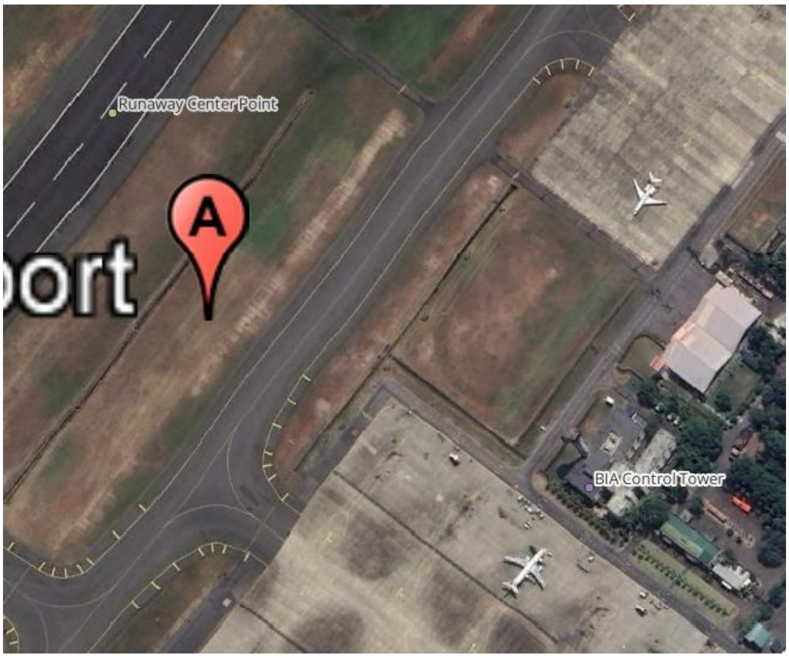


Figure 122 RCP and Control Tower reference points

C.5 PostgreSQL/PostGIS Database

- PostgreSQL connection in QGIS

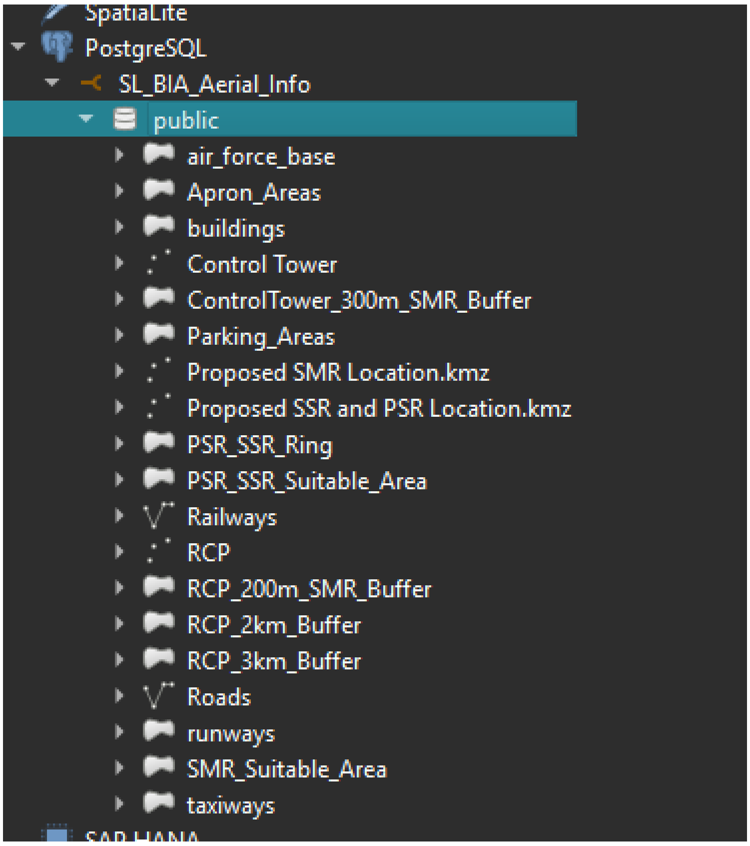


Figure 123 PostgreSQL connection in QGIS

- **SL_BIA_Aerial_Info** database

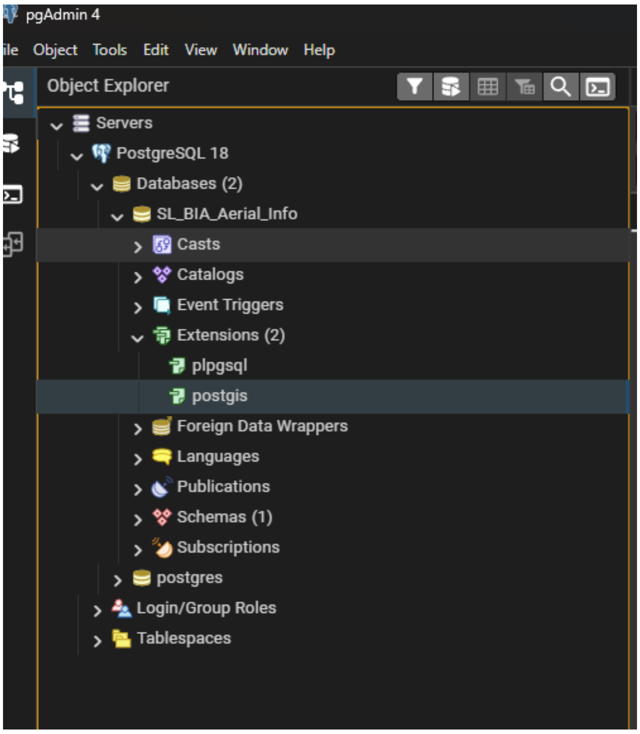


Figure 124 SL_BIA_Aerial_Info database

- Database tables/layers

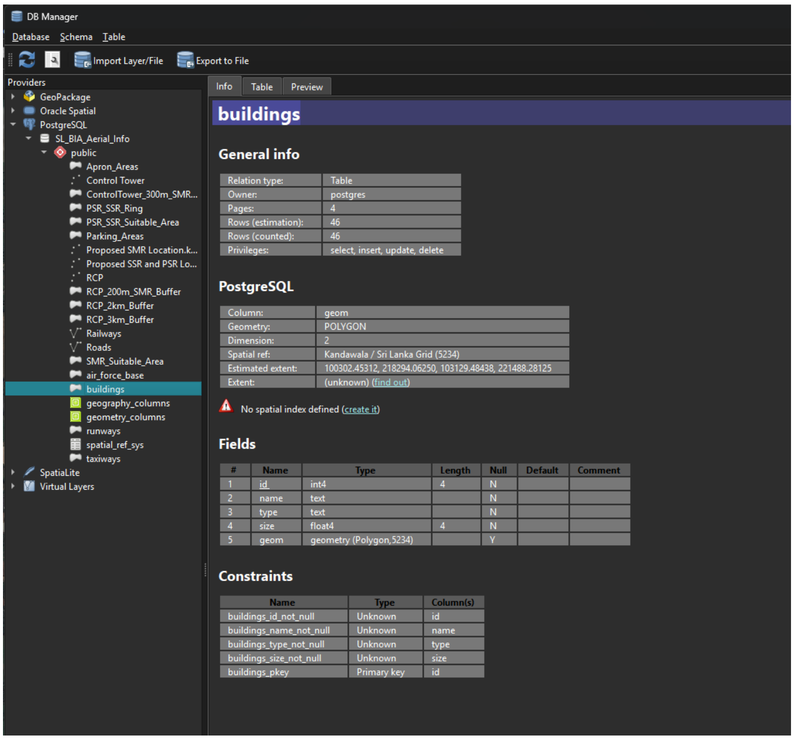


Figure 125 Database tables/layers

C.6 SMR Spatial Analysis

- RCP **200 m buffer**

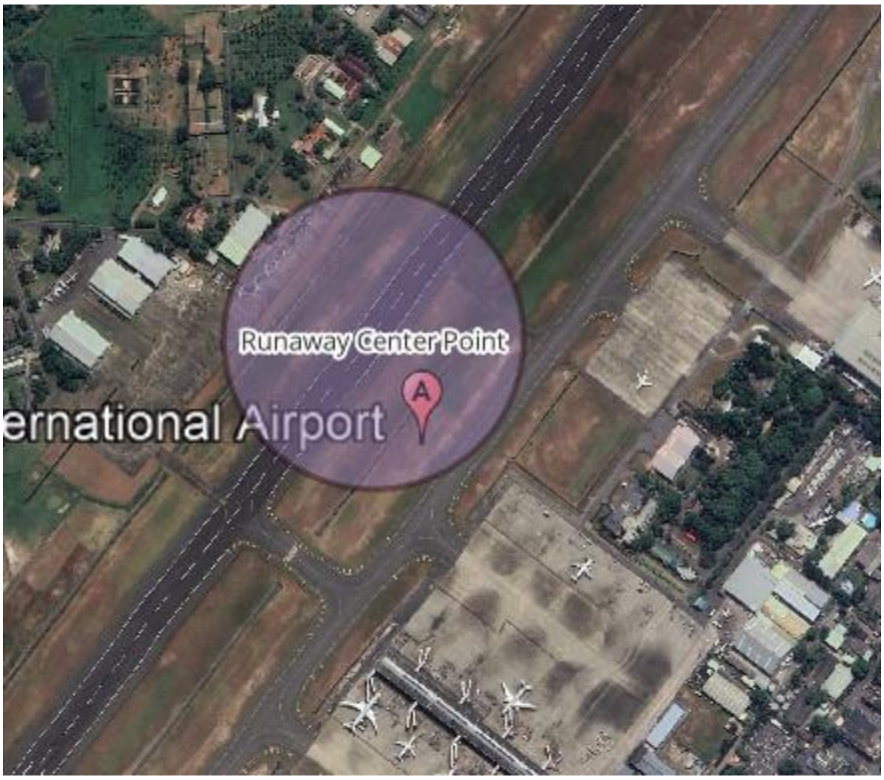


Figure 126 RCP 200 m buffer

- Control Tower **300 m buffer**

Figure 127 Control Tower 300 m buffer

- Intersection of the two buffers

Figure 128 Intersection of the two buffers

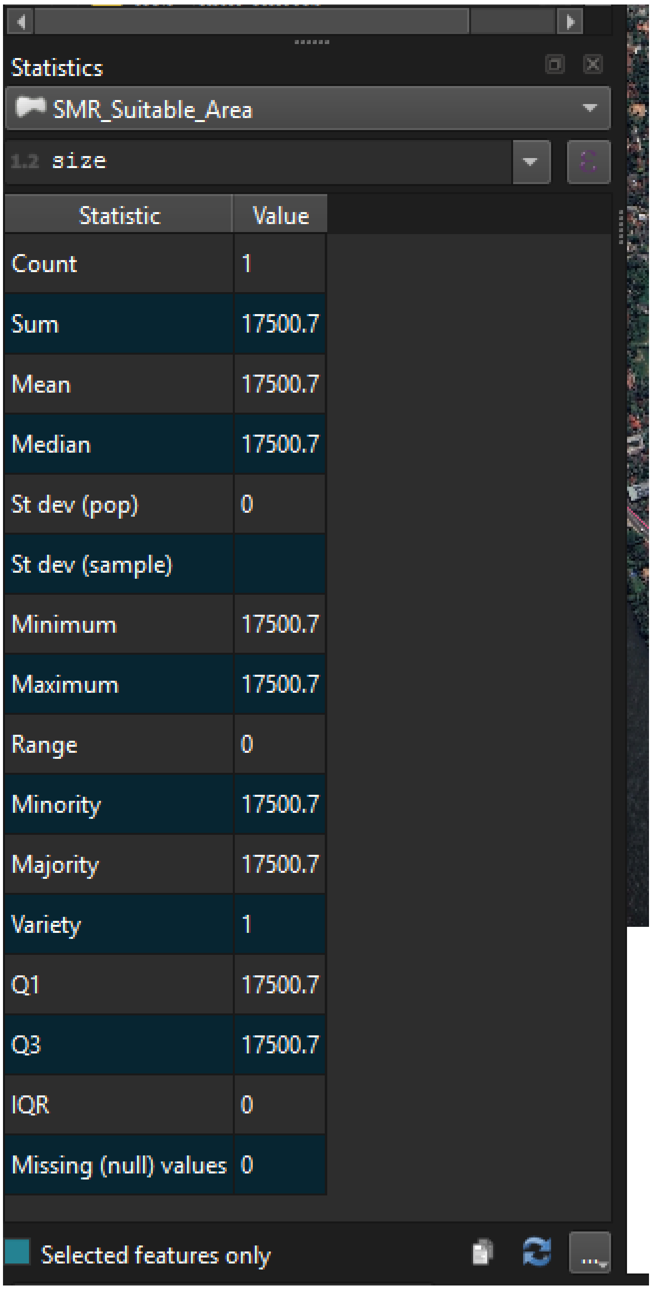
- SMR suitable area

Figure 129 SMR suitable area

- Select by Location/building obstruction result

Figure 130 Select by Location/building obstruction result

- Total Area of the SMR_Suitable_Area



Statistic	Value
Count	1
Sum	17500.7
Mean	17500.7
Median	17500.7
St dev (pop)	0
St dev (sample)	
Minimum	17500.7
Maximum	17500.7
Range	0
Minority	17500.7
Majority	17500.7
Variety	1
Q1	17500.7
Q3	17500.7
IQR	0
Missing (null) values	0

Figure 131 Total Area of the SMR_Suitable_Area

C.7 PSR/SSR Spatial Analysis

- RCP **2 km buffer**

Figure 132 RCP 2 km buffer

- RCP **3 km buffer**

Figure 133 RCP 3 km buffer

- PSR/SSR candidate ring/area

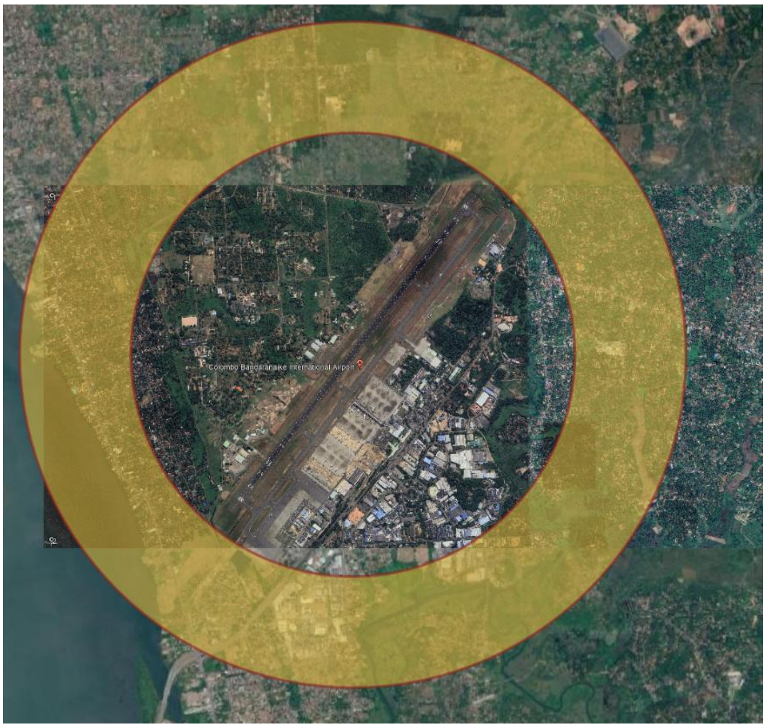


Figure 134 PSR/SSR candidate ring/area

- Difference/exclusion analysis

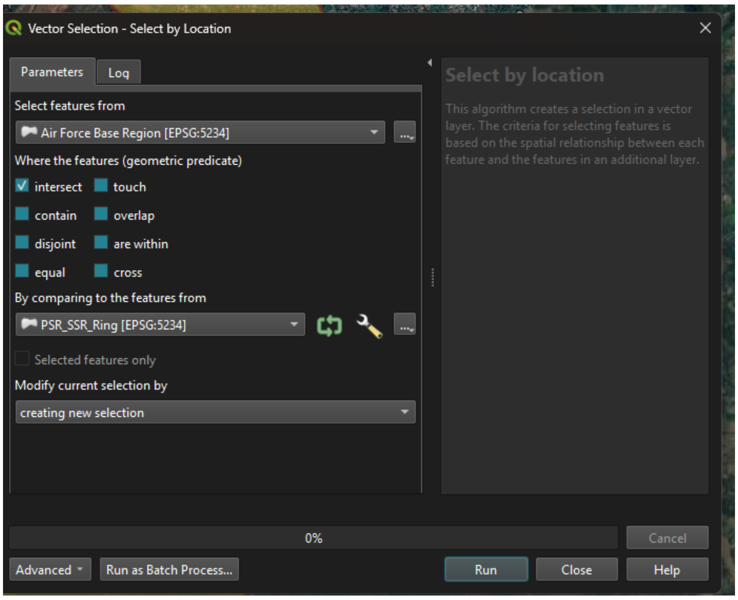


Figure 135 Difference/exclusion analysis

- Selected building within the suitable area

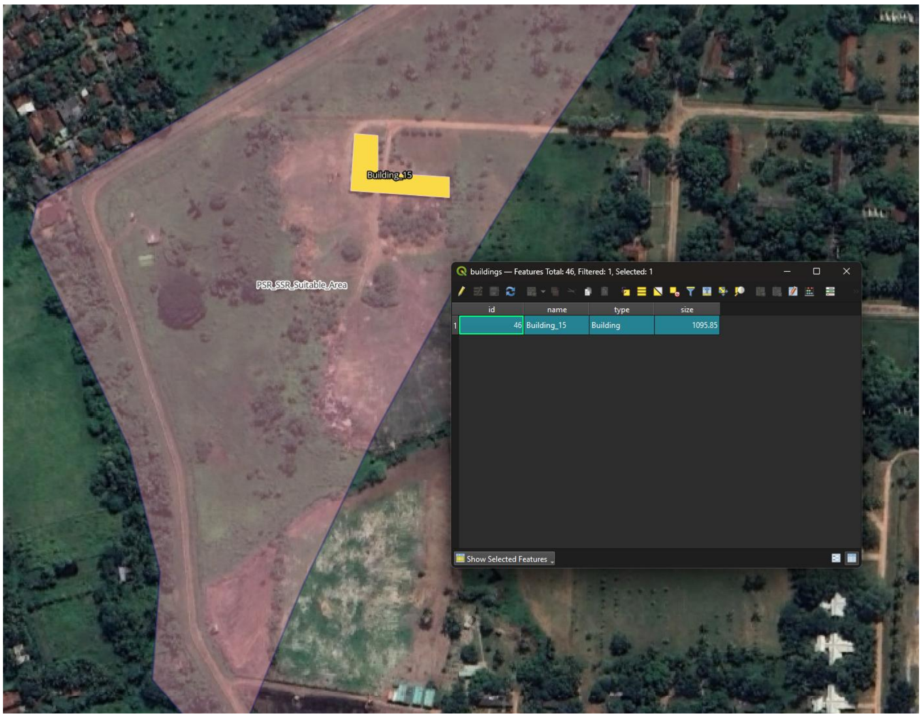
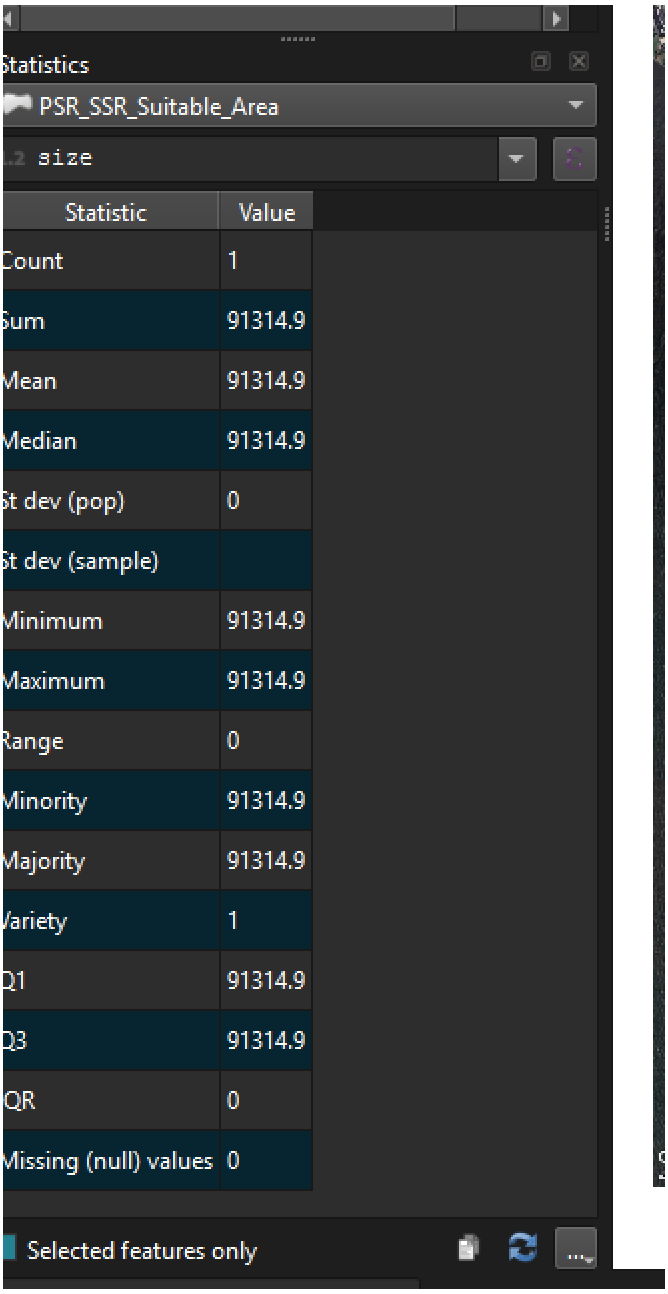


Figure 136 Selected building within the suitable area

- Final PSR/SSR suitable area

Figure 137 Final PSR/SSR suitable area

- Total Area of the PSR_SSR_Suitable_Area



Statistics

PSR_SSR_Suitable_Area

size

Statistic	Value
Count	1
Sum	91314.9
Mean	91314.9
Median	91314.9
St dev (pop)	0
St dev (sample)	
Minimum	91314.9
Maximum	91314.9
Range	0
Minority	91314.9
Majority	91314.9
Variety	1
Q1	91314.9
Q3	91314.9
IQR	0
Missing (null) values	0

Selected features only

Figure 138 Total Area of the PSR_SSR_Suitable_Area

C.8 Radar Location Results

- Proposed PSR/SSR location

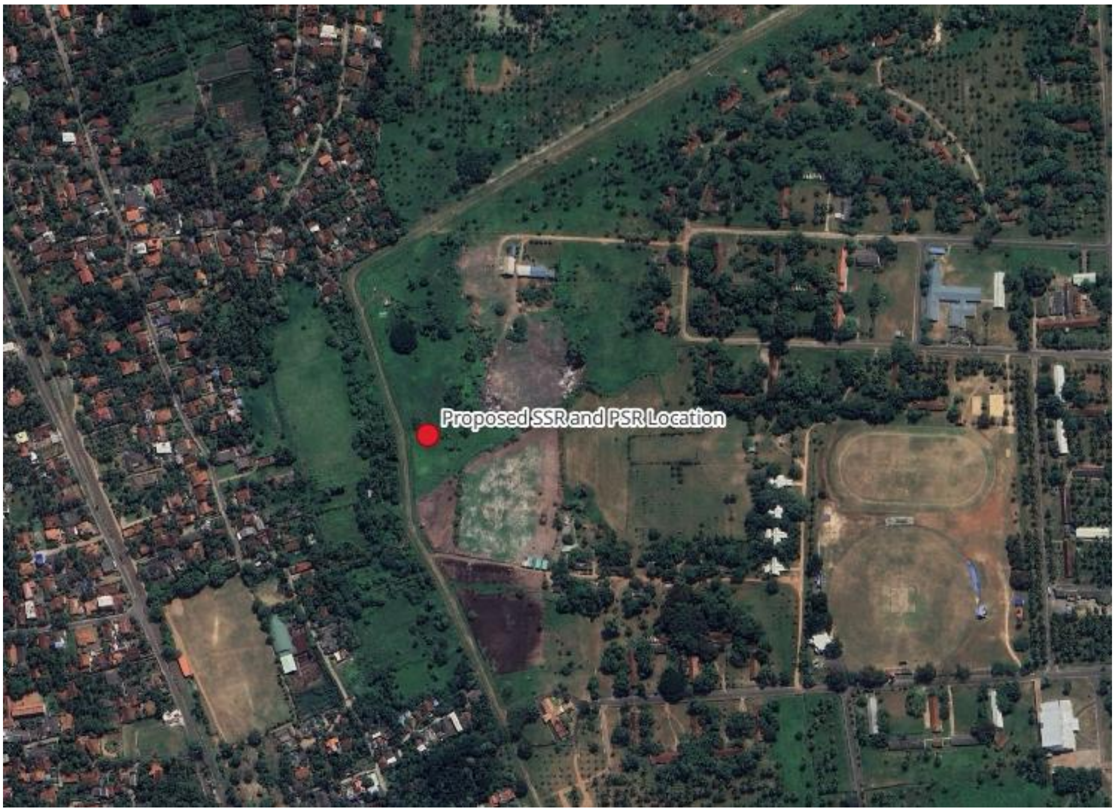


Figure 139 Proposed PSR/SSR location

- Proposed SMR location

Figure 140 Proposed SMR location

C.9 Final Map

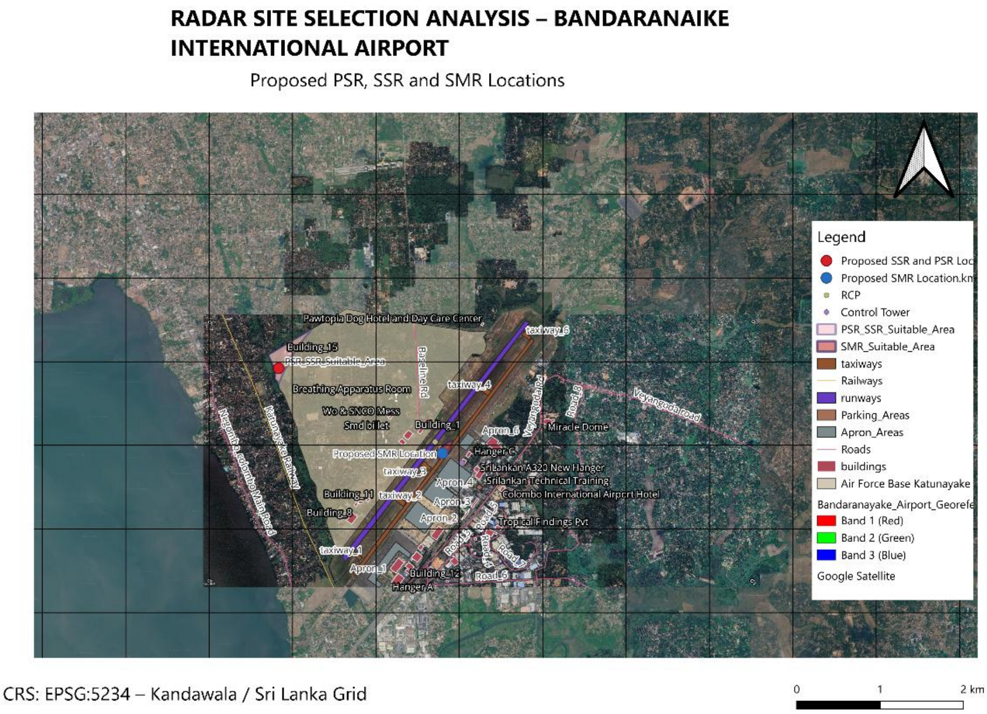


Figure 141 Final Map

APPENDIX D
Task D
Dashboard

Figure 142 BI Dashboard